A Qubit in a Bosonic Bath

Keldysh, Nakajima–Zwanzig and Mori:
ten lectures on one open quantum system

Self-contained lecture notes with worked examples, numerics and exercises

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19 October 2026

Abstract

These notes develop, from scratch and in one consistent notation, the three standard languages for the dynamics of a small quantum system coupled to a large environment: the Keldysh (closed-time-path) formalism, the Nakajima–Zwanzig and time-convolutionless projection-operator master equations, and Mori's projection onto operators. The thread that ties them together is a single model, a two-level system (qubit) coupled linearly to a bath of harmonic oscillators through a general coupling operator $S = \cos\theta\,\sigma_z + \sin\theta\,\sigma_x$, which interpolates between pure dephasing ($\theta = 0$) and the dissipative two-level system of Leggett *et al.* Every formal construction is followed immediately by its evaluation for this model, and a committed Python script produces the figures, so that each approximation can be compared with an exact reference. The last lecture turns the machinery toward quantum sensing. References to the original literature are given throughout.

How to use these notes. Each lecture is meant for one sitting of two to three hours with pen and paper. Boxes labelled *Example* carry the running model; boxes labelled *Key point* are the statements you should be able to reproduce from memory. Exercises at the end of each lecture have solutions or detailed hints in Appendix B. The numerical figures are generated by `figures/spin_boson_numerics.py`; running it takes about a minute and every parameter is listed in Appendix A. These notes are self-contained but build on the shorter working note `keldysh_mori_zwanzig_v2` in the same folder, whose notation they share.

About the references. Entries carried over from `refs.bib` were checked against publisher records on 30 September 2026. Entries added for these notes are marked in the `.bib` file; their bibliographic data were written from memory of the literature and should be verified before being quoted in a publication. All equations were derived independently and the central ones are checked numerically in Appendix A.

Contents

Lecture 0: The map of the course **4**

1 Lecture 1: Liouville space, superoperators and the qubit **6**
1.1 Operators as vectors . . . 6
1.2 Superoperators . . . 6
1.3 Dynamical maps . . . 9
1.4 Exercises . . . 9

2 Lecture 2: The model, its bath, and the two kernels **11**
2.1 The Hamiltonian and its two faces . . . 11
2.2 Bath correlation function; noise and response . . . 12
2.3 The kernels in the frequency domain . . . 12
2.4 Why only $J(\omega)$ matters: Gaussianity and Wick's theorem . . . 15
2.5 Initial state and pictures . . . 16
2.6 Exercises . . . 16

3 Lecture 3: The Keldysh contour is already in the von Neumann equation **17**
3.1 The Dyson series for ρ and the origin of the contour . . . 17
3.2 The Keldysh rotation in Liouville space . . . 18
3.3 Bath contractions and the 2×2 Keldysh structure . . . 19
3.4 Exercises . . . 20

4 Lecture 4: Real-time diagrams for the reduced density matrix **21**
4.1 Feynman rules . . . 21
4.2 Second order: the Born kernel . . . 21
4.3 Pure dephasing to all orders; reducible and irreducible diagrams . . . 22
4.4 What the diagrams are good for . . . 23
4.5 Exercises . . . 24

5 Lecture 5: Nakajima–Zwanzig, and why the memory kernel is a self-energy **25**
5.1 The projection and the exact equation . . . 25
5.2 The kernel is the sum of irreducible diagrams . . . 26
5.2.1 Vocabulary: strings, gaps, cuts, pairings . . . 26
5.2.2 Three lemmas . . . 26
5.2.3 The theorem . . . 27
5.2.4 The converse: the NZ equation as a Dyson resummation . . . 28
5.2.5 Explicit checks at fourth and sixth order . . . 29
5.2.6 What the theorem does and does not say . . . 30
5.3 Second order: Born, Redfield, secular, Bloch . . . 31
5.4 Beyond second order . . . 33
5.5 Correlated initial states and the inhomogeneity . . . 34
5.6 Exercises . . . 34

6 Lecture 6: Time-convolutionless equations and the NZ–TCL map **35**
6.1 Derivation . . . 35
6.2 TCL2 for the qubit: time-dependent rates . . . 35
6.3 The NZ–TCL map . . . 36
6.4 When the generator does not exist . . . 36
6.5 What to take from Figs. 5 and 6 . . . 37
6.6 Exercises . . . 37

7 Lecture 7: Path integrals: the influence functional, the polaron frame and NIBA **38**
7.1 The influence functional as Wick's theorem with a commutative system . . . 38
7.2 Spin paths: blips and sojourns . . . 39
7.3 NIBA is the second-order NZ kernel in the polaron frame . . . 39
7.4 What the Keldysh field theory adds . . . 40

7.5 Exercises . . . 41

8 Lecture 8: Mori's projection, the generalized Langevin equation and the retarded self-energy 42
8.1 The generalized Langevin equation . . . 42
8.2 Mori on the complete basis is the adjoint of Nakajima–Zwanzig . . . 42
8.3 Example: Mori for the qubit's lowering operator . . . 43
8.4 What Mori does *not* give you . . . 44
8.5 GLE $\leftrightarrow$ Keldysh saddle point . . . 44
8.6 Exercises . . . 45

9 Lecture 9: The qubit as a Keldysh field theory 46
9.1 The generating functional and its two kinds of sources . . . 46
9.2 The bath as a Gaussian field; integrating it out . . . 46
9.3 Representing the spin . . . 46
9.4 Markovian limit: the Keldysh action of a Lindblad equation . . . 47
9.5 Self-consistent resummations . . . 48
9.6 Dictionary . . . 48
9.7 Exercises . . . 48

10 Lecture 10: Quantum sensing: which protocol measures which kernel 50
10.1 Relaxometry: the transverse sensor measures $C(\pm\omega_0)$ . . . 50
10.2 Dephasing: the longitudinal sensor measures $S(\omega)$ through a filter . . . 50
10.3 The mixed sensor . . . 51
10.4 Precision scaling: when memory matters . . . 51
10.5 Which formalism for which question . . . 52
10.6 Exercises . . . 53

A The numerical companion 54

B Solutions and hints 55

Lecture 0: The map of the course

The problem. A small quantum system S (here: a qubit) is coupled to a large environment B (here: a bath of harmonic oscillators). The full dynamics is unitary, but we only ever look at the system, so what we want is the *reduced* dynamics: either the reduced density matrix $\rho_S(t) = \mathrm{Tr}_B\, \rho(t)$, or the expectation values and correlation functions of system operators. Three formalisms were invented, by different communities and for different purposes, to do exactly this:

- **Keldysh** (Schwinger [1], Keldysh [2]): write expectation values as a path integral or a diagrammatic series on a closed time contour that runs forward with U and backward with $U^\dagger$. Natural for many-body systems, Green's functions and field theory.
- **Nakajima–Zwanzig** (Nakajima [3], Zwanzig [4]) and its time-convolutionless (TCL) variant [5, 6]: project the full density matrix onto the "relevant" part $\rho_S \otimes \rho_B$ and derive a closed, exact integro-differential equation for $\rho_S(t)$. Natural for small systems and for questions about positivity, memory and master equations.
- **Mori** (Mori [7], Zwanzig [8, Ch. 8]): project *operators* (Heisenberg picture) onto a chosen set of slow variables and derive a generalized Langevin equation. Natural for hydrodynamics, response functions and continued-fraction approximations.

The thesis. The three formalisms are not three theories; they are three bookkeeping schemes for the same perturbation series. The precise statements, proved in the course, are:

The three bridges

1. Ordering superoperators along the real time axis *is* Keldysh contour ordering (Section 3). The "classical/quantum" Keldysh rotation is the pair of superoperators $A^{\mathrm{c}} = \frac{1}{2}\{A, \cdot\}$, $A^{\mathrm{q}} = [A, \cdot]$.
2. The Nakajima–Zwanzig memory kernel is the sum of *irreducible* real-time diagrams of the reduced propagator, and the NZ equation is its Dyson equation (Section 5). TCL is the same Dyson equation solved for a time-local generator (Section 6).
3. Mori's memory function is the *retarded self-energy* of the projected operator; for the full operator basis with the Heisenberg-adjoint inner product it is the adjoint of the NZ kernel (Section 8).

In all three, a Gaussian bath enters only through two kernels: the symmetrized noise $S(t)$ (Keldysh component D^K) and the response $\chi(t)$ (retarded component D^R).

The running example. Everything is worked out for

$$H = \frac{\omega_0}{2}\sigma_z + \sum_k \omega_k b_k^\dagger b_k + \big(\cos\theta\, \sigma_z + \sin\theta\, \sigma_x\big) \otimes \sum_k g_k \big(b_k + b_k^\dagger\big), \tag{1}$$

the spin-boson model with a general linear coupling. The mixing angle θ lets us switch between the two textbook limits without changing the formalism: $\theta = 0$ is pure dephasing (exactly solvable, the sensor's workhorse) and $\theta = \pi/2$ is transverse coupling (relaxation, the damped qubit). Intermediate θ is unitarily equivalent to the biased dissipative two-level system of Leggett *et al.* [9], as we show in Section 2. A Lorentzian spectral density makes the model numerically exact through the pseudomode mapping [10, 11], so each approximation in the course can be compared with the truth (Appendix A).

The route. Sections 1 and 2 set up the language (Liouville space; the model and its bath). Sections 3 and 4 derive the Keldysh structure and the diagram rules from the von Neumann equation. Sections 5 and 6 derive NZ and TCL and identify them with the Dyson equation. Section 7 goes to the path integral: Feynman–Vernon, the polaron frame and the noninteracting-blip approximation. Section 8 does Mori and connects it to the other two. Section 9 rewrites the qubit problem as a Keldysh field theory, so the reader can recognize the structures in the many-body literature. Section 10 turns the machinery to quantum sensing: which protocol measures which bath kernel, and why non-Markovian memory changes precision scaling.

Prerequisites. Quantum mechanics at the level of second quantization for bosons, the interaction picture and the Dyson series; a first exposure to density matrices. No prior knowledge of Keldysh, projection operators or path integrals is assumed. Units: $\hbar = k_B = 1$.

1 Lecture 1: Liouville space, superoperators and the qubit

Open-system theory is linear algebra on the space of operators. This lecture makes that space concrete for a qubit, so that in later lectures every abstract superoperator can be written as a 4×4 matrix and checked by hand.

1.1 Operators as vectors

Let $\mathcal{H}$ be a d-dimensional Hilbert space. The linear operators on $\mathcal{H}$ form a d^2-dimensional vector space, *Liouville space*, with the Hilbert–Schmidt inner product

$$(A, B) \equiv \mathrm{Tr}\left(A^\dagger B\right). \tag{2}$$

We write $|A\rangle\!\rangle$ for an operator regarded as a vector and $\langle\!\langle A|B\rangle\!\rangle = (A, B)$. For a qubit ($d = 2$) an orthonormal basis is

$$|e_0\rangle\!\rangle = \tfrac{1}{\sqrt{2}}\mathbb{1}, \quad |e_1\rangle\!\rangle = \tfrac{1}{\sqrt{2}}\sigma_x, \quad |e_2\rangle\!\rangle = \tfrac{1}{\sqrt{2}}\sigma_y, \quad |e_3\rangle\!\rangle = \tfrac{1}{\sqrt{2}}\sigma_z, \tag{3}$$

since $\mathrm{Tr}(\sigma_i\sigma_j) = 2\delta_{ij}$. Any operator is $X = \frac{1}{2}(x_0\mathbb{1} + \boldsymbol{x}\cdot\boldsymbol{\sigma})$ with $x_\mu = \mathrm{Tr}(\sigma_\mu X)$, and a density matrix is

$$\rho = \tfrac{1}{2}\big(\mathbb{1} + \boldsymbol{r}\cdot\boldsymbol{\sigma}\big), \qquad \boldsymbol{r} = \langle\boldsymbol{\sigma}\rangle, \qquad |\boldsymbol{r}| \le 1, \tag{4}$$

the Bloch vector [12]. Throughout the notes we use the *coefficient convention*: a superoperator $\mathcal{M}$ is represented by the 4×4 matrix $M_{\mu\nu}$ defined by $\mathcal{M}\sigma_\nu = \sum_\mu M_{\mu\nu}\sigma_\mu$, i.e. $M_{\mu\nu} = \frac{1}{2}\,\mathrm{Tr}(\sigma_\mu\,\mathcal{M}\sigma_\nu)$, which is also the matrix in the orthonormal basis $|e_\mu\rangle\!\rangle$. It acts on the coefficient vector (x_0, x_x, x_y, x_z).

A second, equivalent representation is *vectorization*: stack the columns of the $d \times d$ matrix X into a d^2-vector $\mathrm{vec}(X)$. The identity

$$\mathrm{vec}(AXB) = \big(B^T \otimes A\big)\,\mathrm{vec}(X) \tag{5}$$

turns every "sandwich" into a matrix. This is what the numerical script uses (Appendix A); the coefficient convention is what we use on paper. They are related by the unitary change of basis from $\{|i\rangle\langle j|\}$ to $\{\sigma_\mu/\sqrt{2}\}$.

1.2 Superoperators

A *superoperator* is a linear map on Liouville space. The two elementary ones are left and right multiplication by a fixed operator A,

$$A^+X \equiv AX, \qquad A^-X \equiv XA. \tag{6}$$

The superscripts anticipate Section 3: A^+ acts on the ket side of ρ (the forward branch of the Keldysh contour), A^- on the bra side (the backward branch). Left and right multiplications by different operators always commute, $A^+B^- = B^-A^+$, because $(AX)B = A(XB)$. In vectorized form, $A^+ = \mathbb{1}\otimes A$ and $A^- = A^T\otimes\mathbb{1}$ by Eq. (5).

Products of operators. What is the superoperator of a product AB? On the left side the order is preserved, on the right side it is reversed:

$$(AB)^+X = ABX = A^+B^+X, \qquad (AB)^-X = XAB = (XA)B = B^-A^-X, \tag{7}$$

so that

$$\boxed{(AB)^+ = A^+B^+, \qquad (AB)^- = B^-A^-.} \tag{8}$$

The map $A \mapsto A^+$ is an algebra homomorphism, the map $A \mapsto A^-$ is an *anti*-homomorphism (in vectorized form this is just $(AB)^T = B^T A^T$). This innocent-looking reversal is the entire origin of the Keldysh contour: a string of operators acting on the bra side of ρ is applied in the opposite order, which is why the backward branch is anti-time-ordered (Section 3). Two consequences follow immediately: $(A^n)^\pm = (A^\pm)^n$ for powers of a single operator, and the commutators

$$[A^+, B^+] = \big([A,B]\big)^+, \qquad [A^-, B^-] = -\big([A,B]\big)^-, \qquad [A^+, B^-] = 0, \tag{9}$$

the minus sign in the second being the anti-homomorphism again. The Hermitian conjugates with respect to Eq. (2) are $(A^+)^\dagger = (A^\dagger)^+$ and $(A^-)^\dagger = (A^\dagger)^-$, since $\mathrm{Tr}[Y^\dagger AX] = \mathrm{Tr}[(A^\dagger Y)^\dagger X]$ and $\mathrm{Tr}[Y^\dagger XA] = \mathrm{Tr}[(YA^\dagger)^\dagger X]$.

From these we build the two combinations that organize everything that follows:

$$\boxed{A^{\mathrm{c}} \equiv \tfrac{1}{2}\big(A^+ + A^-\big), \quad A^{\mathrm{c}}X = \tfrac{1}{2}\{A, X\}; \qquad A^{\mathrm{q}} \equiv A^+ - A^-, \quad A^{\mathrm{q}}X = [A, X].} \tag{10}$$

Inversely $A^\pm = A^{\mathrm{c}} \pm \frac{1}{2}A^{\mathrm{q}}$. For Hermitian A both A^{c} and A^{q} map Hermitian operators to Hermitian operators (the anticommutator and $i\times$commutator of Hermitian operators are Hermitian; note $A^{\mathrm{q}}X$ is anti-Hermitian for Hermitian X and the i is supplied by the Liouvillian).

Three properties used constantly

For any operator A on $\mathcal{H}$ and any X:

1. $\mathrm{Tr}(A^{\mathrm{q}}X) = \mathrm{Tr}[A, X] = 0$: the quantum superoperator maps into the traceless subspace.
2. $\mathrm{Tr}(A^{\mathrm{c}}X) = \mathrm{Tr}(AX)$: the classical superoperator "measures" A.
3. $[A^{\mathrm{c}}, A^{\mathrm{q}}] = 0$ and $A^{\mathrm{c}}A^{\mathrm{q}}X = \frac{1}{2}\big(A^2X - XA^2\big) = \frac{1}{2}(A^2)^{\mathrm{q}}X$.

The first two are the cyclicity of the trace. The third follows from $A^+A^- = A^-A^+$.

Products in the rotated variables. The product rule Eq. (8) translates into the c/q language as

$$\begin{aligned}(AB)^{\mathrm{c}} &= \tfrac{1}{2}\{A^{\mathrm{c}}, B^{\mathrm{c}}\} + \tfrac{1}{8}\{A^{\mathrm{q}}, B^{\mathrm{q}}\} + \tfrac{1}{4}\big([A^{\mathrm{c}}, B^{\mathrm{q}}] + [A^{\mathrm{q}}, B^{\mathrm{c}}]\big),\\ (AB)^{\mathrm{q}} &= \tfrac{1}{2}\{A^{\mathrm{c}}, B^{\mathrm{q}}\} + \tfrac{1}{2}\{A^{\mathrm{q}}, B^{\mathrm{c}}\} + [A^{\mathrm{c}}, B^{\mathrm{c}}] + \tfrac{1}{4}[A^{\mathrm{q}}, B^{\mathrm{q}}],\end{aligned} \tag{11}$$

obtained by inserting $A^\pm = A^{\mathrm{c}} \pm \frac{1}{2}A^{\mathrm{q}}$ into $(AB)^{\mathrm{c}} = \frac{1}{2}(A^+B^+ + B^-A^-)$ and $(AB)^{\mathrm{q}} = A^+B^+ - B^-A^-$ (Exercise 1.5). When the superoperators of A and B commute, which is the case whenever A and B act on different tensor factors (system and bath) or more generally whenever $[A, B] = 0$, the commutator terms drop and the rule collapses to the familiar Keldysh rule for composite fields,

$$(AB)^{\mathrm{c}} = A^{\mathrm{c}}B^{\mathrm{c}} + \tfrac{1}{4}A^{\mathrm{q}}B^{\mathrm{q}}, \qquad (AB)^{\mathrm{q}} = A^{\mathrm{c}}B^{\mathrm{q}} + A^{\mathrm{q}}B^{\mathrm{c}} \qquad ([A, B] = 0), \tag{12}$$

which is exactly how the interaction vertex $\mathcal{L}_I = -i(S \otimes B)^{\mathrm{q}}$ will acquire its "one classical, one quantum leg" structure in Section 3. For a single operator, $A = B$, Eq. (12) gives $(A^2)^{\mathrm{c}} = (A^{\mathrm{c}})^2 + \frac{1}{4}(A^{\mathrm{q}})^2$ and $(A^2)^{\mathrm{q}} = 2A^{\mathrm{c}}A^{\mathrm{q}}$, which is property 3 of the key box above.

The Liouvillian. The von Neumann equation $\dot\rho = -i[H,\rho]$ reads $\dot\rho = \mathcal{L}\rho$ with

$$\mathcal{L} = -i\big(H^{+} - H^{-}\big) = -iH^{\mathrm{q}}, \qquad \rho(t) = \mathrm{e}^{\mathcal{L}t}\rho(0) = \mathrm{e}^{-iHt}\rho(0)\mathrm{e}^{iHt}. \tag{13}$$

Its adjoint with respect to Eq. (2), defined by $(A, \mathcal{L}X) = (\mathcal{L}^\dagger A, X)$, is $\mathcal{L}^\dagger A = +i[H,A]$, the Heisenberg-picture generator: $\langle A\rangle_t = (A,\rho(t)) = (\mathrm{e}^{\mathcal{L}^\dagger t}A,\rho(0))$. This duality, state picture versus operator picture, is exactly the duality between the Nakajima–Zwanzig projection (Section 5) and the Mori projection (Section 8).

Example 1.1: the free qubit in Bloch form

Take $H_S = \frac{\omega_0}{2}\sigma_z$. Using $[\sigma_z,\sigma_x] = 2i\sigma_y$, $[\sigma_z,\sigma_y] = -2i\sigma_x$,

$$\mathcal{L}_S\sigma_x = -i\tfrac{\omega_0}{2}[\sigma_z,\sigma_x] = \omega_0\sigma_y, \qquad \mathcal{L}_S\sigma_y = -\omega_0\sigma_x, \qquad \mathcal{L}_S\mathbb{1} = \mathcal{L}_S\sigma_z = 0. \tag{14}$$

In the coefficient convention, with rows and columns ordered $(\mathbb{1},\sigma_x,\sigma_y,\sigma_z)$,

$$\mathcal{L}_S = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\omega_0 & 0 \\ 0 & \omega_0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathrm{e}^{\mathcal{L}_S t}: \begin{cases} x_x \mapsto \cos(\omega_0 t)\, x_x - \sin(\omega_0 t)\, x_y, \\ x_y \mapsto \sin(\omega_0 t)\, x_x + \cos(\omega_0 t)\, x_y, \end{cases} \tag{15}$$

so $\dot r_x = -\omega_0 r_y$, $\dot r_y = \omega_0 r_x$: the Bloch vector precesses counter-clockwise about $\hat z$ at ω_0. The eigenvalues of $\mathcal{L}_S$ are $0, 0, \pm i\omega_0$, with eigenvectors $\mathbb{1}, \sigma_z$ (populations) and $\sigma_\mp = \frac{1}{2}(\sigma_x \mp i\sigma_y)$ (coherences): $\mathcal{L}_S\sigma_- = -i\omega_0\sigma_-$. In the basis $\{|0\rangle\langle 0|, |0\rangle\langle 1|, |1\rangle\langle 0|, |1\rangle\langle 1|\}$ the matrix is $\mathrm{diag}(0, -i\omega_0, +i\omega_0, 0)$, which is the "vectorized" form.

Example 1.2: c and q superoperators of the coupling operator

The running model couples the bath to $S = \cos\theta\,\sigma_z + \sin\theta\,\sigma_x$. We need S^{c} and S^{q} as 4×4 matrices. First the building blocks:

$$\begin{aligned} &\sigma_z^{\mathrm{c}}: \mathbb{1}\mapsto\sigma_z,\ \sigma_z\mapsto\mathbb{1},\ \sigma_x\mapsto 0,\ \sigma_y\mapsto 0; \quad \sigma_z^{\mathrm{q}}: \sigma_x\mapsto 2i\sigma_y,\ \sigma_y\mapsto -2i\sigma_x,\ \mathbb{1}\mapsto 0,\ \sigma_z\mapsto 0; \\ &\sigma_x^{\mathrm{c}}: \mathbb{1}\mapsto\sigma_x,\ \sigma_x\mapsto\mathbb{1},\ \sigma_y\mapsto 0,\ \sigma_z\mapsto 0; \quad \sigma_x^{\mathrm{q}}: \sigma_y\mapsto 2i\sigma_z,\ \sigma_z\mapsto -2i\sigma_y,\ \mathbb{1}\mapsto 0,\ \sigma_x\mapsto 0. \end{aligned} \tag{16}$$

(The c map of a Pauli matrix swaps it with the identity; the q map rotates the other two Pauli matrices into each other with a factor $\pm 2i$.) In matrix form, rows and columns ordered $(\mathbb{1},\sigma_x,\sigma_y,\sigma_z)$,

$$S^{\mathrm{c}} = \begin{pmatrix} 0 & \sin\theta & 0 & \cos\theta \\ \sin\theta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \cos\theta & 0 & 0 & 0 \end{pmatrix}, \qquad S^{\mathrm{q}} = 2i\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\cos\theta & 0 \\ 0 & \cos\theta & 0 & -\sin\theta \\ 0 & 0 & \sin\theta & 0 \end{pmatrix}. \tag{17}$$

Read off the three properties: the first row of S^{q} vanishes (traceless-valued), the first row of S^{c} is $(0,\sin\theta,0,\cos\theta) = \frac{1}{2}\,\mathrm{Tr}(S\sigma_\nu)$ (it measures S), and S^{q} is $2i$ times the generator of rotations about the axis $\hat n = (\sin\theta,0,\cos\theta)$ on the Bloch vector, which is what $[S,\cdot]$ should be. Note also S^{q} annihilates S itself (the vector $(0,\sin\theta,0,\cos\theta)$) and the identity. For $\theta = 0$ the matrices reduce to the pure-dephasing case σ_z^{c} = "swap $\mathbb{1}\leftrightarrow\sigma_z$", $\sigma_z^{\mathrm{q}} = 2i\,(\sigma_x\to\sigma_y\to -\sigma_x)$.

1.3 Dynamical maps

For a factorized initial state $\rho(0) = \rho_S(0) \otimes \rho_B$, the reduced state at time t is a linear function of the initial reduced state,

$$\rho_S(t) = \Pi(t)\rho_S(0), \qquad \Pi(t)X \equiv \mathrm{Tr}_B\big[\mathrm{e}^{\mathcal{L}t}\big(X \otimes \rho_B\big)\big]. \tag{18}$$

$\Pi(t)$ is the *dynamical map*; in Liouville space it is a $d^2 \times d^2$ matrix with $\Pi(0) = 1$. Three properties follow directly from Eq. (18): *trace preservation*, $\mathrm{Tr}_S\,\Pi(t)X = \mathrm{Tr}\,X$ (first row of the Bloch matrix is $(1,0,0,0)$); *Hermiticity preservation*; and *complete positivity*, since $\Pi(t)$ is a partial trace of a unitary conjugation [13, Sec. 8.2.3]. Complete positivity is equivalent to the existence of a Kraus form $\Pi X = \sum_k M_k X M_k^\dagger$ [14]. Maps obtained from Eq. (18) with a *correlated* initial state need not be completely positive, or even linear in $\rho_S(0)$ [15]; we stay with factorized initial states except where noted.

What $\Pi(t)$ is *not*, in general, is a semigroup: $\Pi(t+s) \neq \Pi(t)\Pi(s)$. The bath remembers. The whole subject of these notes is how to write equations of motion for $\rho_S(t)$ when the map is not a semigroup, and the two standard answers are: a time-nonlocal (Nakajima–Zwanzig) equation $\dot\rho_S = \mathcal{L}_S\rho_S + \int_0^t K(t-s)\rho_S(s)\mathrm{d}s$, and a time-local (TCL) equation $\dot\rho_S = \mathcal{G}(t)\rho_S(t)$. When $\mathcal{G}(t)$ happens to be constant and of Gorini–Kossakowski–Sudarshan–Lindblad form [16, 17],

$$\mathcal{G}X = -i[H', X] + \sum_j \gamma_j\Big(L_j X L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, X\}\Big), \qquad \gamma_j \geq 0, \tag{19}$$

the map is a completely positive semigroup; this is the Markovian limit.

Example 1.3: the pure-dephasing map in Bloch form

Suppose the only effect of the bath is to multiply the coherence by a real factor: $\rho_{01}(t) = \rho_{01}(0)\,\mathrm{e}^{-i\omega_0 t - \Gamma(t)}$ with $\Gamma(t) \geq 0$ (we will derive this for $\theta = 0$ in Section 4). Then r_z is constant and (r_x, r_y) rotates and shrinks:

$$\Pi(t) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \mathrm{e}^{-\Gamma}\cos\omega_0 t & -\mathrm{e}^{-\Gamma}\sin\omega_0 t & 0 \\ 0 & \mathrm{e}^{-\Gamma}\sin\omega_0 t & \mathrm{e}^{-\Gamma}\cos\omega_0 t & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{20}$$

Its Kraus form has two operators, $M_0 = \sqrt{\frac{1+\mathrm{e}^{-\Gamma}}{2}}\,\mathrm{e}^{-i\omega_0 t\sigma_z/2}$ and $M_1 = \sqrt{\frac{1-\mathrm{e}^{-\Gamma}}{2}}\,\sigma_z\mathrm{e}^{-i\omega_0 t\sigma_z/2}$, so it is completely positive for every $\Gamma \geq 0$. The TCL generator is $\mathcal{G}(t) = \dot\Pi\Pi^{-1} = \mathcal{L}_S - \dot\Gamma(t)\,\frac{1}{4}(\sigma_z^{\mathrm{q}})^2$, where $-\frac{1}{4}(\sigma_z^{\mathrm{q}})^2 X = -\frac{1}{4}[\sigma_z,[\sigma_z,X]] = \frac{1}{2}(\sigma_z X\sigma_z - X)$ is the dephasing dissipator (check with Eq. (17): $(\sigma_z^{\mathrm{q}})^2\sigma_x = \sigma_z^{\mathrm{q}}(2i\sigma_y) = 4\sigma_x$, so $-\frac{1}{4}(\sigma_z^{\mathrm{q}})^2\sigma_x = -\sigma_x$, as it should, while $\mathbb{1}$ and σ_z are annihilated). The generator is of GKSL form with rate $\gamma(t) = \dot\Gamma(t)$, which can be negative at times when Γ decreases: a time-local generator of a completely positive map need not have positive rates. This is the simplest example of "non-Markovian but time-local".

1.4 Exercises

Exercise 1.1. Verify the three properties in the key box above directly from the definitions, and show that for Hermitian A, $(A^{\mathrm{c}})^\dagger = A^{\mathrm{c}}$ and $(A^{\mathrm{q}})^\dagger = -A^{\mathrm{q}}$ with respect to the Hilbert–Schmidt product. Conclude that iA^{q} is Hermitian as a superoperator.

Exercise 1.2. Write $\mathcal{L}_S$ for $H_S = \frac{\omega_0}{2}\sigma_z$ in the vectorized basis $\{|0\rangle\langle 0|, |0\rangle\langle 1|, |1\rangle\langle 0|, |1\rangle\langle 1|\}$ using Eq. (5), and find the unitary 4×4 matrix that maps it to Eq. (15).

Exercise 1.3. Show that the matrix of $\Pi(t)$ in the coefficient convention has the block form $\left(\begin{smallmatrix} 1 & 0 \\ \boldsymbol{c}(t) & T(t) \end{smallmatrix}\right)$ with a 3×3 matrix T and a vector $\boldsymbol{c}$ (the affine Bloch form $\boldsymbol{r} \mapsto T\boldsymbol{r} + \boldsymbol{c}$). Show that positivity requires $|T\boldsymbol{r} + \boldsymbol{c}| \leq 1$ for all $|\boldsymbol{r}| \leq 1$ and that for a unital map ($\boldsymbol{c} = 0$) complete positivity is a stronger condition than this (give the standard counterexample: the transposition $T = \mathrm{diag}(1, -1, 1)$).

Exercise 1.4. Compute $(S^{\mathrm{c}})^2$ and $(S^{\mathrm{q}})^2$ from Eq. (17) and interpret them with property 3 of the key box (use $S^2 = \mathbb{1}$).

Exercise 1.5. Derive Eq. (9) and Eq. (11), and check the latter on $A = \sigma_x$, $B = \sigma_y$ (so $AB = i\sigma_z$) using the matrices of Example 1.2 with $\theta = 0$ and $\theta = \pi/2$: verify that $(i\sigma_z)^{\mathrm{c}} = i\sigma_z^{\mathrm{c}}$ comes out of the right-hand side, including the commutator terms, which do not vanish here because $[\sigma_x, \sigma_y] \neq 0$.

2 Lecture 2: The model, its bath, and the two kernels

2.1 The Hamiltonian and its two faces

The running model is

$$\begin{aligned} H &= H_S + H_B + H_I, \qquad H_S = \frac{\omega_0}{2}\sigma_z, \qquad H_B = \sum_k \omega_k b_k^\dagger b_k, \\ H_I &= S \otimes B, \qquad S = \cos\theta\,\sigma_z + \sin\theta\,\sigma_x, \qquad B = \sum_k g_k\big(b_k + b_k^\dagger\big). \end{aligned} \tag{21}$$

The qubit has energy splitting ω_0; the bath is a collection of harmonic oscillators (phonons, photons, the electromagnetic modes of a cavity or a transmission line, the Gaussian part of a spin bath, etc.); the coupling is linear in the bath coordinates and couples to a fixed Hermitian qubit operator S with $S^2 = \mathbb{1}$. The angle θ is the only new parameter compared with the textbook cases.

Pure dephasing, $\theta = 0$. $H_I = \sigma_z \otimes B$ commutes with H_S. Populations are frozen, coherences decay. This is the model of "T_2" processes, of the filter-function formalism in quantum sensing, and of Ramsey and echo experiments [18, 19]. It is exactly solvable for any Gaussian bath (Sections 4 and 7).

Transverse coupling, $\theta = \pi/2$. $H_I = \sigma_x \otimes B$ flips the qubit. This is the model of "T_1" processes: spontaneous and stimulated emission, absorption, and the associated Lamb shift. In the rotating-wave approximation it becomes the damped Jaynes–Cummings model, which is again exactly solvable for a single excitation.

The dissipative two-level system. For general θ, rotate the qubit about $\hat{y}$ so that S becomes the new σ_z: define $\sigma_z' \equiv S = \cos\theta\sigma_z + \sin\theta\sigma_x$ and $\sigma_x' \equiv \cos\theta\sigma_x - \sin\theta\sigma_z$, which satisfy the Pauli algebra with $\sigma_y' = \sigma_y$. Then $\sigma_z = \cos\theta\,\sigma_z' - \sin\theta\,\sigma_x'$ and

$$H = \frac{\varepsilon}{2}\sigma_z' - \frac{\Delta}{2}\sigma_x' + \sigma_z' \otimes B + H_B, \qquad \varepsilon = \omega_0\cos\theta, \quad \Delta = \omega_0\sin\theta. \tag{22}$$

This is the spin-boson model in the form of Leggett *et al.* [9, Sec. II] and Weiss [20, Ch. 3]: a particle tunnelling with amplitude Δ between two wells with bias ε, whose *position* σ_z' is coupled to the bath. Our θ is the angle between the energy eigenbasis and the bath's pointer basis. Keep both pictures in mind: Eq. (21) is natural for perturbation theory in the coupling g_k (the energy eigenstates are a good starting point), while Eq. (22) is natural for perturbation theory in Δ after a polaron transformation (the pointer states are a good starting point). We will use both, in Section 5 and Section 7 respectively. The two expansions have different domains of validity, and seeing the same kernel emerge in both is one of the pleasures of the subject.

Normalization of the coupling. Leggett *et al.* write the coupling as $\frac{1}{2}\sigma_z' \sum_k c_k x_k$ with oscillator coordinates $x_k = (b_k + b_k^\dagger)/\sqrt{2m_k\omega_k}$, and define $J_{\mathrm{L}}(\omega) = \frac{\pi}{2}\sum_k c_k^2/(m_k\omega_k)\,\delta(\omega-\omega_k)$. With $g_k = c_k/\sqrt{2m_k\omega_k}$ this is $J_{\mathrm{L}}(\omega) = \pi\sum_k g_k^2\delta(\omega-\omega_k)$, which coincides with our $J(\omega)$ of Eq. (24) below, *but* their coupling operator is $\sigma_z'/2$ where ours is σ_z'. All our bath kernels therefore enter with a factor 4 relative to Leggett's formulas when the same J is used; equivalently, our $\theta = 0$ model with J is Leggett's with $4J$. We flag this at each place where it matters.

2.2 Bath correlation function; noise and response

In the interaction picture with respect to H_B, $B(t) = \sum_k g_k(b_k e^{-i\omega_k t} + b_k^\dagger e^{i\omega_k t})$. For a thermal bath $\rho_B = e^{-\beta H_B}/Z_B$ with $\bar{n}_k = (e^{\beta\omega_k} - 1)^{-1}$,

$$C(t) \equiv \langle B(t)B(0)\rangle = \sum_k g_k^2\Big[(\bar{n}_k+1)e^{-i\omega_k t} + \bar{n}_k e^{i\omega_k t}\Big] = \int_0^\infty \frac{d\omega}{\pi}\, J(\omega)\Big[\coth\tfrac{\beta\omega}{2}\cos\omega t - i\sin\omega t\Big], \tag{23}$$

where the *spectral density*

$$J(\omega) \equiv \pi\sum_k g_k^2\,\delta(\omega-\omega_k) \qquad (\omega>0) \tag{24}$$

is the only property of the bath that the system will ever feel (for a Gaussian bath; see below). $C(t)$ is neither real nor even: $C(-t) = C(t)^*$. Its real and imaginary parts are the two kernels of the title:

$$S(t) \equiv \tfrac{1}{2}\langle\{B(t),B(0)\}\rangle = \mathrm{Re}\,C(t) = \int_0^\infty \frac{d\omega}{\pi}\, J(\omega)\coth\tfrac{\beta\omega}{2}\,\cos\omega t, \tag{25}$$

$$\chi(t) \equiv i\,\theta(t)\langle[B(t),B(0)]\rangle = -2\theta(t)\,\mathrm{Im}\,C(t) = 2\theta(t)\int_0^\infty \frac{d\omega}{\pi}\, J(\omega)\sin\omega t, \tag{26}$$

so that $C(t) = S(t) - \frac{i}{2}\chi(t)$ for $t>0$. S is the symmetrized *noise* correlator: it is what a classical measuring device would record as the fluctuations of B. χ is Kubo's *response* function [21]: if a force $f(t)$ is applied through $H \to H - f(t)B$, then $\delta\langle B(t)\rangle = \int ds\,\chi(t-s)f(s)$. The $\theta(t)$ is causality. S depends on temperature through coth; χ does not depend on temperature at all for a harmonic bath.

2.3 The kernels in the frequency domain

With the convention $f(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} f(t)$, the three functions χ, S and C are tied together by

$$\chi''(\omega) = \mathrm{sgn}(\omega)\,J(|\omega|), \quad S(\omega) = \coth\tfrac{\beta\omega}{2}\,\chi''(\omega), \quad C(\omega) = S(\omega)+\chi''(\omega) = 2\chi''(\omega)\big[1+n(\omega)\big], \tag{27}$$

with $\chi'' \equiv \mathrm{Im}\,\chi$ and $n(\omega) = (e^{\beta\omega}-1)^{-1}$. These three lines are used on every page that follows, so we go through them one at a time. The one fact to keep in mind is where the sign of a frequency comes from: in $B(t) = \sum_k g_k(b_k e^{-i\omega_k t} + b_k^\dagger e^{i\omega_k t})$ the annihilation part rotates as $e^{-i\omega_k t}$, so after multiplication by $e^{i\omega t}$ and integration it lands at $\omega = +\omega_k$ and the creation part at $\omega = -\omega_k$. *Positive frequency means that the bath absorbs a quantum.*

(i) The dissipative response is the spectral density. From Eq. (26),

$$\chi(\omega) = \frac{2}{\pi}\int_0^\infty d\omega'\, J(\omega')\int_0^\infty dt\, e^{i\omega t}\sin\omega' t,$$

and the one-sided time integral is a distribution whose imaginary part is $\frac{\pi}{2}[\delta(\omega-\omega') - \delta(\omega+\omega')]$ (the derivation is written out in the solution of Exercise 2.2). Integrating against $J(\omega')$ on $\omega'>0$ picks $J(\omega)$ for $\omega>0$ and $-J(-\omega)$ for $\omega<0$, which is the first line of Eq. (27). Two remarks. χ'' is odd because $\chi(t)$ is real: for a real function $\chi(-\omega) = \chi(\omega)^*$, so the imaginary part is odd and the real part even. And χ'' carries no temperature at all. This is special to a harmonic bath with linear B: the commutator $[B(t),B(0)]$ is then a c-number, and its expectation value cannot depend on the state. For an anharmonic bath χ'' would be temperature dependent, and Eq. (27) would still hold, but with a J that is itself a function of T.

(ii) The fluctuation–dissipation theorem. From Eq. (25), and using $\int \mathrm{d}t\, \mathrm{e}^{i\omega t} \cos\omega' t = \pi[\delta(\omega-\omega') + \delta(\omega+\omega')]$ (a two-sided integral, hence no principal value),

$$S(\omega) = J(|\omega|)\coth\tfrac{\beta|\omega|}{2} = \underbrace{\coth\tfrac{\beta\omega}{2}}_{\text{odd}} \cdot \underbrace{\operatorname{sgn}(\omega) J(|\omega|)}_{\chi''(\omega),\ \text{odd}}, \tag{28}$$

even in ω as the transform of a real even $S(t)$ must be. This is the fluctuation–dissipation theorem in the form of Callen and Welton [22, 23]: the symmetrized noise at frequency ω is fixed by the dissipative response at the same frequency, through a universal factor that depends only on temperature. Two limits are worth memorizing:

$$\begin{aligned} |\omega| \ll T: &\quad S(\omega) \to \frac{2T}{\omega}\,\chi''(\omega) \qquad \text{(classical, Nyquist–Johnson)},\\ |\omega| \gg T: &\quad S(\omega) \to |\chi''(\omega)| = J(|\omega|) \quad \text{(zero-point noise)}. \end{aligned} \tag{29}$$

The noise does not vanish at $T = 0$. In Fig. 1, drawn at $T = 0$, $S(\omega)$ coincides with χ'' for $\omega > 0$ and is its mirror image for $\omega < 0$.

(iii) The full correlator is noise plus response. The first equality in the third line is linearity: $C(t) = S(t) - \frac{i}{2}\chi(t)$ for $t > 0$ and, from $C(-t) = C(t)^*$, $C(t) = S(t) + \frac{i}{2}\chi(-t)$ for $t < 0$; the odd combination $-\frac{i}{2}[\chi(t) - \chi(-t)]$ transforms to $-\frac{i}{2}[\chi(\omega) - \chi(\omega)^*] = \chi''(\omega)$. So S is the *even* part of the spectrum $C(\omega)$ and χ'' is its *odd* part; this is the cleanest way to remember the Keldysh decomposition. The second equality uses the identity $\coth\frac{\beta\omega}{2} = 1 + 2n(\omega)$, valid for all real ω including negative ones, so that $S + \chi'' = (\coth + 1)\chi'' = 2(1+n)\chi''$. Computing $C(\omega)$ directly from Eq. (23) confirms it and exhibits the physics:

$$C(\omega) = 2J(\omega)\big[1 + n(\omega)\big] \quad (\omega > 0), \qquad C(\omega) = 2J(|\omega|)\, n(|\omega|) \quad (\omega < 0), \tag{30}$$

the two cases being covered by the single formula $2\chi''(1+n)$ because $1 + n(\omega) = \mathrm{e}^{\beta\omega}/(\mathrm{e}^{\beta\omega} - 1) = -n(|\omega|)$ for $\omega < 0$, and the two minus signs cancel.

(iv) Absorption, emission and detailed balance. Write $C(\omega)$ in the energy eigenbasis of the bath, $\rho_B = \sum_i p_i |i\rangle\langle i|$:

$$C(\omega) = 2\pi \sum_{i,f} p_i \left|\langle f|B|i\rangle\right|^2 \delta\big(\omega - (E_f - E_i)\big). \tag{31}$$

This is Fermi's golden rule. $C(\omega)$ at $\omega > 0$ is the rate, per unit coupling squared, at which the bath makes a transition *upward* by ω, i.e. absorbs energy that the system must supply; the factor $1 + n$ in Eq. (30) is the "spontaneous plus stimulated" factor of a bosonic mode. At $\omega < 0$ it is the rate at which the bath gives up energy $|\omega|$, which requires a thermally excited mode, hence the factor n. Section 5 uses exactly this: $\Gamma_\downarrow \propto C(\omega_0)$ (the qubit decays, the bath absorbs) and $\Gamma_\uparrow \propto C(-\omega_0)$. Dividing the two cases of Eq. (30),

$$\frac{C(-\omega)}{C(\omega)} = \frac{n(\omega)}{1 + n(\omega)} = \mathrm{e}^{-\beta\omega}, \tag{32}$$

which is detailed balance, and is also the Kubo–Martin–Schwinger condition $\langle B(t)B(0)\rangle = \langle B(0)B(t + i\beta)\rangle$ read in frequency space. At $T = 0$ the negative-frequency side vanishes identically: the bath can only absorb. This asymmetry between $\pm\omega$ is what makes $C(\omega)$ a *quantum* noise spectrum; it is invisible in S, which is even by construction, and a classical noise source always has $C(-\omega) = C(\omega)$. A measurement that resolves the asymmetry, such as $\Gamma_\downarrow \neq \Gamma_\uparrow$ in a T_1 experiment, is seeing quantum noise; see [24, Sec. II] for a careful discussion of the asymmetric spectrum and its measurement.

(v) Kramers–Kronig, and what is independent of what. $\chi(\omega)$ is the transform of a function that vanishes for $t<0$, so it is analytic in the upper half ω-plane and its real and imaginary parts are Hilbert transforms of each other,

$$\chi'(\omega)=\frac{1}{\pi}\,\mathrm{P}\!\int_{-\infty}^{\infty}\mathrm{d}\omega'\,\frac{\chi''(\omega')}{\omega'-\omega}=\frac{2}{\pi}\,\mathrm{P}\!\int_{0}^{\infty}\mathrm{d}\omega'\,\frac{\omega' J(\omega')}{\omega'^2-\omega^2}, \tag{33}$$

the second form following from the oddness of χ'' (it is also what the principal-value part of the integral in Exercise 2.2 gives directly). The static susceptibility $\chi(0)=\frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, J(\omega)/\omega$ is twice the reorganization energy of the bath and will reappear in Sections 7 and 8. Putting (i)–(v) together: in a thermal bath the whole environment enters through the single real function $J(\omega)$ on $\omega>0$. χ'' is J extended oddly, χ' follows by Kramers–Kronig, S follows by the FDT, and $C=S+\chi''$. The two entries of the Keldysh matrix (52) carry no independent information in equilibrium. Out of equilibrium the chain breaks at exactly one link: causality still fixes χ' from χ'', but nothing fixes S from χ''. A driven cavity, a bath with a nonthermal occupation, or a $1/f$ noise source each have their own $S(\omega)$ and $\chi''(\omega)$, and their ratio at a given frequency defines an effective temperature there, $\coth(\beta_{\mathrm{eff}}(\omega)\,\omega/2)\equiv S(\omega)/\chi''(\omega)$, which can differ between frequencies and even be negative (population inversion). Keeping S and χ on separate lines, which is what the Keldysh rotation does, costs nothing and buys this generality: every formula derived with D^K and D^R kept apart remains valid without the FDT, and only the final steps that use $S=\coth\cdot\chi''$ (for instance the Gibbs steady state in Section 5) must be revisited.

Example 2.1: two spectral densities used throughout

Ohmic bath. $J(\omega)=2\pi\alpha\,\omega\,\mathrm{e}^{-\omega/\omega_c}$. With Leggett's coupling normalization (see the warning box) α is the dimensionless Kondo parameter, with the famous values $\alpha=\frac{1}{2}$ (Toulouse point) and $\alpha=1$ (localization transition at $T=0$) [9, Secs. IV–V]. At $T=0$,

$$S(t)=2\alpha\,\omega_c^2\,\frac{1-\omega_c^2t^2}{(1+\omega_c^2t^2)^2},\qquad \chi(t)=\theta(t)\,\frac{8\alpha\,\omega_c^3\,t}{(1+\omega_c^2t^2)^2}, \tag{34}$$

both decaying as power laws on the scale $1/\omega_c$. At temperature T and for $t\gg 1/\omega_c$, $S(t)\simeq -2\alpha(\pi T)^2/\sinh^2(\pi T t)$: the noise correlator becomes negative and decays exponentially on the thermal time $1/(\pi T)$. The static susceptibility $\chi(\omega{=}0)=\frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, J(\omega)/\omega=4\alpha\omega_c$ is twice the reorganization energy.

Lorentzian bath (pseudomode). $J(\omega)=g^2\,\dfrac{\gamma/2}{(\omega-\Omega)^2+\gamma^2/4}$, a single broadened mode at Ω with width γ. If we allow this J to extend to negative frequencies (a good approximation when $\Omega\gg\gamma$, and exactly what the pseudomode construction below assumes), then at $T=0$

$$C(t)=g^2\,\mathrm{e}^{-i\Omega t-\gamma|t|/2},\qquad S(t)=g^2\,\mathrm{e}^{-\gamma|t|/2}\cos\Omega t,\qquad \chi(t)=2g^2\,\theta(t)\,\mathrm{e}^{-\gamma t/2}\sin\Omega t, \tag{35}$$

and $S(\omega)=J(\omega)+J(-\omega)$, $\chi''(\omega)=J(\omega)-J(-\omega)$, $C(\omega)=2J(\omega)$. Figure 1 shows these for the parameters used in all numerical figures, $\Omega=\omega_0$, $\gamma=0.5\,\omega_0$. The limit $\gamma\to\infty$ at fixed g^2/γ is a Markovian (white-noise) bath; the limit $\gamma\to 0$ is a single undamped mode.

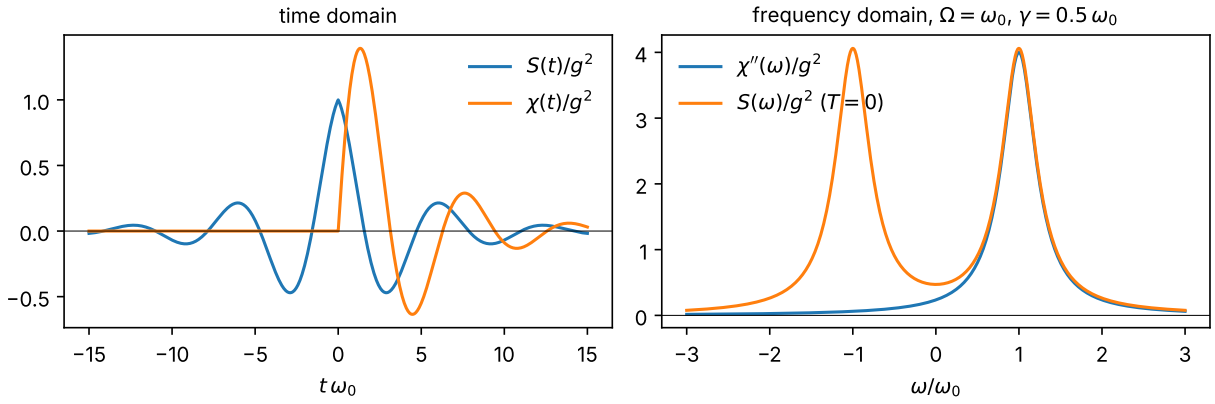


Figure 1: The Lorentzian bath used in the numerical examples. Left: noise kernel $S(t)$ (even in t) and response kernel $\chi(t)$ (causal, odd before the step function), from Eq. (35). Right: $\chi''(\omega)$ and the zero-temperature noise spectrum $S(\omega) = J(\omega) + J(-\omega)$. A T_1 measurement samples these at $\omega = \pm\omega_0$; a dephasing measurement samples S near $\omega = 0$ (Section 10).

2.4 Why only $J(\omega)$ matters: Gaussianity and Wick's theorem

The bath state is Gaussian (thermal state of a quadratic Hamiltonian) and B is linear in the b_k. Hence all higher moments of B factorize into products of two-point functions:

$$\langle T_{\mathcal{C}}\, B(z_1)B(z_2)\cdots B(z_{2n})\rangle = \sum_{\text{pairings}} \prod_{\text{pairs }(ij)} \langle T_{\mathcal{C}}\, B(z_i)B(z_j)\rangle, \qquad \langle T_{\mathcal{C}}\, B(z_1)\cdots B(z_{2n+1})\rangle = 0, \tag{36}$$

for any ordering prescription $T_{\mathcal{C}}$ that is applied consistently inside each pair (Wick's theorem; see [25, Sec. 2.5] for the contour version). Since the two-point function is built from $J(\omega)$ and β alone, so is everything the system can feel. This is the precise sense in which "the bath is characterized by its spectral density". Non-Gaussian baths (spins, anharmonic modes, fermionic reservoirs with interactions) need higher cumulants; the formalism of the following lectures survives with pairings replaced by cumulants [26], but we will not need this.

A useful way to say the same thing: for a Gaussian bath, a *linear* coupling, and a factorized initial state, the reduced dynamics of the system depends on the bath only through $C(t)$. Any two baths with the same $C(t)$ produce identical $\rho_S(t)$. This is the basis of the exact reference solution used in the numerics:

The pseudomode mapping (Garraway [10]; Tamascelli *et al.* [11])

Consider the qubit coupled to a *single* oscillator a of frequency Ω, which is itself damped at rate γ by a Markovian environment:

$$\dot\rho = -i\big[H_S + \Omega a^\dagger a + g\,S\otimes(a+a^\dagger),\rho\big] + \gamma\big(a\rho a^\dagger - \tfrac{1}{2}\{a^\dagger a,\rho\}\big), \qquad \rho(0) = \rho_S(0)\otimes|0\rangle\langle 0|. \tag{37}$$

The "bath" $B_{\rm pm} = g(a+a^\dagger)$ seen by the qubit is Gaussian with $\langle B_{\rm pm}(t)B_{\rm pm}(0)\rangle = g^2 \mathrm{e}^{-i\Omega t-\gamma|t|/2}$, which is exactly the Lorentzian $C(t)$ of Eq. (35). Therefore the qubit's reduced dynamics in Eq. (37) is *identical*, to all orders in g and for all times, to its dynamics in the full Lorentzian bosonic bath. The small Hilbert space (qubit $\otimes$ a few oscillator levels) makes Eq. (37) trivially solvable numerically. The price is the negative-frequency tail of J; the reward is an exact

benchmark for every approximation in these notes.

2.5 Initial state and pictures

We take $\rho(0) = \rho_S(0) \otimes \rho_B$ with ρ_B thermal (so $[H_B, \rho_B] = 0$ and $\langle B \rangle = 0$). Physically this means the coupling is switched on at $t = 0$, or that the qubit was prepared by a fast operation that does not disturb the bath. Correlated initial states (for example the global thermal state of H) can be handled, at the price of an inhomogeneous term in the master equation (Section 5) or an imaginary-time branch of the contour (Section 9).

We will move freely between the Schrödinger picture, the interaction picture with respect to $H_0 = H_S + H_B$, and the Heisenberg picture. For the record, in the interaction picture

$$\tilde{S}(t) = \mathrm{e}^{iH_S t} S \mathrm{e}^{-iH_S t} = \cos\theta\, \sigma_z + \sin\theta \big(\sigma_x \cos\omega_0 t - \sigma_y \sin\omega_0 t\big) = \cos\theta\, \sigma_z + \sin\theta \big(\sigma_+ \mathrm{e}^{i\omega_0 t} + \sigma_- \mathrm{e}^{-i\omega_0 t}\big), \tag{38}$$

with $\sigma_\pm = \tfrac{1}{2}(\sigma_x \pm i\sigma_y)$. Note that the Heisenberg-picture rotation of σ_x is clockwise, opposite to the Schrödinger-picture rotation of the Bloch vector in Eq. (15), as it must be. The $\cos\theta$ part is static, the $\sin\theta$ part oscillates at $\pm\omega_0$: this is why dephasing samples the bath at $\omega \approx 0$ and relaxation samples it at $\omega = \pm\omega_0$.

2.6 Exercises

Exercise 2.1. Derive Eq. (23) and check the single-mode case $J(\omega) = \pi g^2 \delta(\omega - \omega_b)$ against $S(t) = g^2(2\bar{n}+1)\cos\omega_b t$, $\chi(t) = 2g^2\theta(t)\sin\omega_b t$. Verify Eq. (27) for this case.

Exercise 2.2. Prove $\chi''(\omega) = \mathrm{sgn}(\omega) J(|\omega|)$ from Eq. (26) and derive the Ohmic kernels of Example 2.1 at $T = 0$. Then show that at finite T and $\omega_c \to \infty$, $S(t) \to -2\alpha(\pi T)^2/\sinh^2(\pi T t)$ for $t \neq 0$, and explain how a positive-definite $S(\omega)$ can have a negative $S(t)$ at all $t > 0$.

Exercise 2.3. Show that in Eq. (37) the oscillator's two-point function in the absence of the qubit is $\langle a(t) a^\dagger(0)\rangle = \mathrm{e}^{-(i\Omega + \gamma/2)t}$ for $t > 0$ (use the quantum regression theorem, or solve the adjoint equation for a), and that the state remains Gaussian. This is the content of the pseudomode theorem; see [11] for the general statement.

Exercise 2.4. Verify the unitary equivalence Eq. (22), and show that a bath coupled to $S = \cos\theta\sigma_z + \sin\theta\sigma_x$ with correlation $C(t)$ is equivalent to *two independent* baths coupled to σ_z and σ_x only if the cross-correlation $\cos\theta\sin\theta\, C(t)$ is dropped. We will see in Section 5 where this cross term goes.

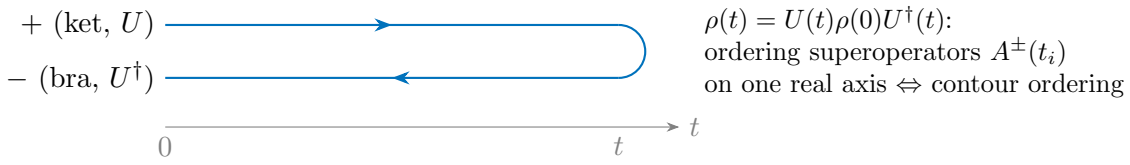


Figure 2: The Keldysh contour [1, 2]. The forward branch carries the operators of $U(t)$ acting on the ket side of $\rho(0)$, the backward branch the operators of $U^\dagger(t)$ acting on the bra side. Contour ordering means: later contour position further from $\rho(0)$.

3 Lecture 3: The Keldysh contour is already in the von Neumann equation

The Keldysh formalism is usually introduced by declaring a closed time contour and then building Green's functions on it. In this lecture we go the other way: we expand the reduced density matrix in powers of H_I and watch the contour, the Keldysh rotation and the 2×2 matrix structure of the bath propagator appear by themselves. Nothing is postulated. At the end we have a set of rules ("contractions") that Section 4 turns into diagrams.

3.1 The Dyson series for ρ and the origin of the contour

Work in the interaction picture with respect to $H_0 = H_S + H_B$. Then $\tilde{\rho}(t) = \tilde{U}(t)\rho(0)\tilde{U}^\dagger(t)$ with

$$\tilde{U}(t) = T \exp\Big[-i\int_0^t \mathrm{d}s\, \tilde{H}_I(s)\Big] = 1 - i\int_0^t \mathrm{d}t_1 \tilde{H}_I(t_1) - \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \tilde{H}_I(t_1)\tilde{H}_I(t_2) + \dots \tag{39}$$

Its adjoint is *anti*-time-ordered: in $\tilde{U}^\dagger$ the later operator stands to the right, $\tilde{U}^\dagger = 1 + i\int \tilde{H}_I - \int_{t_1>t_2} \tilde{H}_I(t_2)\tilde{H}_I(t_1) + \dots$

Example 3.1: the second-order term, written out

Collecting all terms of order H_I^2 in $\tilde{U}\rho(0)\tilde{U}^\dagger$,

$$\begin{aligned}
\tilde{\rho}^{(2)}(t) = &-\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \tilde{H}_I(t_1)\tilde{H}_I(t_2)\,\rho(0) && \text{(both on the ket side; time-ordered)} \\
&-\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \rho(0)\,\tilde{H}_I(t_2)\tilde{H}_I(t_1) && \text{(both on the bra side; anti-time-ordered)} \\
&+\int_0^t \mathrm{d}t_1 \int_0^{t} \mathrm{d}t_2\, \tilde{H}_I(t_1)\,\rho(0)\,\tilde{H}_I(t_2) && \text{(one on each side; } no \text{ mutual ordering).}
\end{aligned} \tag{40}$$

The third term is the one that looks unusual: t_1 and t_2 are integrated independently over $[0, t]$, so there is no time ordering between a ket-side and a bra-side operator. All three terms become a *single* ordering rule once we say that every point on the forward (+) branch is "earlier" than every point on the backward (−) branch, that time runs forward on + and backward on −, and that operators are ordered with later contour positions further from $\rho(0)$. That rule is contour ordering, Fig. 2.

The superoperator language of Section 1 makes the rule automatic. With $A^+X = AX$ and

$A^- X = X A$,

$$\tilde{\rho}^{(2)}(t) = \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \tilde{\mathcal{L}}_I(t_1)\tilde{\mathcal{L}}_I(t_2)\,\rho(0), \qquad \tilde{\mathcal{L}}_I(s) = -i\big[\tilde{H}_I^+(s) - \tilde{H}_I^-(s)\big], \tag{41}$$

and multiplying out the two brackets reproduces the three terms of Eq. (40): H^+H^+ gives the first, H^-H^- gives the second (note $H^-(t_1)H^-(t_2)\rho = \rho H(t_2)H(t_1)$, automatically anti-ordered), and the cross terms $H^+(t_1)H^-(t_2) + H^-(t_1)H^+(t_2)$ give $H(t_1)\rho H(t_2) + H(t_2)\rho H(t_1)$ restricted to $t_1 > t_2$, which is the third term with its two orderings separated. To all orders,

$$\tilde{\rho}(t) = T\exp\Big[\int_0^t \mathrm{d}s\, \tilde{\mathcal{L}}_I(s)\Big]\rho(0) = \sum_n \int_{t>t_1>\cdots>t_n>0} \tilde{\mathcal{L}}_I(t_1)\cdots\tilde{\mathcal{L}}_I(t_n)\,\rho(0), \tag{42}$$

which is just the Dyson series for the superoperator equation $\dot{\tilde{\rho}} = \tilde{\mathcal{L}}_I(t)\tilde{\rho}$.

Key point 3.1

Ordinary time ordering of the superoperators $\tilde{\mathcal{L}}_I(t_i)$ along the single real axis $[0, t]$, with a sum over the branch index $\pm$ at each vertex, *is* Keldysh contour ordering. One never has to think about the contour again: it is encoded in the sign of $\mathcal{L}_I$ and the left/right action of $H^\pm$. Conversely, every Keldysh expression for a one-time expectation value can be read as a Liouville-space string.

3.2 The Keldysh rotation in Liouville space

The $\pm$ basis is natural for deriving the contour but awkward for calculations, because the two branches are strongly correlated (the trace at the final time glues them together). The rotation Eq. (10) separates "what the two branches have in common" (c) from "how they differ" (q). Since S and B act on different tensor factors, $(SB)^\pm = S^\pm B^\pm$ and $S^\pm$ commutes with $B^\pm$. Then with $A^\pm = A^{\mathrm{c}} \pm \frac{1}{2}A^{\mathrm{q}}$,

$$S^+B^+ - S^-B^- = \big(S^{\mathrm{c}} + \tfrac{1}{2}S^{\mathrm{q}}\big)\big(B^{\mathrm{c}} + \tfrac{1}{2}B^{\mathrm{q}}\big) - \big(S^{\mathrm{c}} - \tfrac{1}{2}S^{\mathrm{q}}\big)\big(B^{\mathrm{c}} - \tfrac{1}{2}B^{\mathrm{q}}\big) = S^{\mathrm{c}}B^{\mathrm{q}} + S^{\mathrm{q}}B^{\mathrm{c}}, \tag{43}$$

the $S^{\mathrm{c}}B^{\mathrm{c}}$ and $\frac{1}{4}S^{\mathrm{q}}B^{\mathrm{q}}$ terms cancelling. Hence

$$\boxed{\mathcal{L}_I = -i\big(S^{\mathrm{q}}B^{\mathrm{c}} + S^{\mathrm{c}}B^{\mathrm{q}}\big).} \tag{44}$$

The structure "one c, one q per vertex" is the same as in the Keldysh action, where the interaction term $H_+ - H_-$ is odd in the quantum field and so always contains at least one q leg [25, Sec. 2.7].

The two trace identities. For an arbitrary operator Y on $\mathcal{H}_S \otimes \mathcal{H}_B$ and a bath operator B,

$$\mathrm{Tr}_B\big(B^{\mathrm{q}}Y\big) = \mathrm{Tr}_B\big(BY - YB\big) = 0, \qquad \mathrm{Tr}_B\big(B^{\mathrm{c}}Y\big) = \tfrac{1}{2}\,\mathrm{Tr}_B\big(BY + YB\big) = \mathrm{Tr}_B\big(BY\big), \tag{45}$$

both by cyclicity of the partial trace over the bath factor. The first identity is the superoperator form of the Keldysh statement that a correlation function whose *latest* point is on a quantum leg vanishes; it is what makes $\langle\phi_{\mathrm{q}}\phi_{\mathrm{q}}\rangle = 0$ and $\langle\phi_{\mathrm{c}}\phi_{\mathrm{q}}\rangle$ retarded [25, Sec. 2.6]. The same identities hold with S in place of B and Tr_S in place of Tr_B, and they will be used on both sides.

Example 3.2: what c and q do to a qubit state

From Eq. (17), acting on $\rho = \frac{1}{2}(\mathbb{1} + \boldsymbol{r}\cdot\boldsymbol{\sigma})$:

$$S^{\mathrm{c}}\rho = \tfrac{1}{2}\big[(\hat{n}\cdot\boldsymbol{r})\mathbb{1} + \hat{n}\cdot\boldsymbol{\sigma}\big], \qquad S^{\mathrm{q}}\rho = i\,(\hat{n}\times\boldsymbol{r})\cdot\boldsymbol{\sigma}, \qquad \hat{n} = (\sin\theta, 0, \cos\theta). \tag{46}$$

S^{c} produces a traceful object whose weight is the expectation value $\langle S\rangle = \hat{n}\cdot\boldsymbol{r}$; S^{q} produces a traceless object proportional to the torque $\hat{n}\times\boldsymbol{r}$ that a field along $\hat{n}$ would exert. A state with $\boldsymbol{r}\parallel\hat{n}$ (a pointer state of the coupling) has $S^{\mathrm{q}}\rho = 0$: it does not "differ between the branches". For $\theta = 0$ this is the statement that a diagonal density matrix has no quantum component with respect to σ_z. This is the operator version of "the classical field is the physical configuration, the quantum field measures the difference between the two branches".

3.3 Bath contractions and the 2×2 Keldysh structure

The basic object is the time-ordered two-point function of bath superoperators,

$$D^{ab}(t,s) \equiv \mathrm{Tr}_B\big[T_{\mathcal{L}}\,\tilde{B}^a(t)\tilde{B}^b(s)\,\rho_B\big], \qquad a,b\in\{+,-\}\text{ or }\{\mathrm{c},\mathrm{q}\}, \tag{47}$$

where $T_{\mathcal{L}}$ puts the later superoperator on the left (closer to the trace). In the $\pm$ basis the four components are, by direct evaluation for $t > s$,

component	string ($t > s$)	value	Keldysh name
$D^{++}(t,s)$	$\mathrm{Tr}_B[B(t)B(s)\rho_B]$	$C(t-s)$	time-ordered D^T
$D^{--}(t,s)$	$\mathrm{Tr}_B[\rho_B B(s)B(t)]$	$C(s-t)$	anti-time-ordered $D^{\tilde{T}}$
$D^{+-}(t,s)$	$\mathrm{Tr}_B[B(t)\rho_B B(s)]$	$C(s-t)$	lesser $D^<$
$D^{-+}(t,s)$	$\mathrm{Tr}_B[B(s)\rho_B B(t)]$	$C(t-s)$	greater $D^>$

(For $t < s$ the roles of "later" and "earlier" are exchanged in the string; the $+-$ and $-+$ entries then keep their form while $++$ and $--$ switch to $C(s-t)$ and $C(t-s)$ respectively.) These are the four contour-ordered Green's functions of Keldysh theory up to the overall $-i$ of the usual definition, and the familiar linear relation $D^T + D^{\tilde{T}} = D^< + D^>$ holds entry by entry, which is why the $\pm$ basis is redundant. Now rotate, using $B^{\mathrm{c}} = \frac{1}{2}(B^+ + B^-)$ and $B^{\mathrm{q}} = B^+ - B^-$ on both legs:

$$D^{\mathrm{cc}}(t,s) = \tfrac{1}{4}\big(D^{++} + D^{+-} + D^{-+} + D^{--}\big) = \tfrac{1}{2}\big[C(t-s) + C(s-t)\big] = S(t-s), \tag{48}$$

$$D^{\mathrm{cq}}(t,s) = \tfrac{1}{2}\big(D^{++} - D^{+-} + D^{-+} - D^{--}\big) = \theta(t-s)\langle[B(t),B(s)]\rangle = -i\chi(t-s), \tag{49}$$

$$D^{\mathrm{qc}}(t,s) = \tfrac{1}{2}\big(D^{++} + D^{+-} - D^{-+} - D^{--}\big) = -\theta(s-t)\langle[B(t),B(s)]\rangle = -i\chi(s-t), \tag{50}$$

$$D^{\mathrm{qq}}(t,s) = D^{++} - D^{+-} - D^{-+} + D^{--} = 0. \tag{51}$$

(For the middle two lines evaluate separately for $t > s$ and $t < s$ using the table; the θ functions appear because the "later" entry of the table switches.) In matrix form,

$$\boxed{D(t,s) = \begin{pmatrix} D^{\mathrm{cc}} & D^{\mathrm{cq}} \\ D^{\mathrm{qc}} & D^{\mathrm{qq}} \end{pmatrix} = \begin{pmatrix} S(t-s) & -i\chi(t-s) \\ -i\chi(s-t) & 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} D^K & D^R \\ D^A & 0 \end{pmatrix},} \tag{52}$$

the Larkin–Ovchinnikov structure [27, 28]. The zero in the qq slot and the retardation of the cq slot are both consequences of the first trace identity in Eq. (45): whenever the *later* superoperator

in the string is B^{q}, the trace kills the term. Restricting to $t > s$ and reading off the entries gives the three contraction rules used from now on:

$$\mathrm{Tr}_B[\tilde{B}^{\mathrm{c}}(t)\tilde{B}^{\mathrm{c}}(s)\rho_B] = S(t-s), \qquad \mathrm{Tr}_B[\tilde{B}^{\mathrm{c}}(t)\tilde{B}^{\mathrm{q}}(s)\rho_B] = -i\chi(t-s), \qquad \mathrm{Tr}_B[\tilde{B}^{\mathrm{q}}(t)\,\cdot\,] = 0. \quad (53)$$

Example 3.3: the single-mode bath, and the Lorentzian

For one mode, $C(t) = g^2[(\bar{n}+1)\mathrm{e}^{-i\omega_b t} + \bar{n}\,\mathrm{e}^{i\omega_b t}]$, the table gives $D^{\mathrm{cc}} = g^2(2\bar{n}+1)\cos\omega_b(t-s)$ and $D^{\mathrm{cq}} = -2ig^2\theta(t-s)\sin\omega_b(t-s)$: the noise line carries the occupation, the response line does not. In frequency space $\chi''(\omega) = \pi g^2[\delta(\omega-\omega_b) - \delta(\omega+\omega_b)]$ and $S(\omega) = \coth(\beta\omega/2)\chi''(\omega)$, the FDT checked by hand. For the Lorentzian bath, Eq. (35) gives the same with $\omega_b \to \Omega$, $\bar{n} \to 0$ and an extra damping $\mathrm{e}^{-\gamma|t-s|/2}$ on both lines. In either case the Keldysh matrix Eq. (52) is, in frequency space,

$$D(\omega) = \begin{pmatrix} \coth\frac{\beta\omega}{2}\,\chi''(\omega) & -i\chi(\omega) \\ -i\chi(\omega)^* & 0 \end{pmatrix}, \quad (54)$$

which is Kamenev's $D^K = \coth(\beta\omega/2)(D^R - D^A)$ with $D^R = -i\chi$ [25, Sec. 2.6].

Wick's theorem for superoperators. Because the Liouville string $\mathrm{Tr}_B[\tilde{B}^{a_n}(t_n)\cdots\tilde{B}^{a_1}(t_1)\rho_B]$ with $t_n > \cdots > t_1$ is literally a contour-ordered expectation value (Key point 3.1), the Wick theorem (36) holds for it unchanged, with each pair contributing $D^{a_i a_j}(t_i, t_j)$ from Eq. (52). In particular, every pairing in which the latest superoperator of a pair is B^{q} vanishes.

Two things that are easy to get wrong. (1) The θ function in χ is not an approximation; it comes from the trace identity and is the reason causality is built in. If you ever find a $\chi(t-s)$ with $s > t$ in a system-side diagram, an index has been swapped. (2) The $-i$ in the response line is a convention tied to $\chi = i\theta\langle[B,B]\rangle$. With Kamenev's $D^R = -i\theta\langle[\phi,\phi]\rangle$ the line is $+D^R$. Keep one convention and check it on the single-mode example above.

3.4 Exercises

Exercise 3.1. Write out the third-order term of Eq. (42) in the $\pm$ basis and identify the eight terms with the eight assignments of branches to three contour points. Show that after Tr_B with a Gaussian bath all of them vanish (odd moments).

Exercise 3.2. Derive the middle two lines of the rotated correlator, including the θ functions, from the $\pm$ table. Then show directly from the trace identities that $D^{\mathrm{qq}} = 0$ and that $D^{\mathrm{cq}}(t,s)$ vanishes for $s > t$, without using the table.

Exercise 3.3. Show that S^{c} and S^{q} of Eq. (17) obey the same two trace identities with Tr_S, and compute $\mathrm{Tr}_S[\tilde{S}^{\mathrm{c}}(t)\tilde{S}^{\mathrm{c}}(s)\rho_S]$ and $\mathrm{Tr}_S[\tilde{S}^{\mathrm{c}}(t)\tilde{S}^{\mathrm{q}}(s)\rho_S]$ for $\rho_S = \frac{1}{2}(\mathbb{1} + r_z\sigma_z)$ using Eq. (38). Interpret the second as the qubit's own response function.

Exercise 3.4. Verify Eq. (27) for the Lorentzian bath directly from Eq. (35), and show that $\chi(\omega) = g^2\big[(\Omega - \omega - i\gamma/2)^{-1} + (\Omega + \omega + i\gamma/2)^{-1}\big]$.

4 Lecture 4: Real-time diagrams for the reduced density matrix

We now combine the Dyson series (42), the vertex (44) and the contractions (53) into a diagrammatic expansion of the dynamical map $\Pi(t)$, and evaluate it for the qubit. The second-order result is the Born kernel that every master equation uses; the fourth-order structure introduces the notion of irreducibility on which Section 5 rests.

4.1 Feynman rules

Insert Eq. (44) into Eq. (42) and take Tr_B. Every vertex is either $S^{\mathrm{q}}B^{\mathrm{c}}$ or $S^{\mathrm{c}}B^{\mathrm{q}}$, so the bath index at a vertex is the *opposite* of the system index. Apply Wick's theorem to the bath string and translate Eq. (52) to the system side:

System-side Feynman rules for $\Pi(t)$

A term of order $2n$ is a time-ordered string of $2n$ vertices at $t > t_1 > \cdots > t_{2n} > 0$, with a pairing of the vertices into n lines. Reading from right (earliest) to left:

- each vertex contributes $-i\,\tilde{S}^{\mathrm{q}}(t_i)$ or $-i\,\tilde{S}^{\mathrm{c}}(t_i)$ (interaction picture; in the Schrödinger picture, replace $\tilde{S}^{a}(t_i)$ by S^{a} and insert free propagators $\mathrm{e}^{\mathcal{L}_S(t_i-t_{i+1})}$ between consecutive vertices and $\mathrm{e}^{\mathcal{L}_S t_{2n}}$ before the first, $\mathrm{e}^{\mathcal{L}_S(t-t_1)}$ after the last);
- a *noise line* $S(t_i-t_j)$ connects $S^{\mathrm{q}}(t_i)$ to $S^{\mathrm{q}}(t_j)$ (q–q on the system side is c–c on the bath side);
- a *response line* $-i\chi(t_i-t_j)$ connects the *later* $S^{\mathrm{q}}(t_i)$ to the *earlier* $S^{\mathrm{c}}(t_j)$;
- there is no line with both ends c, and no line whose later end is c.

Consequently the latest vertex of every diagram is S^{q}, which preserves Tr_S (trace preservation of Π), and the earliest vertex of every line can be either.

The last rule has a direct physical reading. A c vertex is where the bath *acts* on the system (a force, through χ); a q vertex is where the system *acts* on the bath or where noise enters. Causality says the bath can only respond to earlier system history, hence the response line points from an earlier c to a later q. For fermionic baths the same rules hold with a sign for each crossing of lines; for non-Gaussian baths, $2n$-point cumulants join more than two vertices [26].

4.2 Second order: the Born kernel

Example 4.1: the second-order term

From Eq. (41) with $\mathcal{L}_I = -i(S^{\mathrm{q}}B^{\mathrm{c}} + S^{\mathrm{c}}B^{\mathrm{q}})$,

$$\mathrm{Tr}_B\left[\tilde{\mathcal{L}}_I(t_1)\tilde{\mathcal{L}}_I(t_2)\,\rho_S\otimes\rho_B\right] = -\Big[\tilde{S}_1^{\mathrm{q}}\tilde{S}_2^{\mathrm{q}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{c}}B_2^{\mathrm{c}}\rho_B]}_{S(t_1-t_2)} + \tilde{S}_1^{\mathrm{q}}\tilde{S}_2^{\mathrm{c}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{c}}B_2^{\mathrm{q}}\rho_B]}_{-i\chi(t_1-t_2)} + \tilde{S}_1^{\mathrm{c}}\tilde{S}_2^{\mathrm{q}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{q}}B_2^{\mathrm{c}}\rho_B]}_{0} + \tilde{S}_1^{\mathrm{c}}\tilde{S}_2^{\mathrm{c}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{q}}B_2^{\mathrm{q}}\rho_B]}_{0}\Big]\rho_S, \tag{55}$$

with the shorthand $\tilde{S}_i^{a} = \tilde{S}^{a}(t_i)$, $B_i^{a} = \tilde{B}^{a}(t_i)$. Only two of the four index combinations survive, exactly as the rules say. In the Schrödinger picture the surviving bracket, with $\tau = t_1 - t_2$,

defines the *second-order kernel*

$$\boxed{K^{(2)}(\tau) = -S^{\mathrm{q}}\, \mathrm{e}^{\mathcal{L}_S \tau}\Big[S(\tau)\, S^{\mathrm{q}} - i\,\chi(\tau)\, S^{\mathrm{c}}\Big],} \tag{56}$$

and the second-order map is $\Pi^{(2)}(t) = \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \mathrm{e}^{\mathcal{L}_S(t-t_1)} K^{(2)}(t_1 - t_2)\, \mathrm{e}^{\mathcal{L}_S t_2}$. Expanding S^{c} and S^{q}, Eq. (56) is the textbook Born kernel $K^{(2)}(\tau)X = -\big[S, \mathrm{e}^{\mathcal{L}_S\tau}\big(C(\tau)SX - C(-\tau)XS\big)\big]$ [29, Sec. 3.3.1], but the rotated form keeps the two bath inputs apart, and that is what we want.

Example 4.2: $K^{(2)}$ for the qubit, in Bloch form

Apply Eq. (56) to the coefficient basis $(\mathbb{1}, \sigma_x, \sigma_y, \sigma_z)$ using the matrices (17) and the rotation (15). Write $c = \cos\theta$, $s = \sin\theta$, $S_\tau = S(\tau)$, $\chi_\tau = \chi(\tau)$, and cos, sin for $\cos\omega_0\tau, \sin\omega_0\tau$. Going through the four input vectors (the computation is written out in the solution to Exercise 4.2) gives

$$K^{(2)}(\tau) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 2cs\,\chi_\tau \sin & -4c^2 S_\tau \cos & 4c^2 S_\tau \sin & 4cs\, S_\tau \cos \\ 2cs\,\chi_\tau(1-\cos) & -4c^2 S_\tau \sin & -4S_\tau(c^2\cos + s^2) & 4cs\, S_\tau \sin \\ -2s^2\chi_\tau \sin & 4cs\, S_\tau \cos & -4cs\, S_\tau \sin & -4s^2 S_\tau \cos \end{pmatrix}, \tag{57}$$

Everything one needs to know about second-order qubit relaxation is in this matrix:

- The first row vanishes: trace preservation (last vertex is S^{q}).
- The first *column* is the only place χ appears. It multiplies the input $\mathbb{1}$, i.e. it is the inhomogeneous term that drives the Bloch vector toward a nonzero steady state. Noise alone (S) would relax the qubit to the maximally mixed state; it is the *response* of the bath that knows about the sign of ω_0 and produces the Boltzmann asymmetry between emission and absorption.
- Populations (z) relax only through $s^2 = \sin^2\theta$: the transverse part of the coupling, sampled at the qubit frequency ($S_\tau \cos\omega_0\tau$).
- The $c^2 = \cos^2\theta$ terms in the coherence block are sampled at zero frequency (S_τ alone, after the rotating-frame average): pure dephasing. The s^2 term in K_{yy} is the "T_1 contributes to T_2" piece.
- The cs cross terms couple populations to coherences. In the rotating frame they oscillate at $\pm\omega_0$ and average out when $\omega_0 \gg$ rates (secular approximation, Section 5); at short times they are real and are the reason Exercise 2.4 has no simple answer.

The matrix is real, as Hermiticity preservation requires in this basis.

4.3 Pure dephasing to all orders; reducible and irreducible diagrams

Example 4.3: $\theta = 0$ and the first irreducible diagram

Take $\theta = 0$, so $S = \sigma_z$ and $S^{\mathrm{c/q}}$ commute with $\mathcal{L}_S$. On the coherence $X = |0\rangle\langle 1|$ we have $S^{\mathrm{q}}X = [\sigma_z, X] = 2X$ and $S^{\mathrm{c}}X = \frac{1}{2}\{\sigma_z, X\} = 0$. Every response line ends on an S^{c} and therefore vanishes on X. What remains is a sum over pairings of noise lines, with every vertex

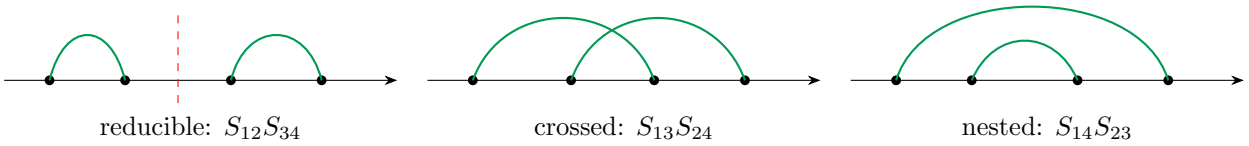


Figure 3: The three fourth-order diagrams. Time runs to the right; vertices are system superoperators; lines are bath contractions (S or χ). Only the first can be cut (red) without cutting a line.

equal to the number $-2i$:

$$\tilde{\rho}_{01}(t) = \rho_{01}(0)\Big[1 - 4\int_{t_1>t_2} S_{12} + 16\int_{t_1>t_2>t_3>t_4} \big[\underbrace{S_{12}S_{34}}_{\text{reducible}} + \underbrace{S_{13}S_{24}}_{\text{crossed}} + \underbrace{S_{14}S_{23}}_{\text{nested}}\big] - \dots\Big], \tag{58}$$

with $S_{ij} \equiv S(t_i - t_j)$, where the sign at order $2n$ is $(-2i)^{2n} = (-4)^n$. Because all vertices commute, the sum over pairings is $\frac{1}{n!}$ times the n-th power of the single contraction, and the series exponentiates:

$$\rho_{01}(t) = \rho_{01}(0)\, \mathrm{e}^{-i\omega_0 t}\, \mathrm{e}^{-\Gamma(t)}, \qquad \Gamma(t) = 4\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, S(t_1 - t_2) = \frac{4}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{S(\omega)}{\omega^2}\big(1 - \cos\omega t\big). \tag{59}$$

(The factor 4 is the "our $S = \sigma_z$ versus Leggett's $\sigma_z/2$" factor of Section 2.) This is the exact pure-dephasing result [29, Sec. 4.2]; we will rederive it from the influence functional in Section 7 and use it in Section 10.

Now look at the three fourth-order pairings. The first, $S_{12}S_{34}$, can be cut between t_2 and t_3 without cutting a line: the bath has returned to ρ_B there, and the diagram is a product of two second-order pieces. It is *reducible*. The crossed and nested pairings cannot be cut: they are *irreducible*, and they are the fourth-order contributions to the memory kernel of Section 5. Summing all orders, the reducible structure is what makes the exponential; the irreducible pieces are what the exponent is made of. The general statement is Section 5.2.

4.4 What the diagrams are good for

The series (42) converges for any finite t but is useless as it stands: at order $2n$ there are $(2n-1)!!$ pairings and the terms grow like $(g^2t^2)^n$. Every practical method is a *resummation*:

- **Dyson resummation** (Sections 5 and 6): sum the reducible structure exactly and approximate the irreducible kernel. This is Nakajima–Zwanzig, and TCL is the same thing organized differently.
- **Exponentiation** (Section 7): when the vertices commute (or are replaced by c-number paths), the pairings exponentiate into an influence functional. The noninteracting-blip approximation is a controlled truncation of that exponent.
- **Self-consistent resummation** (Section 9): replace bare propagators by dressed ones inside the kernel (Kadanoff–Baym, GKBA, DMFT-like schemes).

In all three, the only input from the bath is the pair (S, χ), and the only thing that distinguishes the methods is which subset of diagrams is kept and how it is organized.

4.5 Exercises

Exercise 4.1. From the Feynman rules, show that $\mathrm{Tr}_S\, K^{(2)}(\tau) X = 0$ for every X and interpret it. Show also that $K^{(2)}(\tau)\rho$ is Hermitian for Hermitian ρ (use $(S^{\mathrm{c}})^\dagger = S^{\mathrm{c}}$, $(S^{\mathrm{q}})^\dagger = -S^{\mathrm{q}}$ and the reality of $S(\tau), \chi(\tau)$).

Exercise 4.2. Derive Eq. (57). (Apply Eq. (56) to each of $\mathbb{1}, \sigma_x, \sigma_y, \sigma_z$ in turn; the solution writes out the σ_z and $\mathbb{1}$ columns.) Check the $\theta = 0$ and $\theta = \pi/2$ limits against your expectations for pure dephasing and for a damped qubit.

Exercise 4.3. List the 15 pairings at sixth order and classify them as reducible (cuttable) or irreducible. How many are irreducible? (Answer: 10 of the 15 are irreducible if one counts only cuts between consecutive vertices; the reducible ones are products of lower-order irreducible blocks.)

Exercise 4.4. Evaluate $\Gamma(t)$ in Eq. (59) for the Lorentzian bath at $T = 0$ and show that $\Gamma(t) \simeq 2g^2 t^2$ for $t \ll 1/\Omega$ (Gaussian decay) and $\Gamma(t) \simeq 2S(\omega{=}0)\, t + \text{const}$ for $t \gg 1/\gamma$ (exponential decay). Which value of $S(\omega{=}0)$ enters?

5 Lecture 5: Nakajima–Zwanzig, and why the memory kernel is a self-energy

The Nakajima–Zwanzig (NZ) equation is an exact, closed equation of motion for $\rho_S(t)$. It is derived in five lines with a projection operator, and its memory kernel K looks at first like a formal object. The content of this lecture is that K is exactly the sum of the irreducible diagrams of Section 4, so that the NZ equation is the Dyson equation of the real-time expansion. We then evaluate K at second order for the qubit and recover the Bloch–Redfield description, with every rate and shift expressed through S and χ.

5.1 The projection and the exact equation

Define the Nakajima–Zwanzig projector [3, 4]

$$\mathcal{P}X = \rho_B \otimes \mathrm{Tr}_B\, X, \qquad \mathcal{Q} = 1 - \mathcal{P}, \qquad \mathcal{P}^2 = \mathcal{P}. \tag{60}$$

$\mathcal{P}$ keeps the "relevant" part, the reduced state tensored with the fixed reference bath state. Three facts follow from $[H_B, \rho_B] = 0$ and $\langle B \rangle = 0$:

$$\mathcal{P}\mathcal{L}_S = \mathcal{L}_S\mathcal{P}, \qquad \mathcal{P}\mathcal{L}_B = \mathcal{L}_B\mathcal{P} = 0 \text{ on the range of } \mathcal{P}, \qquad \mathcal{P}\mathcal{L}_I\mathcal{P} = 0. \tag{61}$$

(The last one: $\mathcal{P}\mathcal{L}_I\mathcal{P}X = \rho_B \otimes \mathrm{Tr}_B[-i(SB \otimes \rho_B X' - \dots)] \propto \langle B \rangle = 0$.) Split the von Neumann equation $\dot\rho = \mathcal{L}\rho$ into its $\mathcal{P}$ and $\mathcal{Q}$ components:

$$\partial_t \mathcal{P}\rho = \mathcal{P}\mathcal{L}\mathcal{P}\rho + \mathcal{P}\mathcal{L}\mathcal{Q}\rho, \qquad \partial_t \mathcal{Q}\rho = \mathcal{Q}\mathcal{L}\mathcal{P}\rho + \mathcal{Q}\mathcal{L}\mathcal{Q}\rho. \tag{62}$$

The second is a linear inhomogeneous equation for $\mathcal{Q}\rho$ with the known source $\mathcal{Q}\mathcal{L}\mathcal{P}\rho(s)$:

$$\mathcal{Q}\rho(t) = \mathrm{e}^{\mathcal{Q}\mathcal{L}t}\mathcal{Q}\rho(0) + \int_0^t \mathrm{d}s\, \mathrm{e}^{\mathcal{Q}\mathcal{L}(t-s)}\, \mathcal{Q}\mathcal{L}\mathcal{P}\rho(s). \tag{63}$$

Insert into the first and use Eq. (61):

$$\boxed{\dot\rho_S(t) = \mathcal{L}_S\rho_S(t) + \int_0^t \mathrm{d}s\, K(t-s)\,\rho_S(s) + I(t), \qquad K(\tau)X = \mathrm{Tr}_B\Big[\mathcal{L}_I\, \mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\, \mathcal{Q}\mathcal{L}_I\big(X \otimes \rho_B\big)\Big],} \tag{64}$$

with the inhomogeneity $I(t) = \mathrm{Tr}_B[\mathcal{L}_I \mathrm{e}^{\mathcal{Q}\mathcal{L}t}\mathcal{Q}\rho(0)]$, which vanishes for a factorized initial state ($\mathcal{Q}\rho(0) = 0$). This is the NZ equation [29, Sec. 9.1]. It is exact. Nothing has been assumed about the strength of the coupling, the Gaussianity of the bath or the size of the system. The kernel $K(\tau)$ is a superoperator on the system; it depends on τ only (stationary bath, time-independent H) and it contains the full $\mathcal{L}$ inside the exponential, which is where all the difficulty hides.

Laplace form. With $\hat f(z) = \int_0^\infty \mathrm{d}t\, \mathrm{e}^{-zt} f(t)$ ($\mathrm{Re}\, z > 0$), the convolution becomes a product and Eq. (64) for the map reads

$$\boxed{\hat\Pi(z) = \frac{1}{z - \mathcal{L}_S - \hat K(z)} \quad \Longleftrightarrow \quad \hat\Pi = \hat\Pi_0 + \hat\Pi_0\, \hat K\, \hat\Pi, \qquad \hat\Pi_0 = (z - \mathcal{L}_S)^{-1}.} \tag{65}$$

This is a Dyson equation with $\hat K$ in the role of the self-energy. The next subsection shows that the analogy is an identity.

5.2 The kernel is the sum of irreducible diagrams

This subsection is the heart of the course, so we go slowly. The claim is that the formal object $K(\tau)$ of Eq. (64), with the full Liouvillian inside $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}$, is exactly the sum of the "irreducible" real-time diagrams of Section 4, and that the NZ equation is the Dyson resummation of the reducible ones. We first fix the vocabulary, then prove three lemmas, then the theorem, then its converse (the Dyson resummation), and finally check everything explicitly at fourth and sixth order.

5.2.1 Vocabulary: strings, gaps, cuts, pairings

Throughout, $\mathcal{L}_0 = \mathcal{L}_S + \mathcal{L}_B$ and $\tilde{\mathcal{L}}_I(t) = \mathrm{e}^{-\mathcal{L}_0 t}\mathcal{L}_I \mathrm{e}^{\mathcal{L}_0 t}$ is the interaction-picture vertex, $\tilde{\mathcal{L}}_I(t) = -i[\tilde{S}^{\mathrm{q}}(t)\tilde{B}^{\mathrm{c}}(t) + \tilde{S}^{\mathrm{c}}(t)\tilde{B}^{\mathrm{q}}(t)]$ by Eq. (44).

Definition 5.1 (Wick string)**.** For ordered times $t_0 > t_1 > \cdots > t_k$ and a system operator X, the *Wick string* is the superoperator

$$W(t_0, \ldots, t_k)X \equiv \mathrm{Tr}_B\Big[\tilde{\mathcal{L}}_I(t_0)\tilde{\mathcal{L}}_I(t_1)\cdots\tilde{\mathcal{L}}_I(t_k)\,(X \otimes \rho_B)\Big]. \tag{66}$$

Its k *gaps* are the intervals (t_{j+1}, t_j) between consecutive vertices, $j = 0, \ldots, k-1$. By Wick's theorem (36) applied to the bath factors (every vertex carries exactly one bath superoperator), W is a sum over *pairings* π of the $k+1$ vertices (so $k+1$ must be even, otherwise $W = 0$):

$$W(t_0, \ldots, t_k) = \sum_{\pi} W_\pi(t_0, \ldots, t_k), \qquad W_\pi = \prod_{\text{vertices}} \big(-i\,\tilde{S}^{a_j}(t_j)\big) \prod_{(ij)\in\pi} D^{\bar{a}_i \bar{a}_j}(t_i, t_j), \tag{67}$$

where $a_j \in \{\mathrm{c}, \mathrm{q}\}$ is the system index at vertex j, $\bar{a}_j$ the opposite index, and the sum over the a_j is understood. W_π is what we called a *diagram* in Section 4; the Feynman rules there are just Eq. (67) with Eq. (52) inserted (recall $D^{\mathrm{qq}} = 0$ and $D^{\mathrm{qc}}(t_i, t_j) = 0$ for $t_i > t_j$, which kills many assignments).

Definition 5.2 (Cuts, reducibility)**.** A pairing π *crosses* gap j if some pair $(il) \in \pi$ has $i \le j < l$, i.e. one vertex before the gap and one after it. A gap not crossed by π is a *cut* of π. Write $U(\pi) \subseteq \{0, \ldots, k-1\}$ for the set of cuts of π. A pairing is *irreducible* if $U(\pi) = \varnothing$ (every gap is crossed by at least one line) and *reducible* otherwise.

At a cut, the vertices to the left and to the right are paired only among themselves; the bath, which was in ρ_B at the start, is "back in ρ_B" there in the sense that no bath correlation connects the two sides. The first lemma makes this precise.

5.2.2 Three lemmas

Lemma 5.1 (A $\mathcal{P}$ in a gap factorizes the string)**.** *Inserting $\mathcal{P}$ in gap j of a Wick string gives the product of the two sub-strings,*

$$\mathrm{Tr}_B\Big[\tilde{\mathcal{L}}_I(t_0)\cdots\tilde{\mathcal{L}}_I(t_j)\;\mathcal{P}\;\tilde{\mathcal{L}}_I(t_{j+1})\cdots\tilde{\mathcal{L}}_I(t_k)\,(X \otimes \rho_B)\Big] = W(t_0, \ldots, t_j)\,W(t_{j+1}, \ldots, t_k)\,X, \tag{68}$$

and therefore equals $\sum_{\pi:\, j\in U(\pi)} W_\pi$, the sum over pairings that have a cut at gap j. More generally, inserting $\mathcal{P}$ in every gap of a set $\mathcal{C}$ gives $\sum_{\pi:\, \mathcal{C}\subseteq U(\pi)} W_\pi$.

Proof. By definition $\mathcal{P}Y = \rho_B \otimes \mathrm{Tr}_B\, Y$. Apply it to $Y = \tilde{\mathcal{L}}_I(t_{j+1}) \cdots \tilde{\mathcal{L}}_I(t_k)(X \otimes \rho_B)$: $\mathrm{Tr}_B\, Y = W(t_{j+1}, \ldots, t_k)X$ is a system operator, and $\mathcal{P}Y = \big(W(t_{j+1}, \ldots, t_k)X\big) \otimes \rho_B$. The remaining string acts on this factorized operator, which gives the first statement. For the second, apply Wick's theorem to each factor: the left factor is a sum over pairings of $\{0, \ldots, j\}$, the right factor a sum over pairings of $\{j+1, \ldots, k\}$, and their product is the sum over pairings of $\{0, \ldots, k\}$ that pair no vertex on the left with one on the right, i.e. those with a cut at j. The value of each such pairing is the same whether computed in the full string or in the product, because the contraction $D^{ab}(t_i, t_l)$ of a pair depends only on the two times and indices, and the time ordering is the same. The statement for a set $\mathcal{C}$ follows by induction on $|\mathcal{C}|$. □

Lemma 5.2 (Expansion of $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}$)**.** *With $\mathcal{P}$ of Eq. (60), $[\mathcal{P}, \mathcal{L}_0] = 0$, and for every s*

$$\mathrm{e}^{\mathcal{Q}\mathcal{L}_0 s} = \mathcal{P} + \mathcal{Q}\,\mathrm{e}^{\mathcal{L}_0 s}, \qquad \textit{hence} \qquad \mathrm{e}^{\mathcal{Q}\mathcal{L}_0 s}\,\mathcal{Q} = \mathcal{Q}\,\mathrm{e}^{\mathcal{L}_0 s} = \mathrm{e}^{\mathcal{L}_0 s}\,\mathcal{Q}. \tag{69}$$

Consequently

$$\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\,\mathcal{Q} = \sum_{m\geq 0} \int_{\tau > s_1 > \cdots > s_m > 0} \mathcal{Q}\,\mathrm{e}^{\mathcal{L}_0(\tau - s_1)}\,\mathcal{Q}\mathcal{L}_I\,\mathrm{e}^{\mathcal{L}_0(s_1 - s_2)}\,\mathcal{Q}\mathcal{L}_I \cdots \mathrm{e}^{\mathcal{L}_0(s_{m-1} - s_m)}\,\mathcal{Q}\mathcal{L}_I\,\mathrm{e}^{\mathcal{L}_0 s_m}\,\mathcal{Q}, \tag{70}$$

i.e. the ordinary Dyson series in $\mathcal{L}_I$ with free propagation $\mathrm{e}^{\mathcal{L}_0 \Delta s}$ between vertices and a factor $\mathcal{Q}$ in every gap (and at both ends).

Proof. $[\mathcal{P}, \mathcal{L}_S] = 0$ because $\mathcal{L}_S$ acts on the system factor only: $\mathcal{P}\mathcal{L}_S Y = \rho_B \otimes \mathrm{Tr}_B(-i[H_S, Y]) = -i[H_S, \rho_B \otimes \mathrm{Tr}_B\, Y] = \mathcal{L}_S \mathcal{P} Y$. And $\mathcal{P}\mathcal{L}_B = 0$ (trace of a bath commutator) while $\mathcal{L}_B \mathcal{P} Y = -i[H_B, \rho_B] \otimes \mathrm{Tr}_B\, Y = 0$ (stationary ρ_B); so $\mathcal{P}\mathcal{L}_B = \mathcal{L}_B \mathcal{P} = 0$. Hence $\mathcal{Q}$ commutes with $\mathcal{L}_0$ and $(\mathcal{Q}\mathcal{L}_0)^n = \mathcal{Q}^n \mathcal{L}_0^n = \mathcal{Q}\mathcal{L}_0^n$ for $n \geq 1$, which gives $\mathrm{e}^{\mathcal{Q}\mathcal{L}_0 s} = 1 + \mathcal{Q}(\mathrm{e}^{\mathcal{L}_0 s} - 1) = \mathcal{P} + \mathcal{Q}\mathrm{e}^{\mathcal{L}_0 s}$. Multiplying by $\mathcal{Q}$ on the right and using $\mathcal{P}\mathcal{Q} = 0$, $\mathcal{Q}^2 = \mathcal{Q}$ gives Eq. (69). For Eq. (70), write $\mathcal{Q}\mathcal{L} = \mathcal{Q}\mathcal{L}_0 + \mathcal{Q}\mathcal{L}_I$ and expand $\mathrm{e}^{(\mathcal{Q}\mathcal{L}_0 + \mathcal{Q}\mathcal{L}_I)\tau}$ in the usual Dyson series with "free" part $\mathrm{e}^{\mathcal{Q}\mathcal{L}_0 s}$ and "perturbation" $\mathcal{Q}\mathcal{L}_I$; every $\mathrm{e}^{\mathcal{Q}\mathcal{L}_0 \Delta s}$ is adjacent to a $\mathcal{Q}$ (from a neighbouring $\mathcal{Q}\mathcal{L}_I$ or from the final $\mathcal{Q}$), so Eq. (69) replaces it by $\mathrm{e}^{\mathcal{L}_0 \Delta s}$ with a $\mathcal{Q}$ kept in the gap. □

Lemma 5.3 (Inclusion–exclusion)**.** *For any finite set U, $\sum_{\mathcal{C} \subseteq U} (-1)^{|\mathcal{C}|} = \delta_{U,\varnothing}$.*

Proof. $\sum_{\mathcal{C} \subseteq U} (-1)^{|\mathcal{C}|} = \prod_{u \in U} (1 - 1) = 0^{|U|}$, which is 1 for $U = \varnothing$ and 0 otherwise. □

5.2.3 The theorem

Theorem 5.4 (NZ kernel = irreducible diagrams)**.** *For the projector (60) and a Gaussian bath with $\langle B \rangle = 0$,*

$$K(\tau) = \mathrm{e}^{\mathcal{L}_S \tau} \sum_{m \geq 0} \int_{\tau > s_1 > \cdots > s_m > 0} \sum_{\pi \text{ irreducible}} W_\pi(\tau, s_1, \ldots, s_m, 0), \tag{71}$$

where the inner sum runs over the irreducible pairings of the $m + 2$ vertices at $\tau, s_1, \ldots, s_m, 0$ (so only even m contribute). In words: $K(\tau)$ is the sum of all diagrams with outermost vertices at 0 and τ that cannot be cut between two consecutive vertices without cutting a line, dressed with the free system propagation that brings the interaction-picture string back to the Schrödinger picture.

Proof. Step 1: reduce K to a Wick string with $\mathcal{Q}$'s in the gaps. Insert Eq. (70) into $K(\tau)X = \mathrm{Tr}_B[\mathcal{L}_I \mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\,\mathcal{Q}\mathcal{L}_I(X \otimes \rho_B)]$. The two end factors of $\mathcal{Q}$ can be dropped: on the right, $\mathcal{Q}\mathcal{L}_I\mathcal{P} = \mathcal{L}_I\mathcal{P} -$

$\mathcal{P}\mathcal{L}_I\mathcal{P} = \mathcal{L}_I\mathcal{P}$ by Eq. (61); on the left, $\mathrm{Tr}_B[\mathcal{L}_I\mathcal{Q}Y] = \mathrm{Tr}_B[\mathcal{L}_I Y] - \mathrm{Tr}_B[\mathcal{L}_I\mathcal{P}Y]$ and $\mathrm{Tr}_B[\mathcal{L}_I(\rho_B\otimes Z)] = -i\,\mathrm{Tr}_B[(SB\otimes\rho_B Z - \dots)] \propto \langle B\rangle = 0$. What remains is

$$K(\tau)X = \sum_{m\geq 0}\int_{\tau>s_1>\cdots>s_m>0} \mathrm{Tr}_B\Big[\mathcal{L}_I\, \mathrm{e}^{\mathcal{L}_0(\tau-s_1)}\,\mathcal{Q}\mathcal{L}_I\, \mathrm{e}^{\mathcal{L}_0(s_1-s_2)}\,\mathcal{Q}\mathcal{L}_I\cdots\mathcal{Q}\mathcal{L}_I\, \mathrm{e}^{\mathcal{L}_0 s_m}\,\mathcal{Q}\,\mathcal{L}_I\,(X\otimes\rho_B)\Big]. \tag{72}$$

Note that there is one $\mathcal{Q}$ in each of the $m+1$ gaps between the $m+2$ vertices (the two $\mathcal{Q}$'s that were adjacent in the first gap, $\mathcal{L}_I\mathcal{Q}\mathrm{e}^{\mathcal{L}_0(\tau-s_1)}\mathcal{Q}\mathcal{L}_I$, merge into one because $\mathcal{Q}$ commutes with $\mathrm{e}^{\mathcal{L}_0 s}$ and $\mathcal{Q}^2=\mathcal{Q}$; we then dropped the outer one).

Step 2: go to the interaction picture. Write $\mathrm{e}^{\mathcal{L}_0(s_{j-1}-s_j)} = \mathrm{e}^{\mathcal{L}_0 s_{j-1}}\mathrm{e}^{-\mathcal{L}_0 s_j}$ and move all $\mathrm{e}^{\pm\mathcal{L}_0 s}$ next to the vertices, using $[\mathcal{Q},\mathrm{e}^{\mathcal{L}_0 s}]=0$: the string becomes $\mathrm{e}^{\mathcal{L}_0\tau}\,\tilde{\mathcal{L}}_I(\tau)\,\mathcal{Q}\,\tilde{\mathcal{L}}_I(s_1)\,\mathcal{Q}\cdots\mathcal{Q}\,\tilde{\mathcal{L}}_I(s_m)\,\mathcal{Q}\,\tilde{\mathcal{L}}_I(0)$. The leftover $\mathrm{e}^{\mathcal{L}_0\tau}$ sits directly under Tr_B, and $\mathrm{Tr}_B[\mathrm{e}^{\mathcal{L}_0\tau}Y] = \mathrm{e}^{\mathcal{L}_S\tau}\,\mathrm{Tr}_B[\mathrm{e}^{\mathcal{L}_B\tau}Y] = \mathrm{e}^{\mathcal{L}_S\tau}\,\mathrm{Tr}_B\,Y$ because $\mathrm{e}^{\mathcal{L}_B\tau}$ is a unitary conjugation on the bath factor, which the partial trace does not see. Hence

$$K(\tau)X = \mathrm{e}^{\mathcal{L}_S\tau}\sum_{m\geq 0}\int_{\tau>s_1>\cdots>s_m>0} \mathrm{Tr}_B\Big[\tilde{\mathcal{L}}_I(\tau)\,\mathcal{Q}\,\tilde{\mathcal{L}}_I(s_1)\,\mathcal{Q}\,\tilde{\mathcal{L}}_I(s_2)\cdots\mathcal{Q}\,\tilde{\mathcal{L}}_I(s_m)\,\mathcal{Q}\,\tilde{\mathcal{L}}_I(0)\,(X\otimes\rho_B)\Big], \tag{73}$$

a Wick string of $m+2$ vertices with $\mathcal{Q}=1-\mathcal{P}$ in each of its $m+1$ gaps.

Step 3: expand the $\mathcal{Q}$'s. $\prod_{j=0}^{m}(1-\mathcal{P}_j) = \sum_{\mathcal{C}\subseteq\{0,\dots,m\}}(-1)^{|\mathcal{C}|}\prod_{j\in\mathcal{C}}\mathcal{P}_j$, where $\mathcal{P}_j$ denotes $\mathcal{P}$ inserted in gap j. By Lemma 5.1, the term with the set $\mathcal{C}$ equals $\sum_{\pi:\,\mathcal{C}\subseteq U(\pi)} W_\pi$. Therefore

$$\mathrm{Tr}_B\left[\cdots\mathcal{Q}\cdots\mathcal{Q}\cdots\right] = \sum_{\mathcal{C}}(-1)^{|\mathcal{C}|}\sum_{\pi:\,\mathcal{C}\subseteq U(\pi)} W_\pi = \sum_{\pi} W_\pi \sum_{\mathcal{C}\subseteq U(\pi)}(-1)^{|\mathcal{C}|} = \sum_{\pi} W_\pi\,\delta_{U(\pi),\varnothing}, \tag{74}$$

where we exchanged the two finite sums and used Lemma 5.3. Only pairings with no cut survive, which is Eq. (71). □

Remark. The three ingredients are worth separating. (i) $\mathcal{P}$ *factorizes* a string (Lemma 5.1); this uses only the form $\mathcal{P}=\rho_B\otimes\mathrm{Tr}_B$. (ii) The exponential $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}$ puts a $\mathcal{Q}$ in every gap (Lemma 5.2); this uses only $[\mathcal{P},\mathcal{L}_0]=0$. (iii) Inclusion–exclusion converts "$\mathcal{Q}$ in every gap" into "no cut anywhere"; this is pure combinatorics. Gaussianity enters only in identifying the factorized strings with sums over pairings. For a non-Gaussian bath the same three steps go through with pairings replaced by partitions into cumulants, and the result is that K is the sum of diagrams in which every gap is crossed by at least one cumulant block [26]. For fermionic reservoirs each W_π acquires the sign of the permutation that brings paired operators next to each other; nothing else changes.

5.2.4 The converse: the NZ equation as a Dyson resummation

The theorem identifies K. The complementary statement is that the NZ equation rebuilds the *full* map from K by summing the reducible diagrams, which is what one means by a Dyson equation.

Proposition 5.5 (Unique decomposition into irreducible blocks)**.** *Every pairing π of the vertices $t_0>t_1>\cdots>t_k$ decomposes uniquely as a sequence of irreducible pairings of consecutive blocks of vertices, the blocks being separated by the cuts of π. Conversely, concatenating irreducible pairings of consecutive blocks gives a pairing whose cuts are exactly the block boundaries.*

Proof. Let $U(\pi)=\{j_1<\cdots<j_r\}$. The vertices between consecutive cuts form blocks; no line connects different blocks (that is what a cut means), so π restricted to a block is a pairing of that block; and that restriction has no cut, because a cut inside a block would be a cut of π not in $U(\pi)$. Uniqueness: the blocks are determined by $U(\pi)$, which is determined by π. The converse is immediate. □

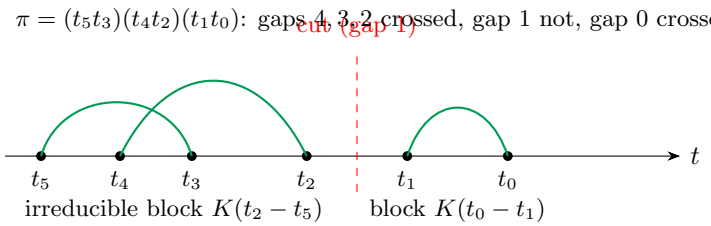


Figure 4: A sixth-order pairing with exactly one cut. By Proposition 5.5 it is the product of a fourth-order irreducible (crossed) block and a second-order block, with free propagation $\mathrm{e}^{\mathcal{L}_S(t_1-t_2)}$ across the cut. In the NZ equation this term is generated by $\iint \Pi_0 K^{(4)} \Pi_0 K^{(2)} \Pi_0$; in the kernel it is removed by the $\mathcal{P}$ inserted in gap 1.

Now expand the full map $\Pi(t) = \mathrm{Tr}_B[\mathrm{e}^{\mathcal{L}t}(\cdot \otimes \rho_B)]$ in the Dyson series (42) and apply Wick's theorem: in the Schrödinger picture, every term is a product of free propagators $\mathrm{e}^{\mathcal{L}_S \Delta t}$ and diagrams W_π. Group the terms by the cuts of π using Proposition 5.5: a term with r cuts is a product of $r+1$ irreducible blocks, each of which is (after the time integrations over its interior vertices) a factor $K(\tau_\ell)$ with τ_ℓ the duration of the block, separated by free propagation across the cuts. Summing over the number of blocks and over the positions of the block boundaries,

$$\Pi(t) = \Pi_0(t) + \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \Pi_0(t-t_1) K(t_1-t_2) \Pi_0(t_2) + \iiint \Pi_0 K \Pi_0 K \Pi_0 + \dots, \qquad \Pi_0(t) = \mathrm{e}^{\mathcal{L}_S t}, \tag{75}$$

which is the iterated solution of $\Pi(t) = \Pi_0(t) + \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \Pi_0(t-t_1) K(t_1-t_2) \Pi(t_2)$, i.e. of the NZ equation $\dot{\Pi} = \mathcal{L}_S \Pi + \int K\, \Pi$. In Laplace space this is Eq. (65), $\hat{\Pi} = (z - \mathcal{L}_S - \hat{K})^{-1} = \hat{\Pi}_0 + \hat{\Pi}_0 \hat{K} \hat{\Pi}_0 + \hat{\Pi}_0 \hat{K} \hat{\Pi}_0 \hat{K} \hat{\Pi}_0 + \dots$, a geometric series in $\hat{K}\hat{\Pi}_0$. The geometric series is precisely the statement that reducible diagrams are products of irreducible ones with free propagation in between; the time integrations over block boundaries turn the products into convolutions, and convolutions into products of Laplace transforms. One point deserves emphasis: a given term of the series (75) with $r+1$ blocks is not "of order $g^{2(r+1)}$" in any useful sense, because each K contains all orders. The expansion parameter of Eq. (75) is the number of returns of the bath to ρ_B, not the coupling.

5.2.5 Explicit checks at fourth and sixth order

Example 5.1: fourth order

Take $m = 2$ in Eq. (73): vertices at $\tau > s_1 > s_2 > 0$, three gaps, and $\prod_{j=0}^{2}(1 - \mathcal{P}_j)$ has eight terms. Most vanish identically, for a reason that is general: a $\mathcal{P}$ in gap j splits the string into a left part with $j+1$ vertices and a right part with $m+1-j$ vertices, and each part is a Wick string of its own, which vanishes if it has an odd number of vertices. Hence $\mathcal{P}_0$ and $\mathcal{P}_2$ (one vertex on one side) give zero, and so does any set $\mathcal{C}$ containing them. Only $\mathcal{C} = \varnothing$ and $\mathcal{C} = \{1\}$

survive:

$$K^{(4)}(\tau) = \mathrm{e}^{\mathcal{L}_S\tau}\int_{\tau>s_1>s_2>0}\Big\{W(\tau,s_1,s_2,0) - W(\tau,s_1)\,W(s_2,0)\Big\}$$
$$= \mathrm{e}^{\mathcal{L}_S\tau}\int_{\tau>s_1>s_2>0}\Big\{\big[W_{(\tau s_1)(s_2 0)} + W_{(\tau s_2)(s_1 0)} + W_{(\tau 0)(s_1 s_2)}\big] - W_{(\tau s_1)(s_2 0)}\Big\}$$
$$= \mathrm{e}^{\mathcal{L}_S\tau}\int_{\tau>s_1>s_2>0}\Big\{W_{\text{crossed}} + W_{\text{nested}}\Big\}, \tag{76}$$

exactly the two irreducible fourth-order diagrams of Fig. 3. The subtraction of the reducible pairing is not an approximation imposed by hand; it is the single $\mathcal{P}$ that survives in the middle gap. For $\theta = 0$ on the coherence, with all vertices equal to $-2i$ and every line a noise line, this is $16\int_{\tau>s_1>s_2>0}[S(\tau-s_2)S(s_1) + S(\tau)S(s_1-s_2)]$, the crossed and nested integrals of Eq. (58). The identity can be checked numerically to the letter. In the pseudomode representation (Appendix A) both $K(\tau)$, as $\mathcal{P}\mathcal{L}_I\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}\mathcal{L}_I\mathcal{P}$, and $K^{(4)}(\tau)$, as the double integral of $\mathcal{P}\mathcal{L}_I\mathrm{e}^{\mathcal{L}_0\cdot}\mathcal{Q}\mathcal{L}_I\mathrm{e}^{\mathcal{L}_0\cdot}\mathcal{Q}\mathcal{L}_I\mathrm{e}^{\mathcal{L}_0\cdot}\mathcal{Q}\mathcal{L}_I\mathcal{P}$, are finite matrices. For the Lorentzian bath at $\theta = \pi/4$ one finds, e.g. at $\tau = 1/\omega_0$, $\max|K - K^{(2)}| = 4.1\times10^{-4}$ and $\max|K - K^{(2)} - K^{(4)}| = 4\times10^{-6}$ at $g = 0.1$, and 2.6×10^{-5} versus 6×10^{-7} at $g = 0.05$: the residual is of order g^6 (and dominated by the quadrature error), as it must be.

Example 5.2: sixth order, and the count of irreducible diagrams

With $m = 4$ there are five gaps, numbered 0 to 4 from the latest vertex. The parity argument kills every $\mathcal{C}$ that contains gap 0, 2 or 4 (a $\mathcal{P}$ there leaves an odd number of vertices on one side). The surviving sets are $\varnothing$, $\{1\}$, $\{3\}$ and $\{1,3\}$:

$$K^{(6)} = \mathrm{e}^{\mathcal{L}_S\tau}\int\Big\{W_6 - W_2W_4 - W_4W_2 + W_2W_2W_2\Big\}, \tag{77}$$

where W_6 is the full six-vertex string (15 pairings), W_2W_4 the pairings with a cut after the first two vertices ($1\times3 = 3$ pairings), W_4W_2 those with a cut before the last two (3 pairings), and $W_2W_2W_2$ the single pairing with both cuts. Counting: $15-3-3+1 = 10$ irreducible pairings, which matches the sequence $1, 2, 10, 74, \ldots$ of irreducible chord diagrams quoted in Exercise 4.3. The reducible ones are $5 = 15-10$: the pairing with two cuts, and the $2+2$ pairings with exactly one cut (an irreducible fourth-order block next to a second-order one, as in Fig. 4). Note that the inclusion–exclusion signs are needed to avoid double-subtracting the pairing with two cuts, which appears in both W_2W_4 and W_4W_2.

5.2.6 What the theorem does and does not say

- **Irreducibility is relative to $\mathcal{P}$.** A cut is a moment at which the bath is *exactly* back in the reference state ρ_B. Change the reference state (a correlated projector, a projector onto populations only as in Section 7, Mori's operator projector of Section 8) and both the "free" propagation and the set of diagrams that count as reducible change with it. The kernel is not a property of the model alone but of the model and the projector.
- $K(0) = K^{(2)}(0)$ **exactly.** At $\tau = 0$ there is no room for internal vertices, so only $m = 0$ contributes. This fixes the short-time behaviour of every quantity to all orders (used in Section 10).

- **K is small; Π is not.** K starts at order g^2 while Π starts at order 1. Truncating K and then solving Eq. (64) resums an infinite class of diagrams of Π: all products of the kept irreducible blocks. This is why a second-order kernel can describe long-time exponential decay, which no finite-order truncation of Π itself can (a polynomial in t does not decay).
- **Truncating K is not the same as truncating $\mathcal{G}$.** The TCL generator of Section 6 resums a *different* class of diagrams at the same nominal order; the two agree at second order and differ from fourth order on [30, 31].
- **The theorem is representation-independent.** It is the real-time, Liouville-space version of "the self-energy is one-particle irreducible". In the fermionic reservoir context the identification is due to Schoeller and Schön [32, 33]; it is the starting point of the real-time renormalization group [26, 34], and the explicit fourth-order comparison with the NZ expansion is in [30, 31]. Historically, Zwanzig [35] already showed that the projection-operator master equation coincides with the resummed perturbation series of Prigogine's school and of van Hove; the present diagrammatic form is the modern version of that identity.

5.3 Second order: Born, Redfield, secular, Bloch

At order $\mathcal{L}_I^2$ we drop $\mathcal{Q}$ in Eq. (64) (because $\mathcal{P}\mathcal{L}_I\mathcal{P}=0$) and replace $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\to\mathrm{e}^{\mathcal{L}_0\tau}$; the result is $K^{(2)}$ of Eq. (56), now identified as the first term of the irreducible series. The approximations one layers on top are:

1. **Born (NZ2).** Keep $K\approx K^{(2)}$ in Eq. (64), as a time-nonlocal equation.
2. **Redfield (time-local, nonsecular).** Replace $\rho_S(s)\to\mathrm{e}^{-\mathcal{L}_S(t-s)}\rho_S(t)$ under the integral (valid when ρ_S changes little, in the interaction picture, over the bath correlation time) and extend the upper limit to ∞:

$$\dot\rho_S=\big[\mathcal{L}_S+\mathcal{G}^{(2)}_\infty\big]\rho_S,\qquad \mathcal{G}^{(2)}_\infty=\int_0^\infty \mathrm{d}\tau\, K^{(2)}(\tau)\,\mathrm{e}^{-\mathcal{L}_S\tau}. \tag{78}$$

   This is Redfield's equation [36–38]; we will see in Section 6 that it is the $t\to\infty$ limit of the second-order TCL generator.
3. **Secular.** Drop the terms of $\mathcal{G}^{(2)}_\infty$ that oscillate at $\pm\omega_0$ or $\pm2\omega_0$ in the interaction picture. The result is of GKSL form (19) and generates a completely positive semigroup (Davies' theorem [39]; see [29, Sec. 3.3.1] for the general construction). Without the secular step, Redfield can violate positivity at short times.

Example 5.3: the Bloch equations for the qubit

Apply Eq. (78) to the kernel (57). The rotation $\mathrm{e}^{-\mathcal{L}_S\tau}$ acts on the *input*: $(x_x,x_y)\mapsto(\cos x_x+\sin x_y,\,-\sin x_x+\cos x_y)$. Multiplying out and keeping the secular terms (the τ-integrals of products such as $S_\tau\cos\omega_0\tau\cos\omega_0\tau$ contain a non-oscillating half; the cs cross terms and the $\cos^2$-type oscillating halves drop), one finds the Bloch equations

$$\dot r_z=-\Gamma_1\big(r_z-r_z^{\mathrm{eq}}\big),\qquad \dot r_\pm=\big(\pm i\tilde\omega_0-\Gamma_2\big)r_\pm,\qquad r_\pm\equiv r_x\pm i r_y, \tag{79}$$

with

$$\Gamma_1 = \Gamma_\downarrow + \Gamma_\uparrow = 2\sin^2\theta\, S(\omega_0), \qquad \Gamma_\downarrow = 2\sin^2\theta\, J(\omega_0)\big[1 + n(\omega_0)\big], \quad \Gamma_\uparrow = 2\sin^2\theta\, J(\omega_0)\, n(\omega_0), \tag{80}$$

$$r_z^{\mathrm{eq}} = -\frac{\chi''(\omega_0)}{S(\omega_0)} = -\tanh\frac{\beta\omega_0}{2}, \tag{81}$$

$$\Gamma_2 = \tfrac{1}{2}\Gamma_1 + \Gamma_\varphi, \qquad \Gamma_\varphi = 2\cos^2\theta\, S(\omega{=}0), \tag{82}$$

$$\tilde{\omega}_0 = \omega_0 + \delta\omega, \qquad \delta\omega = 2\sin^2\theta \int_0^\infty \mathrm{d}\tau\, S(\tau)\sin\omega_0\tau = \frac{\sin^2\theta}{\pi}\,\mathrm{P}\!\int_{-\infty}^{\infty}\mathrm{d}\omega\,\frac{S(\omega)}{\omega_0-\omega}. \tag{83}$$

Each line is worth a comment.

- Equation (80): the population column of $K^{(2)}$ gives $\int_0^\infty K_{zz}\mathrm{d}\tau = -4\sin^2\theta\int_0^\infty S(\tau)\cos\omega_0\tau\,\mathrm{d}\tau = -2\sin^2\theta S(\omega_0)$. The *sum* of the rates is set by the noise at ω_0. Using the FDT, $S(\omega_0) = J(\omega_0)[1 + 2n(\omega_0)]$, which splits into $\Gamma_\downarrow + \Gamma_\uparrow$ as written. This is Fermi's golden rule with the $C(\pm\omega_0)$ of Eq. (27).
- Equation (81): the inhomogeneous term is $\int_0^\infty K_{z\mathbb{1}}\mathrm{d}\tau = -2\sin^2\theta\int_0^\infty\chi(\tau)\sin\omega_0\tau\,\mathrm{d}\tau = -2\sin^2\theta\,\chi''(\omega_0)$. The *difference* of the rates, $\Gamma_\downarrow - \Gamma_\uparrow = 2\sin^2\theta\chi''(\omega_0)$, is set by the response. The ratio $\chi''/S = \tanh(\beta\omega_0/2)$ is the FDT, and it makes the steady state the Gibbs state of H_S: detailed balance is the fluctuation–dissipation theorem seen from the system. Out of equilibrium (a driven or nonthermal bath), S and χ'' are independent and r_z^{eq} defines an effective temperature.
- Equation (82): pure dephasing samples the noise at zero frequency and comes with $\cos^2\theta$; relaxation contributes half its rate to the coherence decay. For $S = \sigma_z/2$ one would have $\Gamma_\varphi = \frac{1}{2}\cos^2\theta S(0)$.
- Equation (83): the Lamb shift is the Hilbert transform of the *noise* spectrum. At $T = 0$, $S(\omega) = J(|\omega|)$ and the integral splits into the rotating-wave term $\mathrm{P}\int_0^\infty J(\omega)/(\omega_0 - \omega)$ and the counter-rotating (Bloch–Siegert) term $\int_0^\infty J(\omega)/(\omega_0 + \omega)$. The response χ' drops out of the shift because it is even in ω (Exercise 5.2).

For the Lorentzian bath of Section 2 at $T = 0$, with $\Omega = \omega_0 = 1$, $\gamma = 0.5$, $\theta = \pi/4$, $g = 0.3$: $\Gamma_\downarrow = 0.360$, $\Gamma_\uparrow = 0.0055$ (from the negative-frequency tail), $\Gamma_\varphi = 0.0424$, $\Gamma_2 = 0.225$, $\delta\omega = g^2\sin^2\theta\,\frac{2\Omega}{4\Omega^2+\gamma^2/4} = 0.022$. The script of Appendix A computes $\mathcal{G}_\infty^{(2)}$ numerically and reproduces all of these to three digits (the nonsecular generator gives a precession frequency 1.0219 versus 1.0222 secular).

Figure 5 compares the three second-order schemes with the exact solution. Three lessons: (i) at weak coupling all agree, and the Markovian rates of Example 5.3 are quantitatively right after a transient of duration $\sim 1/\gamma$; (ii) the Born kernel used *nonlocally* (NZ2) is the *worst* of the three at stronger coupling, overshooting to unphysical states, a well-known pathology [40]; (iii) the exact steady state is not the Gibbs state of H_S but the reduced state of the global equilibrium, which the second-order schemes can only approximate.

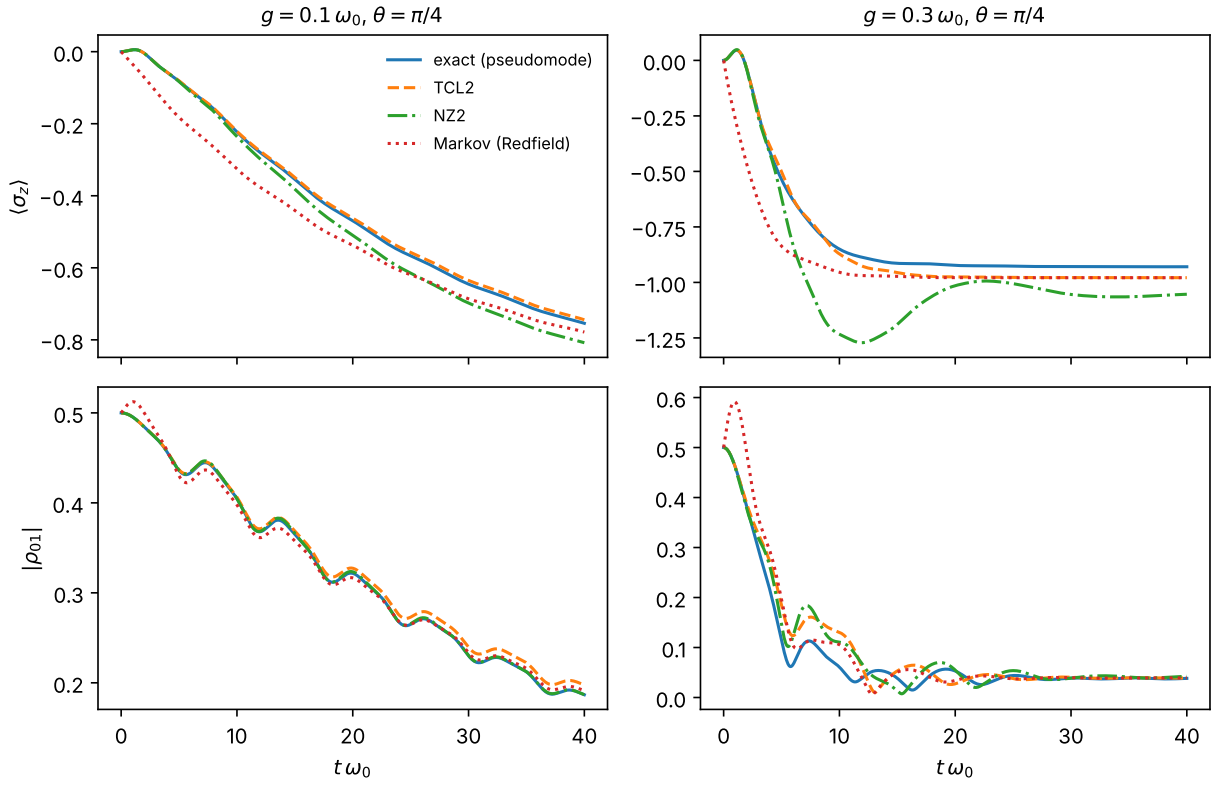


Figure 5: Qubit dynamics in the Lorentzian bath ($\Omega = \omega_0$, $\gamma = 0.5\omega_0$, $T = 0$, $\theta = \pi/4$), starting from $|+x\rangle$, for weak (left) and moderate (right) coupling. Exact (pseudomode) versus the three second-order schemes of this lecture and the next: NZ2 (Born, time-nonlocal), TCL2 (time-local, time-dependent rates) and the Markovian Redfield limit. At $g = 0.3$ the Born kernel drives $\langle\sigma_z\rangle$ below -1, a violation of positivity; TCL2 stays physical and tracks the exact solution; Markov misses the initial slip and the coupling-induced shift of the steady state.

5.4 Beyond second order

Example 5.1 gives the fourth-order kernel as the crossed and nested diagrams of Fig. 3, each with its full vertex structure (all assignments of c/q consistent with the rules, i.e. qq lines carrying S and qc lines carrying $-i\chi$). In the Schrödinger picture, with $V_t \in \{-iS^{\mathrm{q}}, -iS^{\mathrm{c}}\}$ the vertex at time t and free propagators $\mathrm{e}^{\mathcal{L}_S\Delta t}$ between vertices,

$$K^{(4)}(\tau) = \int_{\tau>t_1>t_2>0} \Big\{ V_\tau\, \mathrm{e}^{\mathcal{L}_S(\tau-t_1)} V_{t_1} \mathrm{e}^{\mathcal{L}_S(t_1-t_2)} V_{t_2} \mathrm{e}^{\mathcal{L}_S t_2} V_0 \Big\}_{\text{crossed } (\tau t_2)(t_1 0) \;+\; \text{nested } (\tau 0)(t_1 t_2)} . \tag{84}$$

For $\theta = 0$ on the coherence it reduces to the crossed and nested integrals of Eq. (58) times 16. Explicit fourth-order kernels for the spin-boson model were worked out in the TCL form by Breuer, Kappler and Petruccione [41]; in the NZ form the bookkeeping is easiest in Laplace space [30]. Figure 6 (left) shows the exact kernel of the Lorentzian model, extracted numerically from the pseudomode representation as $\mathcal{P}\mathcal{L}_I \mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}\mathcal{L}_I\mathcal{P}$ (which is Eq. (64) evaluated literally, with the pseudomode's Markovian Liouvillian playing the role of $\mathcal{L}$), against $K^{(2)}$: the higher orders shift the oscillation frequency inside the kernel (a self-consistent Lamb shift) and change its damping, but $K(0) = K^{(2)}(0)$ exactly, as Theorem 5.4 requires.

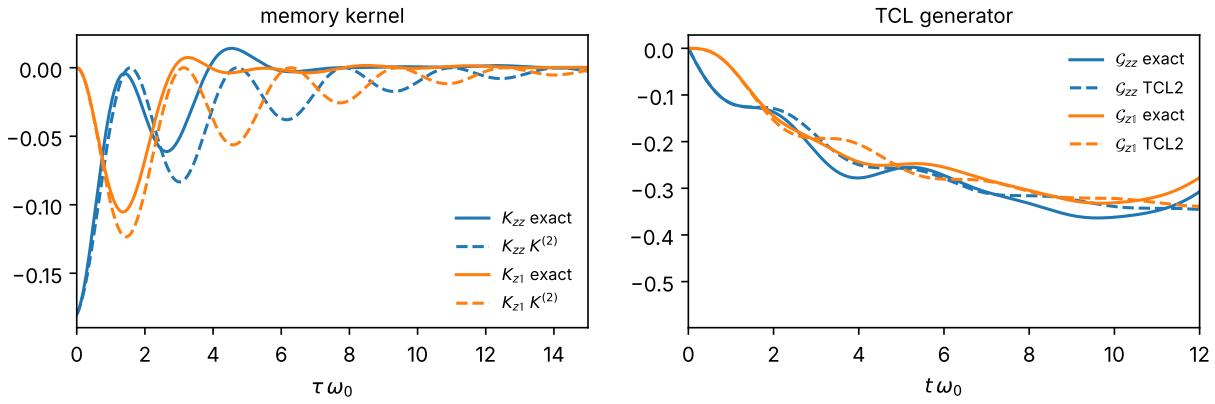


Figure 6: Left: the exact NZ memory kernel of the Lorentzian model at $g = 0.3$, $\theta = \pi/4$ (elements K_{zz} and $K_{z\mathbb{1}}$ in the Bloch basis), compared with the Born kernel $K^{(2)}$. Right: the exact TCL generator $\mathcal{G}(t) = \dot{\Pi}\Pi^{-1} - \mathcal{L}_S$ compared with the TCL2 generator $\mathcal{G}^{(2)}(t)$ (Section 6). Both agree at short times by construction; the exact generator develops large excursions near $t\omega_0 \approx 12$ where $\Pi(t)$ becomes nearly singular.

5.5 Correlated initial states and the inhomogeneity

If $\rho(0)$ is not factorized, $I(t)$ in Eq. (64) is nonzero. It decays on the bath correlation time if the initial correlations are short-ranged, and it can be absorbed into a modified kernel in some cases. In the Keldysh language the same information lives on the imaginary-time (Matsubara) branch attached to the contour at $t = 0$ [42, Ch. 4]. One should not expect $\Pi(t)$ to be a completely positive map of $\rho_S(0)$ in that case [15].

5.6 Exercises

Exercise 5.1. Carry out the derivation of Eq. (64) with a general projector $\mathcal{P}$ (not necessarily of the form (60)) and identify which steps use $\mathcal{P}\mathcal{L}_I\mathcal{P} = 0$ and $[\mathcal{P}, \mathcal{L}_0] = 0$. What changes if $\langle B\rangle \neq 0$? (Hint: absorb $S\langle B\rangle$ into H_S.)

Exercise 5.2. Derive the Bloch equations (79)–(83) from Eq. (57). Show that the real part of the response, χ', cancels from $\delta\omega$ and explain why with a parity argument. Then obtain the same rates from the operator form $\mathcal{G}X = -i[H_S + H_{\mathrm{LS}}, X] + \sum_\omega \gamma(\omega)\big(A(\omega)XA(\omega)^\dagger - \tfrac{1}{2}\{A(\omega)^\dagger A(\omega), X\}\big)$ with $A(\pm\omega_0) = \sin\theta\,\sigma_\mp$, $A(0) = \cos\theta\,\sigma_z$, $\gamma(\omega) = C(\omega)$ [29, Sec. 3.3.1].

Exercise 5.3. Show that the secular generator of Exercise 5.2 is of GKSL form with $L_1 = \sigma_-$, $L_2 = \sigma_+$, $L_3 = \sigma_z$ and nonnegative rates. Then take the nonsecular Redfield generator for $\theta = \pi/2$ and find an initial state and a time at which $\mathrm{Tr}\,\rho_S^2 > 1$ for a Lorentzian bath with $\gamma \gtrsim \omega_0$.

Exercise 5.4. Prove $K(0) = K^{(2)}(0)$ directly from Eq. (64). Then show that in Laplace space $\hat{K}(z) = z - \mathcal{L}_S - \hat{\Pi}(z)^{-1}$, so that the exact kernel can be *extracted* from the exact map (this is the route of Shi and Geva [43] and of Cohen and Rabani [44]). Use it to show that for pure dephasing ($\theta = 0$) the coherence kernel $\hat{K}_{01}(z)$ is an infinite series in g^2 even though the exact solution (59) is "second order" in a different sense.

6 Lecture 6: Time-convolutionless equations and the NZ–TCL map

A time-nonlocal equation is exact but awkward: it needs the whole history of ρ_S, and its truncations behave badly (Fig. 5). The time-convolutionless (TCL) technique [5, 6, 45] trades the convolution for a time-dependent generator, $\dot\rho_S = \mathcal{G}(t)\rho_S(t)$. This lecture derives it, computes it for the qubit, shows how it is related to the NZ kernel, and discusses when it fails.

6.1 Derivation

Work in the interaction picture, $\dot{\tilde\rho} = \tilde{\mathcal{L}}_I(t)\tilde\rho$, and let $\tilde G(t,s)$ be the full propagator from s to t (so $\tilde\rho(t) = \tilde G(t,s)\tilde\rho(s)$) and $\tilde G(t,s)^{-1} = \tilde G(s,t)$ the backward propagator. The trick is to express the *past* in terms of the *present*: $\tilde\rho(s) = \tilde G(s,t)\tilde\rho(t)$. Start from the $\mathcal{Q}$-equation solved as in Section 5.1,

$$\mathcal{Q}\tilde\rho(t) = \hat G(t,0)\mathcal{Q}\tilde\rho(0) + \int_0^t \mathrm{d}s\, \hat G(t,s)\, \mathcal{Q}\tilde{\mathcal{L}}_I(s)\, \mathcal{P}\, \tilde\rho(s), \qquad \hat G(t,s) = T\exp\Big[\int_s^t \mathcal{Q}\tilde{\mathcal{L}}_I\Big], \tag{85}$$

and substitute $\tilde\rho(s) = \tilde G(s,t)[\mathcal{P}+\mathcal{Q}]\tilde\rho(t)$. Defining

$$\Sigma(t) \equiv \int_0^t \mathrm{d}s\, \hat G(t,s)\, \mathcal{Q}\tilde{\mathcal{L}}_I(s)\, \mathcal{P}\, \tilde G(s,t), \tag{86}$$

which is of order $\mathcal{L}_I$ and vanishes at $t=0$, we get $[1-\Sigma(t)]\mathcal{Q}\tilde\rho(t) = \hat G(t,0)\mathcal{Q}\tilde\rho(0) + \Sigma(t)\mathcal{P}\tilde\rho(t)$. Wherever $1-\Sigma(t)$ is invertible (certainly for short times or weak coupling),

$$\boxed{\partial_t \mathcal{P}\tilde\rho(t) = \tilde{\mathcal{G}}(t)\,\mathcal{P}\tilde\rho(t) + \tilde I(t), \qquad \tilde{\mathcal{G}}(t) = \mathcal{P}\tilde{\mathcal{L}}_I(t)\big[1-\Sigma(t)\big]^{-1}\mathcal{P},} \tag{87}$$

with $\tilde I(t) = \mathcal{P}\tilde{\mathcal{L}}_I(t)[1-\Sigma(t)]^{-1}\hat G(t,0)\mathcal{Q}\tilde\rho(0)$, zero for factorized initial states [29, Sec. 9.2]. The equation is exact, time-local, and its generator has a perturbative expansion obtained by expanding $[1-\Sigma]^{-1} = \sum_n \Sigma^n$:

$$\tilde{\mathcal{G}}^{(2)}(t) = \int_0^t \mathrm{d}t_1\, \mathcal{P}\tilde{\mathcal{L}}_I(t)\tilde{\mathcal{L}}_I(t_1)\mathcal{P}, \tag{88}$$

$$\begin{aligned}\tilde{\mathcal{G}}^{(4)}(t) = \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \int_0^{t_2} \mathrm{d}t_3 \Big[&\langle \tilde{\mathcal{L}}_I(t)\tilde{\mathcal{L}}_I(t_1)\tilde{\mathcal{L}}_I(t_2)\tilde{\mathcal{L}}_I(t_3)\rangle - \langle \tilde{\mathcal{L}}_I(t)\tilde{\mathcal{L}}_I(t_1)\rangle\langle \tilde{\mathcal{L}}_I(t_2)\tilde{\mathcal{L}}_I(t_3)\rangle \\ &- \langle \tilde{\mathcal{L}}_I(t)\tilde{\mathcal{L}}_I(t_2)\rangle\langle \tilde{\mathcal{L}}_I(t_1)\tilde{\mathcal{L}}_I(t_3)\rangle - \langle \tilde{\mathcal{L}}_I(t)\tilde{\mathcal{L}}_I(t_3)\rangle\langle \tilde{\mathcal{L}}_I(t_1)\tilde{\mathcal{L}}_I(t_2)\rangle\Big],\end{aligned} \tag{89}$$

where $\langle\cdots\rangle \equiv \mathcal{P}\cdots\mathcal{P}$ restricted to the system (the "ordered cumulants" of van Kampen; see [29, Sec. 9.2.4] and [41]). Compare with the fourth-order NZ kernel: there, only the *reducible* pairing $(t\,t_1)(t_2t_3)$ is subtracted (by the $\mathcal{Q}$'s), while here all three products of pair contractions are subtracted, including the crossed and nested ones. TCL4 is *not* NZ4 with the convolution removed; it is a different reorganization, which happens to contain NZ4 plus terms generated by the time-locality.

6.2 TCL2 for the qubit: time-dependent rates

In the Schrödinger picture, Eq. (88) reads

$$\dot\rho_S = \big[\mathcal{L}_S + \mathcal{G}^{(2)}(t)\big]\rho_S(t), \qquad \mathcal{G}^{(2)}(t) = \int_0^t \mathrm{d}\tau\, K^{(2)}(\tau)\, \mathrm{e}^{-\mathcal{L}_S\tau}, \tag{90}$$

i.e. Redfield's generator (78) with the upper limit at t instead of ∞. All the Markovian rates of Example 5.3 become running integrals:

Example 6.1: the running rates

Repeating the secular reduction of Example 5.3 with $\int_0^t$ instead of $\int_0^\infty$,

$$\Gamma_1(t) = 4\sin^2\theta \int_0^t \mathrm{d}\tau\, S(\tau)\cos\omega_0\tau, \qquad \Gamma_1(t)\, r_z^{\mathrm{eq}}(t) = -2\sin^2\theta \int_0^t \mathrm{d}\tau\, \chi(\tau)\sin\omega_0\tau,$$
$$\Gamma_\varphi(t) = 4\cos^2\theta \int_0^t \mathrm{d}\tau\, S(\tau), \qquad \delta\omega(t) = 2\sin^2\theta \int_0^t \mathrm{d}\tau\, S(\tau)\sin\omega_0\tau. \tag{91}$$

Three features are general. (i) All rates vanish at $t = 0$ as t^1 (since $K(0) \neq 0$): decay is initially quadratic in time, the Zeno regime. (ii) They approach their Markov values on the bath correlation time τ_B ($= 2/\gamma$ for the Lorentzian, $1/\omega_c$ or $1/\pi T$ for the Ohmic bath). (iii) In between they can *overshoot* and even become negative: for the Lorentzian bath, $\Gamma_1(t) \propto \int_0^t S(\tau)\cos\omega_0\tau\,\mathrm{d}\tau$ with $S(\tau)\cos\omega_0\tau = \frac{g^2}{2}\mathrm{e}^{-\gamma\tau/2}[1+\cos 2\omega_0\tau]$ at resonance is monotonic, but $\Gamma_\varphi(t) \propto \int_0^t S(\tau)\mathrm{d}\tau = g^2\int_0^t \mathrm{e}^{-\gamma\tau/2}\cos\Omega\tau\,\mathrm{d}\tau$ oscillates and is negative in windows when $\Omega > \gamma/2$. A negative rate in a time-local generator is the standard signature of information flowing back from the bath [46].

For pure dephasing ($\theta = 0$) TCL2 is *exact*: the generator on the coherence is $-i\omega_0 - \dot\Gamma(t)$ with Γ of Eq. (59), and $\dot\Gamma(t) = 4\int_0^t S(\tau)\mathrm{d}\tau = \Gamma_\varphi(t)$. All higher ordered cumulants (89) vanish because the vertices commute and the full fourth-order moment equals the sum of the three pair products. This is the cleanest demonstration that "order n" means different things in different schemes: the same model is second order in TCL and infinite order in NZ (Exercise 5.4).

6.3 The NZ–TCL map

Both equations describe the same map, so the kernel and the generator determine each other. Writing the NZ equation for $\Pi(t)$ and multiplying by $\Pi(t)^{-1}$,

$$\boxed{\mathcal{G}(t) = \mathcal{L}_S + \int_0^t \mathrm{d}s\, K(t-s)\,\Pi(s)\,\Pi(t)^{-1}, \qquad \mathcal{G}(t) = \dot\Pi(t)\Pi(t)^{-1},} \tag{92}$$

an exact relation wherever $\Pi(t)$ is invertible [47]. Conversely, given $\mathcal{G}$ one reconstructs $\Pi = T\exp\int\mathcal{G}$ and then K from the Laplace relation of Exercise 5.4. Nestmann, Bruch and Wegewijs [48] turned Eq. (92) into a fixed-point iteration, $\mathcal{G} \mapsto \mathcal{L}_S + \int K(t-s)\,T\mathrm{e}^{\int_t^s \mathcal{G}}\mathrm{d}s$, that converges to the exact generator and makes explicit how a time-nonlocal kernel "generates" a time-local evolution: the generator at time t is the kernel averaged over the past with the propagator as weight. Inserting $K \approx K^{(2)}$ and $\Pi \approx \Pi_0$ gives back $\mathcal{G}^{(2)}$; inserting $K^{(2)} + K^{(4)}$ with Π to the required order reproduces Eq. (89) and exhibits the extra subtractions as the $\Pi^{(2)}\Pi_0^{-1}$ corrections (Exercise 6.3). Reimer *et al.* [49] compare these and three further exact formulations with attention to complete positivity and divisibility.

6.4 When the generator does not exist

$\mathcal{G}(t)$ is defined only where $\Pi(t)$ is invertible. For the transverse model ($\theta = \pi/2$) in the rotating-wave approximation and a Lorentzian bath at $T = 0$, the single-excitation sector is exactly solvable:

the excited amplitude obeys $\ddot{c}_e + \frac{\gamma}{2}\dot{c}_e + g^2 c_e = 0$ at resonance, so

$$c_e(t) = \mathrm{e}^{-\gamma t/4}\Big[\cosh\tfrac{dt}{4} + \tfrac{\gamma}{d}\sinh\tfrac{dt}{4}\Big], \qquad d = \sqrt{\gamma^2 - 16g^2}, \tag{93}$$

and $\rho_{ee}(t) = |c_e(t)|^2\rho_{ee}(0)$, $\rho_{eg}(t) = c_e(t)\rho_{eg}(0)$. The exact generator is of GKSL form with the single time-dependent rate $\gamma(t) = -2\,\mathrm{Re}[\dot{c}_e/c_e]$ and shift $-2\,\mathrm{Im}[\dot{c}_e/c_e]$ [29, 50, Sec. 10.1]. For $16g^2 > \gamma^2$ the amplitude oscillates and crosses zero at finite times ($t \approx 7.3/\omega_0$ for our $g = 0.3$, $\gamma = 0.5$), where $\gamma(t) \to \infty$: the map is still perfectly well defined and completely positive, but it is not invertible, and no time-local generator exists there. The NZ kernel has no such problem. "Local versus nonlocal" is therefore a statement about the representation, not about the physics [47]. Without the rotating-wave approximation the zero of c_e is lifted (the counter-rotating and σ_z terms feed the coherence from other components), but the near-singularity survives as the large excursion of the exact generator in Fig. 6 (right) near $t\omega_0 \approx 12$, where the smallest singular value of $\Pi(t)$ drops to 10^{-2}.

6.5 What to take from Figs. 5 and 6

The numerical comparison illustrates a hierarchy that holds quite generally for Gaussian baths and weak-to-moderate coupling: Markov $<$ NZ2 $\lesssim$ TCL2 in accuracy at short times, and Markov $\approx$ TCL2 $>$ NZ2 at long times, with NZ2 prone to unphysical overshoots. The reason is structural: TCL2 is exact to second order in the coupling *for the generator*, and errors in the generator stay bounded; NZ2 is exact to second order for the kernel but the convolution with the exact history amplifies the missing fourth-order terms. Smirne and Vacchini [40] give a careful side-by-side for the two-level system; the lesson has been known since the early comparisons of the two schemes in chemical physics (see the references in [41]).

6.6 Exercises

Exercise 6.1. Show that $\Sigma(0) = 0$, that Σ is of first order in $\mathcal{L}_I$, and derive Eqs. (88) and (89) from Eq. (87). Explain why odd orders vanish.

Exercise 6.2. Evaluate the running rates (91) in closed form for the Lorentzian bath at $T = 0$ and find the time windows in which $\Gamma_\varphi(t) < 0$. Plot $\Gamma_\varphi(t)$, $\Gamma_1(t)$ and $r_z^{\mathrm{eq}}(t)$ for the parameters of Fig. 5 and compare with the script's output.

Exercise 6.3. Expand Eq. (92) to fourth order using $\Pi = \Pi_0 + \Pi^{(2)} + \dots$ and $K = K^{(2)} + K^{(4)} + \dots$, and show that the extra terms relative to $\int K^{(4)}\Pi_0\Pi_0^{-1}$ are exactly the crossed and nested subtractions in Eq. (89).

Exercise 6.4. For the rotating-wave model of Section 6.4, compute the exact NZ kernel $\hat{K}_{ee}(z)$ for the population and show that it is finite and analytic for $\mathrm{Re}\, z > -\gamma/4$ even when c_e has zeros. Where are the singularities of $\hat{\mathcal{G}}$, if one tries to define it by Laplace transform?

7 Lecture 7: Path integrals: the influence functional, the polaron frame and NIBA

So far we have organized the perturbation series by what the bath does to the *system superoperators*. The path integral does the opposite: it replaces the system operators by c-number paths, so that the bath can be integrated out *exactly* into an exponential, the Feynman–Vernon influence functional [51]. For a qubit the paths are discrete, which leads to the blip–sojourn expansion of Leggett *et al.* [9] and to the noninteracting-blip approximation (NIBA), the workhorse of the dissipative two-level system. We show that NIBA is nothing but the second-order Nakajima–Zwanzig kernel in the polaron frame.

7.1 The influence functional as Wick's theorem with a commutative system

Suppose the system superoperators at different times commuted. Then the sum over all pairings of $2n$ vertices would be $\frac{1}{n!}$ times the n-th power of the single contraction, and the whole Dyson series (42) would exponentiate. Operators do not commute, but in a path integral they are replaced by c-number paths: on the forward branch $S \to s^+(t)$, on the backward branch $S \to s^-(t)$, and in rotated variables $s^{\mathrm{c}} = \frac{1}{2}(s^+ + s^-)$, $s^{\mathrm{q}} = s^+ - s^-$. Repeating Section 4.1 with c-number vertices, the sum over pairings exponentiates to

$$\boxed{\Phi[s^{\mathrm{c}}, s^{\mathrm{q}}] = -\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \left[s^{\mathrm{q}}(t_1)\, S(t_1 - t_2)\, s^{\mathrm{q}}(t_2) - i\, s^{\mathrm{q}}(t_1)\, \chi(t_1 - t_2)\, s^{\mathrm{c}}(t_2) \right],} \tag{94}$$

and the reduced density matrix is

$$\rho_S(t) = \int \mathcal{D}s^+ \mathcal{D}s^-\, \mathrm{e}^{\,iA_S[s^+] - iA_S[s^-] + \Phi[s^{\mathrm{c}}, s^{\mathrm{q}}]}\ (\text{boundary terms}), \tag{95}$$

with A_S the system action. In the original $\pm$ variables and with $C = S - \frac{i}{2}\chi$, Eq. (94) is Feynman and Vernon's $\Phi = -\int_{t_1 > t_2} \left(s_1^+ - s_1^-\right)\left[C(t_1 - t_2) s_2^+ - C^*(t_1 - t_2) s_2^-\right]$, which the reader should check (Exercise 7.1). The rotated form Eq. (94) says: the *noise* kernel couples the quantum path to itself, the *response* kernel couples the quantum path to the earlier classical path, and there is no c–c term. Equivalently, $\Phi = -\frac{1}{2}\iint s^a D^{ab} s^b$ with D the Larkin–Ovchinnikov matrix (52): this is how the influence functional appears when a Gaussian bath is integrated out of the Keldysh action [25, Sec. 3.2], [52, Secs. 3–4], [20, Ch. 5]. For a single Gaussian bath, Eq. (94) contains *all* the information that any method uses; they differ only in how they handle the non-commutativity that the path integral pushes into the measure.

Example 7.1: reading the influence functional

(i) Pure dephasing. With $S = \sigma_z$ and the system restricted to the coherence $|0\rangle\langle 1|$, the path variables are frozen at $s^+ = +1$, $s^- = -1$, so $s^{\mathrm{c}} = 0$ and $s^{\mathrm{q}} = 2$. Then $\Phi = -4\int_{t_1 > t_2} S(t_1 - t_2) = -\Gamma(t)$ and $\rho_{01}(t) = \rho_{01}(0)\mathrm{e}^{-i\omega_0 t - \Gamma(t)}$, reproducing Eq. (59) in one line. The coherence is a path on which the two branches maximally disagree, and only the noise sees it.
(ii) Semiclassical limit and Langevin equation. Φ is at most quadratic in s^{q}. Decouple the quadratic term with an auxiliary real field ξ (Hubbard–Stratonovich), $\exp[-\frac{1}{2}\iint s^{\mathrm{q}} S s^{\mathrm{q}}] = \int \mathcal{D}\xi\, \exp[-\frac{1}{2}\iint \xi S^{-1}\xi + i\int \xi s^{\mathrm{q}}]$, and expand the system action to first order in s^{q} (exact for a harmonic system). The s^{q} integral then produces a delta functional enforcing an equation of motion for s^{c} with damping kernel χ and a Gaussian force ξ of correlator S: the quantum

Langevin equation [53, 54]. The q field has become the Martin–Siggia–Rose response field [55], the qq = 0 rule of Eq. (52) is the statement that the response field has no self-correlation (causality), and in the high-temperature limit $S(\omega) \to 2T\chi''(\omega)/\omega$ is the classical FDT [25, Ch. 4]. Section 8 returns to this from Mori's side.

7.2 Spin paths: blips and sojourns

For a qubit there is no continuous coordinate, but Eq. (94) still applies if we expand in the part of H_S that does not commute with S. Use the Leggett form (22): $S = \sigma_z'$ is diagonal, the bias ε is diagonal, and the tunnelling $-\frac{\Delta}{2}\sigma_x'$ flips the spin. Expanding e^{-iHt} and e^{iHt} in Δ, each branch becomes a piecewise-constant path $s^{\pm}(t) = \pm 1$ with flips at the vertices (a factor $i\Delta/2$ each, and the bias phase $\mathrm{e}^{\mp i\varepsilon s^{\pm}\mathrm{d}t/2}$ along the way). The pair (s^+, s^-) then alternates between *sojourns*, $s^+ = s^-$ (so $s^{\mathrm{q}} = 0$, $s^{\mathrm{c}} = \pm 1$), and *blips*, $s^+ = -s^-$ (so $s^{\mathrm{q}} = \pm 2$, $s^{\mathrm{c}} = 0$) [9, Sec. IV], [20, Ch. 21]. In Keldysh language a sojourn is a segment where the two branches agree, a classical configuration, and a blip is where they differ, a quantum excursion. Inserting such paths in Eq. (94), two integrations by parts convert the kernels into the functions

$$Q_2(\tau) = \frac{4}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{J(\omega)}{\omega^2}\coth\tfrac{\beta\omega}{2}\,(1-\cos\omega\tau), \qquad Q_1(\tau) = \frac{4}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{J(\omega)}{\omega^2}\sin\omega\tau, \tag{96}$$

with $\ddot{Q}_2 = 4S$ and $\ddot{Q}_1 = -2\chi$ for $\tau > 0$ (the factor 4 is again "σ_z versus $\sigma_z/2$"; with Leggett's coupling the prefactor is $1/\pi$). The influence functional becomes a sum of pairwise interactions: blip–blip through Q_2 (noise), and blip–sojourn through Q_1 (response), with the sojourn *preceding* the blip, as causality requires.

The noninteracting-blip approximation

Keep, for each blip of duration τ, only its self-interaction $\mathrm{e}^{-Q_2(\tau)}$ and its interaction with the immediately preceding sojourn, $\mathrm{e}^{\mp iQ_1(\tau)}$; drop all interactions between different blips and between a blip and earlier sojourns. The resulting series for the population $P(t) \equiv \langle\sigma_z'\rangle_t$ sums to the integro-differential equation

$$\dot{P}(t) = -\int_0^t \mathrm{d}\tau\Big[K^s(\tau)\,P(t-\tau) + K^a(\tau)\Big], \qquad \begin{aligned} K^s(\tau) &= \Delta^2\,\mathrm{e}^{-Q_2(\tau)}\cos Q_1(\tau)\cos\varepsilon\tau, \\ K^a(\tau) &= \Delta^2\,\mathrm{e}^{-Q_2(\tau)}\sin Q_1(\tau)\sin\varepsilon\tau. \end{aligned} \tag{97}$$

This is the NIBA of Leggett *et al.* [9, Sec. V]; see [20, Ch. 21] for its many extensions. It is exact to leading order in Δ; it is also exact for the Ohmic bath at $\alpha = \frac{1}{2}$ (Toulouse) and very accurate at high temperature or strong damping, where blips are short and dilute. It fails for a biased system at low temperature and weak damping, where the long-time Boltzmann weights come out wrong (Exercise 7.4); the reason will be transparent below.

7.3 NIBA is the second-order NZ kernel in the polaron frame

The blip language suggests a unitary transformation: a blip is a segment in which the bath is displaced in opposite directions on the two branches. Displace it once and for all with the polaron transformation [56]

$$U = \exp\Big[-\sigma_z'\sum_k \frac{g_k}{\omega_k}\big(b_k^\dagger - b_k\big)\Big], \qquad U^\dagger b_k U = b_k - \sigma_z'\frac{g_k}{\omega_k}, \tag{98}$$

which gives $U^\dagger(H_B+\sigma_z' B)U = H_B - \sum_k g_k^2/\omega_k$: the diagonal coupling is removed at the price of dressing the tunnelling,

$$H' = U^\dagger H U = \frac{\varepsilon}{2}\sigma_z' + H_B - \frac{\Delta}{2}\big(\sigma_+' B_+ + \sigma_-' B_-\big) + \text{const}, \qquad B_\pm = \exp\Big[\pm 2\sum_k \frac{g_k}{\omega_k}\big(b_k^\dagger - b_k\big)\Big]. \tag{99}$$

Now treat $V = -\frac{\Delta}{2}(\sigma_+' B_+ + \sigma_-' B_-)$ as the perturbation and use the NZ equation with Zwanzig's *diagonal* projector $\mathcal{P}_d X = \sum_{i=\uparrow\downarrow} |i\rangle\langle i| \otimes \rho_B \,\mathrm{Tr}_B\langle i|X|i\rangle$, which keeps only the populations. Since V is off-diagonal, $\mathcal{P}_d V \mathcal{P}_d = 0$ exactly and the second-order kernel needs no $\mathcal{Q}$:

$$\dot P_\uparrow(t) = \int_0^t \mathrm{d}\tau \Big[w_{\uparrow\leftarrow\downarrow}(\tau) P_\downarrow(t-\tau) - w_{\downarrow\leftarrow\uparrow}(\tau) P_\uparrow(t-\tau)\Big], \qquad w_{\uparrow\leftarrow\downarrow}(\tau) = \frac{\Delta^2}{2}\,\mathrm{Re}\Big[\mathrm{e}^{i\varepsilon\tau}\langle B_-(0)B_+(\tau)\rangle\Big], \tag{100}$$

and $w_{\downarrow\leftarrow\uparrow}(\tau) = \frac{\Delta^2}{2}\,\mathrm{Re}[\mathrm{e}^{-i\varepsilon\tau}\langle B_+(0)B_-(\tau)\rangle]$. The dressing operators are displacement operators, and the thermal expectation of a product of two displacements is a Gaussian (Exercise 7.3):

$$\langle B_-(0)B_+(\tau)\rangle = \langle B_+(0)B_-(\tau)\rangle = \mathrm{e}^{-Q_2(\tau)+iQ_1(\tau)}, \tag{101}$$

with exactly the $Q_{1,2}$ of Eq. (96). Hence $w_{\downarrow\leftarrow\uparrow} = \frac{\Delta^2}{2}\mathrm{e}^{-Q_2}\cos(\varepsilon\tau - Q_1)$, $w_{\uparrow\leftarrow\downarrow} = \frac{\Delta^2}{2}\mathrm{e}^{-Q_2}\cos(\varepsilon\tau + Q_1)$, and with $P = P_\uparrow - P_\downarrow$, $P_{\uparrow\downarrow} = \frac{1}{2}(1 \pm P)$, the two rate equations combine into Eq. (97) with $K^s = w_{\downarrow\leftarrow\uparrow} + w_{\uparrow\leftarrow\downarrow}$ and $K^a = w_{\downarrow\leftarrow\uparrow} - w_{\uparrow\leftarrow\downarrow}$. This identification is due to Aslangul, Pottier and Saint-James [57] and Dekker [58].

Key point 7.1

NIBA = Born approximation (second-order NZ) for the populations, in the polaron frame, with the full (unsubtracted) Franck–Condon correlator (101). Interblip interactions are the fourth- and higher-order irreducible diagrams of Theorem 5.4 in that frame. The two expansions of the course, in g (Section 5) and in Δ (this lecture), are second-order NZ kernels with respect to two different projectors and two different "free" dynamics.

Example 7.2: where NIBA goes wrong, and why

The steady state of Eq. (97) is $P_\infty = -\hat K^a(0)/\hat K^s(0)$. At weak coupling expand $\mathrm{e}^{-Q_2}\cos Q_1 \approx 1 - Q_2$, $\mathrm{e}^{-Q_2}\sin Q_1 \approx Q_1$; the τ-integrals pick out $J(\varepsilon)$ and give

$$\hat K^s(0) \simeq \frac{2\Delta^2}{\varepsilon^2} J(\varepsilon)\coth\frac{\beta\varepsilon}{2}, \qquad \hat K^a(0) \simeq \frac{2\Delta^2}{\varepsilon^2} J(\varepsilon), \qquad P_\infty \simeq -\tanh\frac{\beta\varepsilon}{2}. \tag{102}$$

The individual rates are $w_{\downarrow\leftarrow\uparrow} \simeq 2\Delta^2 J(\varepsilon)[1+n(\varepsilon)]/\varepsilon^2$ and $w_{\uparrow\leftarrow\downarrow} \simeq 2\Delta^2 J(\varepsilon) n(\varepsilon)/\varepsilon^2$: golden-rule rates with detailed balance, but with the bath sampled at the *bias* ε and weighted by $(\Delta/\varepsilon)^2$. Compare Eq. (80): the correct weak-coupling rates sample the bath at the *level splitting* $\omega_0 = \sqrt{\varepsilon^2+\Delta^2}$ with weight $\sin^2\theta = (\Delta/\omega_0)^2$, and the correct P_∞ is the Gibbs state of H_S, $-\cos\theta\tanh(\beta\omega_0/2)$ along σ_z'. The two agree only for $\Delta \ll \varepsilon$. The second-order-in-Δ kernel cannot know that the true eigenstates are superpositions of the pointer states; that information sits in the interblip diagrams. Conversely, the expansion in g cannot describe the strong-coupling regime where Q_2 is large and tunnelling is incoherent, which NIBA handles well. Each projector has its domain.

7.4 What the Keldysh field theory adds

Reading Eq. (97) from the Keldysh side: K^s is built from the real part of the dressed propagator e^{-Q} (the Keldysh component), K^a from the imaginary part (the retarded component). The full

blip–sojourn series is a Keldysh diagrammatics in which blips are quantum lines and sojourns are classical ones, and a renormalization-group treatment of it is the real-time RG of Keil and Schoeller [34]. We return to the field-theory formulation of the qubit in Section 9.

7.5 Exercises

Exercise 7.1. Show that the Feynman–Vernon form $\Phi = -\int_{t_1>t_2}(s_1^+ - s_1^-)[C_{12}s_2^+ - C_{12}^* s_2^-]$ is identical to Eq. (94), and derive Eq. (94) from the Wick sum by induction on the number of lines.

Exercise 7.2. For piecewise-constant paths, perform the two integrations by parts that convert Eq. (94) into a sum of blip–blip and blip–sojourn terms, and verify $\ddot{Q}_2 = 4S$, $\ddot{Q}_1 = -2\chi$. Write the full influence functional for one blip (duration τ) preceded by one sojourn and recover the factor $\mathrm{e}^{-Q_2(\tau)\mp iQ_1(\tau)}$.

Exercise 7.3. Derive Eq. (101). Use $D(\beta)D(\gamma) = \mathrm{e}^{(\beta\gamma^* - \beta^*\gamma)/2}D(\beta+\gamma)$ for displacement operators and $\langle D(\mu)\rangle_\beta = \mathrm{e}^{-|\mu|^2(2\bar{n}+1)/2}$ for a thermal mode. Check the sign of Q_1 by verifying the detailed-balance relation of Example 7.2.

Exercise 7.4. Solve Eq. (97) in Laplace space, $\hat{P}(z) = [P(0) - \hat{K}^a(z)/z]/[z + \hat{K}^s(z)]$, and analyze (a) $\varepsilon = 0$, J Ohmic at $T = 0$, where $Q_2 = \alpha\ln(1+\omega_c^2\tau^2)$, $Q_1 = 2\alpha\arctan\omega_c\tau$ and $K^s(\tau) = \Delta^2(1+\omega_c^2\tau^2)^{-\alpha}\cos[2\alpha\arctan\omega_c\tau]$ in Leggett's normalization (replace $\alpha \to 4\alpha$ for ours): find the renormalized tunnelling frequency $\Delta_r \propto \Delta(\Delta/\omega_c)^{\alpha/(1-\alpha)}$ and show that the dynamics is incoherent for $\alpha \geq \frac{1}{2}$; (b) the weak-coupling biased case of Example 7.2.

8 Lecture 8: Mori's projection, the generalized Langevin equation and the retarded self-energy

Nakajima–Zwanzig projects *states*. Mori [7] projects *operators*: pick a few "slow" Heisenberg operators, and derive an exact equation of motion for them in which everything else appears as a memory term and a random force. The output is a generalized Langevin equation (GLE), and its Laplace transform is again a Dyson equation, now for correlation functions. This lecture derives the GLE, works it out for the qubit's lowering operator, and establishes two precise relations: Mori on the *complete* operator basis is the Heisenberg-picture adjoint of NZ, and Mori's memory function for a single operator is the retarded self-energy of that operator's Green's function.

8.1 The generalized Langevin equation

Work in the Heisenberg picture: $\dot{A}(t) = i[H, A(t)] \equiv i\mathsf{L}A(t)$, so $A(t) = \mathrm{e}^{i\mathsf{L}t}A$. Choose an inner product (X, Y) on operators, a set of operators $\{A_j\}$ (one, for simplicity; the matrix generalization is immediate) and the projector

$$\mathsf{P}X = A\,(A,A)^{-1}(A,X), \qquad \mathsf{Q} = 1 - \mathsf{P}. \tag{103}$$

The derivation parallels Section 5.1 with the Dyson-type identity $\mathrm{e}^{i\mathsf{L}t} = \mathrm{e}^{i\mathsf{Q}\mathsf{L}t} + \int_0^t \mathrm{d}s\, \mathrm{e}^{i\mathsf{L}(t-s)}\, i\mathsf{P}\mathsf{L}\, \mathrm{e}^{i\mathsf{Q}\mathsf{L}s}$ applied to $i\mathsf{L}A$ [8, Ch. 8], [59, Secs. 5.2–5.3]. The result is exact:

$$\boxed{\dot{A}(t) = i\Omega\, A(t) - \int_0^t \mathrm{d}s\, M(t-s)\, A(s) + F(t), \qquad F(t) = \mathrm{e}^{i\mathsf{Q}\mathsf{L}t}\,\mathsf{Q}\, i\mathsf{L}A,} \tag{104}$$

with the *frequency* $i\Omega = (A, i\mathsf{L}A)(A,A)^{-1}$ and the *memory function*

$$M(t) = \big(F(0), F(t)\big)\,(A,A)^{-1}. \tag{105}$$

The random force lives in the orthogonal complement, $(A, F(t)) = 0$ for all t, which is what makes the correlation function $C(t) \equiv (A, A(t))$ obey the closed equation

$$\dot{C}(t) = i\Omega\, C(t) - \int_0^t \mathrm{d}s\, M(t-s)\, C(s) \quad \Longrightarrow \quad \hat{C}(z) = \frac{C(0)}{z - i\Omega + \hat{M}(z)}. \tag{106}$$

Equation (105), the statement that the memory kernel is the autocorrelation of the random force, is Kubo's *second fluctuation–dissipation theorem* [23]. Everything depends on the inner product. The two standard choices are the *Kubo–Mori* product $(X,Y) = \beta^{-1}\int_0^\beta \mathrm{d}\lambda \langle X^\dagger \mathrm{e}^{-\lambda H} Y \mathrm{e}^{\lambda H}\rangle$, for which $C(t)$ is the canonical (Kubo) correlation and L is self-adjoint, and the *symmetrized* product $(X,Y) = \frac{1}{2}\langle\{X^\dagger, Y\}\rangle$, which is simpler for fermions and spins; at the order we work, the two agree. In both, $\langle\cdot\rangle$ is the thermal average in the *correlated* global equilibrium state, not the factorized state of Section 5; this difference will show up.

8.2 Mori on the complete basis is the adjoint of Nakajima–Zwanzig

Proposition 8.1. *Let $\{E_\mu\}_{\mu=1}^{d^2}$ be an orthonormal basis of system operators, $\mathrm{Tr}_S(E_\mu^\dagger E_\nu) = \delta_{\mu\nu}$, and take the inner product $(X,Y)_B \equiv \mathrm{Tr}\left[X^\dagger Y\,(\mathbb{1}_S \otimes \rho_B)\right]$ on operators of $S \otimes B$. Mori's projector onto the set $\{E_\mu \otimes \mathbb{1}_B\}$ is the Hilbert–Schmidt adjoint of the NZ projector (60): $\mathsf{P} = \mathcal{P}^\dagger$, with $\mathcal{P}^\dagger Y = \mathrm{Tr}_B[(\mathbb{1}_S \otimes \rho_B)Y] \otimes \mathbb{1}_B$. Consequently the Mori memory matrix for this set is the adjoint of the NZ kernel, $M(t) = -K(t)^\dagger$ (for the Heisenberg generator $\mathsf{L}^\dagger$ in place of L), and the GLE (104) for the Heisenberg operators $E_\mu(t)$ is the adjoint of the NZ equation.*

Proof. $\mathsf{P}X = \sum_\mu E_\mu\,(E_\mu, X)_B = \sum_\mu E_\mu \operatorname{Tr}[(E_\mu^\dagger \otimes \rho_B)X] = \sum_\mu E_\mu \operatorname{Tr}_S\big[E_\mu^\dagger \operatorname{Tr}_B[(\mathbb{1}\otimes\rho_B)X]\big] = \operatorname{Tr}_B[(\mathbb{1}\otimes\rho_B)X]\otimes\mathbb{1}_B$ by completeness, and $(Y, \mathcal{P}X)_{\mathrm{HS}} = \operatorname{Tr}[Y^\dagger(\rho_B\otimes\operatorname{Tr}_B X)] = \operatorname{Tr}_S\big[\operatorname{Tr}_B[(\mathbb{1}\otimes\rho_B)Y]^\dagger \operatorname{Tr}_B X\big] = (\mathcal{P}^\dagger Y, X)_{\mathrm{HS}}$. The rest is the general fact that taking the adjoint of Eq. (64) (with $\mathcal{L}^\dagger = i\mathsf{L}$ and $\mathcal{P}^\dagger = \mathsf{P}$) gives Eq. (104) for the full basis, since the derivations are identical line by line. □

So the two projection techniques are not merely analogous: with the *same* reference state and the *complete* set of system operators, they are the same equation written in the two pictures. The novelty of Mori's scheme is in the two freedoms it adds: a *reduced* set of slow variables (one does not need all d^2 of them), and an inner product that weights with the *interacting* equilibrium state, so that the reference state already knows about the system–bath correlations. The first is what makes Mori useful for hydrodynamics and for Green's functions; the second is what makes its results differ from factorized-initial-state NZ even at second order, as we now see.

8.3 Example: Mori for the qubit's lowering operator

Example 8.1: $A = \sigma_-$ and the retarded self-energy

Take $A = \sigma_-$ with the symmetrized product in the state $\rho_S^{\mathrm{eq}}\otimes\rho_B$ (the lowest-order approximation to the correlated equilibrium state), $\rho_S^{\mathrm{eq}} \propto \mathrm{e}^{-\beta H_S}$. Then $(\sigma_-,\sigma_-) = \frac{1}{2}\langle\{\sigma_+,\sigma_-\}\rangle = \frac{1}{2}$. Using $[\sigma_z,\sigma_-] = -2\sigma_-$ and $[\sigma_x,\sigma_-] = -\sigma_z$,

$$i\mathsf{L}\sigma_- = i[H,\sigma_-] = -i\omega_0\sigma_- - i\big(2\cos\theta\,\sigma_- + \sin\theta\,\sigma_z\big)B. \tag{107}$$

The first term is along A, so $i\Omega = -i\omega_0$; the second is orthogonal to A (it has $\langle B\rangle = 0$) and is the random force, $F(0) = -iX_-B$ with $X_- \equiv 2\cos\theta\sigma_- + \sin\theta\sigma_z$. To second order, $F(t)$ evolves freely: $F(t) = -iX_-(t)B(t)$ with $X_-(t) = 2\cos\theta\sigma_-\mathrm{e}^{-i\omega_0 t} + \sin\theta\sigma_z$. The memory function (105) is $M(t) = 2(F(0),F(t)) = \langle\{X_+B, X_-(t)B(t)\}\rangle$ with $X_+ = X_-^\dagger$. The system and bath averages factorize, $\langle X_+BX_-(t)B(t)\rangle = \langle X_+X_-(t)\rangle\,C(-t)$ and $\langle X_-(t)B(t)X_+B\rangle = \langle X_-(t)X_+\rangle\,C(t)$; with $\langle\sigma_+\sigma_-\rangle = p_e$, $\langle\sigma_-\sigma_+\rangle = p_g$ (excited and ground populations) and $\langle\sigma_z^2\rangle = 1$,

$$M(t) = 2\sin^2\theta\,S(t) + 4\cos^2\theta\,\mathrm{e}^{-i\omega_0 t}\Big[S(t) - \tfrac{i}{2}\tanh\tfrac{\beta\omega_0}{2}\,\chi(t)\Big] \qquad (t>0), \tag{108}$$

where we used $p_gC(t) + p_eC(-t) = S - \frac{i}{2}(p_g-p_e)\chi$ and $p_g - p_e = \tanh(\beta\omega_0/2)$.
Now the Green's function. With $G^R(t) = -i\theta(t)\langle\{\sigma_-(t),\sigma_+\}\rangle = -2i\theta(t)C(t)$, the Laplace transform of Eq. (106) at $z = -i\omega + 0^+$ gives

$$G^R(\omega) = \frac{1}{\omega - \omega_0 - \Sigma^R(\omega)}, \qquad \boxed{\Sigma^R(\omega) = -i\hat{M}(-i\omega + 0^+) = -i\int_0^\infty \mathrm{d}t\,\mathrm{e}^{i\omega t}M(t).} \tag{109}$$

Mori's memory function *is* the retarded self-energy of the spin-flip propagator. Evaluating at $\omega = \omega_0$ with Eq. (108):

$$-\operatorname{Im}\Sigma^R(\omega_0) = \operatorname{Re}\int_0^\infty \mathrm{e}^{i\omega_0 t}M(t)\,\mathrm{d}t = \sin^2\theta\,S(\omega_0) + 2\cos^2\theta\,S(0) = \tfrac{1}{2}\Gamma_1 + \Gamma_\varphi = \Gamma_2, \tag{110}$$

$$\operatorname{Re}\Sigma^R(\omega_0) = \operatorname{Im}\int_0^\infty \mathrm{e}^{i\omega_0 t}M(t)\,\mathrm{d}t = \underbrace{2\sin^2\theta\int_0^\infty S(t)\sin\omega_0 t\,\mathrm{d}t}_{\delta\omega \text{ of Eq. (83)}} - 2\cos^2\theta\tanh\tfrac{\beta\omega_0}{2}\,\chi(\omega{=}0). \tag{111}$$

The decay rate is exactly the Bloch–Redfield Γ_2 of Eq. (82): the transverse part of the memory sampled at ω_0 gives $\Gamma_1/2$ (the $(\sigma_- \mathrm{e}^{-i\omega_0 t})$ term has been shifted to zero frequency by the $\mathrm{e}^{i\omega_0 t}$, which is why it yields $S(0)$), and the longitudinal part gives Γ_φ. The real part contains the Lamb shift $\delta\omega$ of Section 5 *plus* a term that Section 5 did not have: a static shift $-2\cos^2\theta\tanh(\beta\omega_0/2)\chi(0)$, with $\chi(0) = \frac{2}{\pi}\int_0^\infty J(\omega)/\omega\,\mathrm{d}\omega = 2\lambda$ twice the reorganization energy. It is a Franck–Condon (Stokes) shift: in the *equilibrium* state the bath is displaced in opposite directions for the two qubit levels, so a vertical transition from the thermally more populated level costs $\omega_0 \mp 4\lambda$, and the thermal average of the two line positions is $\omega_0 - 4\lambda\tanh(\beta\omega_0/2)$. Mori sees it because its reference state is correlated; NZ from a factorized initial state sees it only dynamically, through the $\cos\theta\sin\theta\,\chi$ cross terms in the first column of Eq. (57) that the secular approximation discards. Same physics, different bookkeeping of the initial condition.

Key point 8.1

For a single projected operator A with (A, A) normalized, Mori's memory function is $-i$ times the retarded self-energy of the Green's function $G^R_{AA^\dagger}$, and the frequency Ω is the static (Hartree–Fock-like) part of that self-energy. Iterating the projection (project F onto itself, etc.) produces Mori's continued fraction, which is the recursion method for G^R [8, Ch. 8], [59, Sec. 5.3]; for the resonant-level model the construction reproduces the exact hybridization self-energy (see the companion working note).

8.4 What Mori does *not* give you

Equation (106) determines the retarded sector. In equilibrium the FDT then fixes the Keldysh component, $G^K = \coth(\beta\omega/2)(G^R - G^A)$ for bosonic operators and $(1-2f)(G^R - G^A)$ for fermionic ones, so nothing is lost. Out of equilibrium, occupations, currents and noise live in $G^<$ (equivalently G^K), which the retarded function alone does not determine. Three routes supply the missing information:

1. *Density-matrix projection* (NZ/TCL, Sections 5 and 6), which evolves the one-time state and therefore carries occupations by construction; by Proposition 8.1 this is Mori on the complete basis with a factorized reference state.
2. *Kadanoff–Baym equations* [42, 60] for the two-time $G^{\lessgtr}(t, t')$, closed by the generalized Kadanoff–Baym ansatz (GKBA) [61], $G^{\lessgtr}(t, t') \approx i[G^R(t, t')G^{\lessgtr}(t', t') - G^{\lessgtr}(t, t)G^A(t, t')]$, which plays the role of a projection onto one-time quantities.
3. *Time-dependent Mori projections* with nonstationary reference states [62, Ch. 2], which produce two-time memory kernels and are closer in spirit to Kadanoff–Baym.

8.5 GLE ↔ Keldysh saddle point

For a linear bath coupled to a continuous coordinate x (Caldeira–Leggett), Example 7.1(ii) produced from the influence functional the quantum Langevin equation $m\ddot{x}^{\mathrm{c}} + V'(x^{\mathrm{c}}) - \int^t \chi(t-s)x^{\mathrm{c}}(s)\mathrm{d}s = \xi(t)$, $\langle\xi\xi\rangle = S$. This is Mori's Eq. (104) for $A = x$ (with the momentum as a second variable), with $M \leftrightarrow \chi$ and $F \leftrightarrow \xi$; Kubo's second FDT (105) and the equilibrium relation $D^K = \coth(\beta\omega/2)(D^R - D^A)$ between the bath's Keldysh and retarded components are the same statement [25, Sec. 3.2], [63, Ch. 3]. For the qubit there is no such classical limit, but the structure is unchanged: the memory function (108) is a combination of S and χ with the system's own thermal weight $\tanh(\beta\omega_0/2)$ multiplying the response.

8.6 Exercises

Exercise 8.1. Derive Eq. (104) from the Dyson identity and prove $(A, F(t)) = 0$. Show that $M(t) = (F, F(t))(A, A)^{-1}$ requires L to be self-adjoint for the chosen product, and check this for the Kubo–Mori and symmetrized products in a thermal state.

Exercise 8.2. Complete the derivation of Eq. (108) (the four anticommutator terms) and verify the rate and shift formulas. Then repeat with $A = \sigma_z - \langle \sigma_z \rangle$ and show that the memory function reproduces Γ_1 (use the FDT) and that the inhomogeneous term is absent (why?). What goes wrong if one projects on σ_z itself?

Exercise 8.3. Prove Proposition 8.1 in detail for a qubit using the basis $E_\mu = \sigma_\mu/\sqrt{2}$, and show that the Mori memory matrix at second order is $-K^{(2)}(t)^T$ in the coefficient convention (the transpose, because the matrices are real).

Exercise 8.4. Mori's continued fraction: project the random force of Example 8.1 onto itself and write the second-level memory function. For the Lorentzian bath, show that the continued fraction terminates after two levels in the rotating-wave approximation and reproduces the exact $c_e(t)$ of Section 6.4.

9 Lecture 9: The qubit as a Keldysh field theory

Everything in Sections 3 and 4 was "Keldysh" in content but Liouville-space in form. This lecture translates the same statements into the language of the many-body literature (generating functionals, actions, Green's functions and self-energies), so that the reader can open Kamenev [25], Rammer and Smith [28] or Sieberer, Buchhold and Diehl [64] and recognize each object. The lecture is deliberately brief on technique and long on dictionary.

9.1 The generating functional and its two kinds of sources

The partition function on the closed contour is trivially normalized,

$$Z = \mathrm{Tr}\left[U_{\mathcal{C}}\,\rho(0)\right] = \mathrm{Tr}\left[U^{\dagger}(t)\,U(t)\,\rho(0)\right] = 1\,: \tag{112}$$

there is no vacuum-to-vacuum denominator to divide out, which is the reason the Keldysh technique has no disconnected diagrams and no Matsubara-to-real-time continuation. Information is extracted by adding sources to H on the two branches separately, $H_{\pm} \to H_{\pm} - h_{\pm}(t)S$, and in rotated variables $h^{\mathrm{c}} = \tfrac{1}{2}(h^{+} + h^{-})$, $h^{\mathrm{q}} = h^{+} - h^{-}$. Then

$$\begin{aligned} \frac{\delta Z}{\delta h^{\mathrm{q}}(t)}\Big|_{h=0} &= i\langle S(t)\rangle, \qquad \frac{\delta^2 Z}{\delta h^{\mathrm{q}}(t)\,\delta h^{\mathrm{c}}(t')}\Big|_{h=0} = i\,\chi_{SS}(t-t') \ \text{(retarded)}, \\ \frac{\delta^2 Z}{\delta h^{\mathrm{q}}(t)\,\delta h^{\mathrm{q}}(t')}\Big|_{h=0} &= -\tfrac{1}{2}\langle\{S(t), S(t')\}\rangle, \end{aligned} \tag{113}$$

while $\delta^2 Z/\delta h^{\mathrm{c}}\delta h^{\mathrm{c}} = 0$ identically ($Z = 1$ for any classical source, since $h^{+} = h^{-}$ is a physical Hamiltonian). In Liouville-space terms: a quantum source inserts S^{q} (it is $[S,\cdot]$, the generator of the physical perturbation), a classical source inserts S^{c} (it measures S), and the three statements are the trace identities (45) applied to the system. The retarded response of S to a force coupling to S is the object Mori computes (Section 8), the symmetrized correlator is the object a sensor measures (Section 10), and $Z = 1$ for classical sources is the normalization of probability.

9.2 The bath as a Gaussian field; integrating it out

For each mode, the Keldysh action is $A_B = \int \mathrm{d}t \sum_{\pm} \pm\big[\bar{b}_{\pm}(i\partial_t - \omega_k)b_{\pm}\big]$, and after the rotation its inverse is the Larkin–Ovchinnikov matrix. For the collective coordinate $B = \sum_k g_k(b_k + b_k^{\dagger})$ that couples to the system, the Gaussian integral gives exactly the influence functional (94), i.e. $-\tfrac{1}{2}\iint s^a D^{ab} s^b$ with $D^{R}(\omega) = -i\chi(\omega)$, $D^{A} = (D^{R})^{*}$, $D^{K}(\omega) = -i\coth(\beta\omega/2)\,2\chi''(\omega)$ in Kamenev's normalization [25, Secs. 2.5–2.7 and 3.2]. The bath enters the system's problem only as this 2×2 propagator, exactly as in Section 3. If the bath is itself of interest (a cavity, a transmission line: the pseudomode of Section 2 is literally a cavity mode), its dressed propagator obeys a Dyson equation with a self-energy from the system, $\mathbf{D} = \mathbf{D}_0 + \mathbf{D}_0\mathbf{\Pi}\mathbf{D}$, whose retarded component is what input–output theory measures as reflection or transmission and whose Keldysh component is the emitted noise [24].

9.3 Representing the spin

A qubit has no canonical field, so one must choose a representation. Four are in use.

(a) Liouville space. What we did: treat ρ as a vector and $H^{\pm}$ as superoperators. This is the natural "Keldysh field theory" of any finite-dimensional system and is the formalism of the real-time diagrammatics of Schoeller and Schön [32], König *et al.* [65] and the real-time RG [26, 34]. Its drawback is that the system "propagator" $\mathrm{e}^{\mathcal{L}_S\tau}$ is a $d^2 \times d^2$ matrix rather than a single-particle Green's function; its advantage is that it is exact and needs no constraint.

(b) Spin-coherent-state path integral. Paths on the Bloch sphere with a Berry-phase term, on both branches [66, Ch. 4]. Elegant for large spins and semiclassics; for a single qubit the measure is awkward and the blip–sojourn expansion of Section 7 is the practical form.

(c) Fermionic representations. Abrikosov pseudofermions [67], $\sigma_i = \sum_{ab} f_a^\dagger (\sigma_i)_{ab} f_b$ with the constraint $\sum_a f_a^\dagger f_a = 1$, or Schwinger bosons [68]. Standard Wick's theorem applies, at the cost of projecting onto the physical subspace (a chemical potential sent to $-\infty$, or Popov–Fedotov phases). These are the representations in which the Kondo problem is usually treated.

(d) Majorana representation. Three Majorana fermions η_x, η_y, η_z with $\{\eta_i, \eta_j\} = 2\delta_{ij}$ and $\sigma_i = -\frac{i}{2}\epsilon_{ijk}\eta_j\eta_k$ reproduce the Pauli algebra with *no constraint* [69]. The object $\Theta = -i\eta_x\eta_y\eta_z$ commutes with every σ_i and with H, satisfies $\Theta^2 = 1$ and $\sigma_i = \Theta\eta_i$, so spin–spin correlators reduce to Majorana two-point functions, $\langle T\sigma_i(t)\sigma_i(0)\rangle = \langle T\eta_i(t)\eta_i(0)\rangle$. For the dissipative two-level system this gives a genuine single-particle Keldysh diagrammatics: the Majorana propagator $\mathbf{G}_\eta$ obeys a Dyson equation, its retarded self-energy at second order is built from one bath line D and one Majorana line, and the resulting $\Sigma^R_{\eta_-}(\omega)$ for the combination $\eta_x - i\eta_y$ is exactly the Mori self-energy of Example 8.1 (the spin-flip operator $\sigma_- \propto \Theta(\eta_x - i\eta_y)$).

Key point 9.1

Whichever representation is chosen, the second-order retarded self-energy of the spin-flip propagator is the Mori memory function (108), and the second-order Liouville-space kernel is $K^{(2)}$ of Eq. (56). They are different projections of the same two diagrams (one noise line, one response line). The representation decides how convenient the *higher* orders and the *resummations* are.

9.4 Markovian limit: the Keldysh action of a Lindblad equation

When the kernel is time-local and the generator is of GKSL form (19), the Liouville-space evolution $\mathrm{e}^{\mathcal{G}t}$ can be written as a Keldysh path integral with the action [64]

$$A_K = \int \mathrm{d}t \left[\bar\psi_+ i\partial_t \psi_+ - \bar\psi_- i\partial_t \psi_- - H_+ + H_- - i\sum_j \gamma_j \Big(L_{j+}\bar L_{j-} - \tfrac{1}{2}\big(\bar L_{j+}L_{j+} + \bar L_{j-}L_{j-}\big)\Big)\right], \quad (114)$$

where $\psi_\pm$ are the (bosonic or fermionic) fields representing the system on the two branches and $L_{j\pm}$ the jump operators evaluated on them. The term with mixed branches, $L_+\bar L_-$, is the "jump" $L\rho L^\dagger$; the branch-diagonal terms are the anticommutator. In rotated variables the quadratic part has a *frequency-independent* Σ^K and a *time-local* Σ^R: a Lindblad equation is a Keldysh theory with white noise and instantaneous damping. The three Lindblad terms of the secular Bloch equations (Exercise 5.3) translate into $\Sigma^R = -i\Gamma_2$ on the spin-flip propagator and a constant Σ^K fixed by $\Gamma_\downarrow$ and $\Gamma_\uparrow$, with the ratio $\Sigma^K/(\Sigma^R - \Sigma^A)$ equal to $\coth(\beta\omega_0/2)$ only if the rates obey detailed balance: a Lindblad equation whose rates do not obey the FDT describes a nonthermal (effective-temperature) steady state, and the Keldysh language makes this visible at a glance [64, Sec. 4].

9.5 Self-consistent resummations

Beyond strict perturbation theory, the field theory offers resummations that are awkward in Liouville space:

- **Kadanoff–Baym.** Replace the free system propagator inside the self-energy by the dressed one, $\Sigma = \Sigma[\mathbf{G}]$, and solve the two-time equations self-consistently [42, 60]. In Liouville-space terms this dresses the propagators *between* vertices inside K with the full Π, which NZ never does. The GKBA reduces the two-time problem to a one-time one; it is to Kadanoff–Baym what TCL is to NZ.
- **Real-time RG.** Integrate out bath modes shell by shell in the Liouville-space diagrammatics, generating a flowing effective Liouvillian [26, 34]. For the spin-boson model this reproduces the renormalized tunnelling $\Delta_r \sim \Delta(\Delta/\omega_c)^{\alpha/(1-\alpha)}$ of Exercise 7.4.
- **Large-N, DMFT.** For many coupled open systems the Keldysh action is the starting point of mean-field and dynamical mean-field schemes, e.g. open-system DMFT [70], where the local problem is precisely an impurity in a self-consistent bath, i.e. the problem of these notes with a $\mathbf{D}$ that is determined self-consistently.

9.6 Dictionary

Keldysh / NEGF	**Nakajima–Zwanzig / TCL**	**Mori**
Contour branches $\pm$; c/q rotation	Left/right superoperators; $A^{\mathrm{c}} = \frac{1}{2}\{A, \cdot\}$, $A^{\mathrm{q}} = [A, \cdot]$	—
Quantum source h^{q} / classical source h^{c}	S^{q} vertex (perturbation) / S^{c} vertex (measurement)	force term / observable
D^R (bath response)	χ on q–c lines	damping kernel in M
D^K (bath noise)	S on q–q lines	random-force statistics $(F, F(t))$
Dyson: $\mathbf{G} = \mathbf{G}_0 + \mathbf{G}_0\Sigma\mathbf{G}$	$\hat{\Pi} = \hat{\Pi}_0 + \hat{\Pi}_0\hat{K}\hat{\Pi}$	$\hat{C} = [z - i\Omega + \hat{M}]^{-1}C(0)$
Irreducible self-energy	NZ kernel K (irreducible diagrams)	memory function $M = i\Sigma^R$
Static self-energy (Hartree–Fock)	secular Lamb shift + reaction field	frequency Ω
Single-particle space	Liouville space of ρ_S (dimension d^2)	span of chosen operators
Retarded *and* Keldysh sectors	full one-time state	retarded sector only (out of eq.)
Kadanoff–Baym $\to$ GKBA	NZ $\to$ TCL (one-time closure)	time-dependent projections
Matsubara leg (initial correlations)	inhomogeneous term $I(t)$	correlated reference state in $(\cdot, \cdot)$
Saddle point + Hubbard–Stratonovich $\Rightarrow$ Langevin	—	GLE (104)
Local Σ^R, constant Σ^K	GKSL / Lindblad	Markovian $M(t) \propto \delta(t)$

9.7 Exercises

Exercise 9.1. Prove the three source derivatives at the top of the lecture from the Liouville-space Dyson series, and show that $\delta^n Z/\delta h^{\mathrm{c}} \cdots \delta h^{\mathrm{c}} = 0$ to all orders.

Exercise 9.2. Verify the Majorana representation: show that $\sigma_i = -\frac{i}{2}\epsilon_{ijk}\eta_j\eta_k$ obey the Pauli algebra, that $\Theta = -i\eta_x\eta_y\eta_z$ commutes with all σ_i, and that $\sigma_i = \Theta\eta_i$. Write H of Eq. (21) in Majoranas and identify the interaction vertex (one bath line, two Majorana lines).

Exercise 9.3. Starting from the secular GKSL generator of Exercise 5.3 for $\theta = \pi/2$, write the Keldysh action (114) in the Majorana representation and read off Σ^R and Σ^K of the spin-flip propagator. Verify the detailed-balance statement in the text.

10 Lecture 10: Quantum sensing: which protocol measures which kernel

A qubit sensor is a two-level system whose coupling to the quantity of interest is exactly Eq. (21): the "signal" is a classical or quantum field B, the "noise" is its fluctuating part, and the sensor's decoherence is the measurement. The formalism of the previous lectures tells us precisely what is measured. This lecture collects the results, with references to the sensing literature [71], and ends with the one place where non-Markovian memory changes the metrological scaling.

10.1 Relaxometry: the transverse sensor measures $C(\pm\omega_0)$

For $\theta = \pi/2$ (or the transverse component of a general coupling) the population dynamics is governed by the two golden-rule rates of Eq. (80),

$$\Gamma_\downarrow = C(\omega_0), \qquad \Gamma_\uparrow = C(-\omega_0), \qquad \Gamma_1 = \Gamma_\downarrow + \Gamma_\uparrow = 2S(\omega_0), \qquad \Gamma_\downarrow - \Gamma_\uparrow = 2\chi''(\omega_0), \tag{115}$$

(for $\sin\theta = 1$). Measuring the decay rate of the population *and* its steady-state value therefore gives *both* bath kernels at the qubit frequency: the noise $S(\omega_0)$ from Γ_1 and the response $\chi''(\omega_0)$ from the polarization $r_z^{\rm eq} = -\chi''(\omega_0)/S(\omega_0)$. In a thermal bath the ratio is $\tanh(\beta\omega_0/2)$ and the qubit is a thermometer; in a driven or nonthermal environment the ratio defines an effective temperature at ω_0, and the asymmetry $C(\omega_0) \neq C(-\omega_0)$ is the hallmark of *quantum* noise, as emphasized by Schoelkopf *et al.* [72] and reviewed in [24, Sec. II.B]. Sweeping ω_0 (by a magnetic field, a flux, a gate voltage) maps $J(\omega)$: this is T_1 relaxometry. In Keldysh language, Γ_1 comes from the qq (Keldysh) component of the bath and $\Gamma_\downarrow - \Gamma_\uparrow$ from the qc (retarded) component; the first column of Eq. (57) says this directly.

At finite times the rates are the running integrals (91). Written in frequency space,

$$\Gamma_\downarrow(t) = \sin^2\theta \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{\pi}\, C(\omega)\, \frac{\sin[(\omega-\omega_0)t]}{\omega-\omega_0}, \tag{116}$$

the bath spectrum is sampled through a filter of width $1/t$ around ω_0. Modulating the qubit (frequency sweeps, pulses) reshapes the filter: this is the Kofman–Kurizki universal formula for controlled decay [73, 74], and the Zeno and anti-Zeno effects are its short-time consequences.

10.2 Dephasing: the longitudinal sensor measures $S(\omega)$ through a filter

For $\theta = 0$ the coherence is given exactly by Eq. (59), $\rho_{01}(t) = \rho_{01}(0)\mathrm{e}^{-i\omega_0 t - \Gamma(t)}$. A pulse sequence of π-pulses at times $t_1 < t_2 < \dots$ multiplies the integrand by the sign function $y(s) = \pm 1$ that flips at each pulse, and the exponent becomes

$$\boxed{\Gamma(t) = \frac{4}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{S(\omega)}{\omega^2}\, F(\omega,t), \qquad F(\omega,t) = \frac{\omega^2}{2}\left|\int_0^t \mathrm{d}s\, y(s)\, \mathrm{e}^{i\omega s}\right|^2,} \tag{117}$$

with $F = 1-\cos\omega t$ for free induction (Ramsey [18]), $F = 8\sin^4(\omega t/4)$ for the Hahn echo [19], and comb-like filters for CPMG and Uhrig sequences [75]; the general formalism is due to Cywiński *et al.* [76] (for Leggett's coupling normalization replace $4/\pi \to 1/\pi$; normalizations of F differ between papers). Three statements follow from the structure of the formula.

- Only the *noise* kernel enters. A pure-dephasing sensor is blind to χ: every response line in the diagrams ends on an $S^{\rm c}$, which annihilates the coherence (Example 4.3). It therefore cannot distinguish a quantum bath from a classical Gaussian noise with the same $S(\omega)$, and it cannot measure the temperature of the bath except through the coth in S.

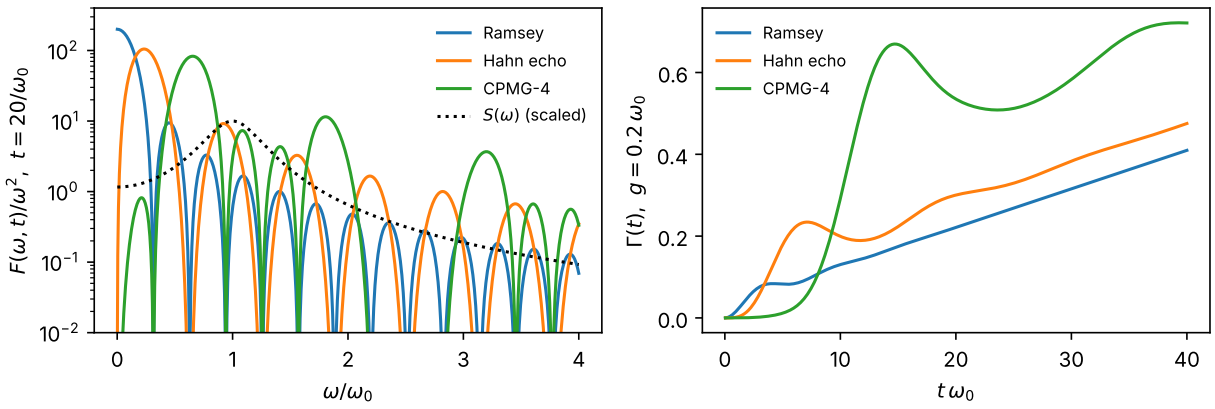


Figure 7: Left: filter functions $F(\omega,t)/\omega^2$ at $t = 20/\omega_0$ for free induction, Hahn echo and CPMG with four pulses, against the Lorentzian noise spectrum of the running example. Right: the resulting $\Gamma(t)$ (computed for the coupling $\frac{1}{2}\sigma_z \otimes B$, i.e. with $1/\pi$ in Eq. (117)) at $g = 0.2\omega_0$. The CPMG curve shows a bump near $t\omega_0 \approx 4\pi$, where the comb's first tooth $\pi/\tau = 4\pi/t$ sweeps through the bath resonance at $\Omega = \omega_0$: the sequence has detected the spectral line.

- The filter is set by the control, the spectrum by the bath. With n equally spaced pulses (CPMG, spacing $\tau = t/n$) the filter approaches a comb at $\omega_k = k\pi/\tau$, k odd, and

$$\Gamma(t) \simeq \frac{16\,t}{\pi^2} \sum_{k\ \text{odd}} \frac{S(k\pi/\tau)}{k^2} \qquad (n \gg 1), \tag{118}$$

  so that the decay rate measured as a function of τ is $S(\pi/\tau)$ up to harmonics. This is noise spectroscopy by dynamical decoupling [77, 78], realized e.g. in [79]; the general transfer-function viewpoint is in [80].

- The exponent is a quadratic functional of y: this is the statement that the influence functional (94) is quadratic in s^{q} for a Gaussian bath. For a non-Gaussian bath, higher cumulants add terms of higher order in y, and the decoupling sequences no longer act as linear filters.

10.3 The mixed sensor

For general θ both channels are open. In the secular (weak-coupling, long-time) regime the Ramsey signal decays at $\Gamma_2 = \frac{1}{2}\Gamma_1 + \Gamma_\varphi$ with $\Gamma_1 \propto \sin^2\theta\, S(\omega_0)$ and $\Gamma_\varphi \propto \cos^2\theta\, S(0)$, so a Ramsey measurement alone cannot separate low-frequency from resonant noise; a T_1 measurement is needed to subtract the $\Gamma_1/2$ part. Dynamical decoupling filters only the $\cos^2\theta$ part: π-pulses about $\hat{x}$ commute with σ_x, so the transverse coupling is not refocused, and the echo decay saturates at $\mathrm{e}^{-\Gamma_1 t/2}$. Outside the secular regime the cross terms $\cos\theta\sin\theta$ of Eq. (57) mix populations and coherences, and the pseudomode numerics of Fig. 5 is the honest way to design a protocol.

10.4 Precision scaling: when memory matters

Frequency estimation by Ramsey interferometry with N qubits and total time T: each run of duration t yields a signal $\frac{1}{2}[1 + \mathrm{e}^{-\Gamma(t)}\cos\omega_0 t]$, with Fisher information $t^2\mathrm{e}^{-2\Gamma(t)}$ per qubit per run

at the optimal phase [81]. With T/t runs the Cramér–Rao bound reads

$$\delta\omega_0^2 \;\geq\; \frac{1}{N\,T\,t\,\mathrm{e}^{-2\Gamma(t)}} \qquad \text{(product state)}, \qquad \delta\omega_0^2 \;\geq\; \frac{1}{N^2\,T\,t\,\mathrm{e}^{-2N\Gamma(t)}} \qquad \text{(GHZ state)}, \tag{119}$$

the GHZ state acquiring the phase $N\omega_0 t$ but, for independent baths, also N times the dephasing exponent. Optimizing over t:

	product state	GHZ state
Markov, $\Gamma = \gamma t$	$t^* = \frac{1}{2\gamma},\ \delta\omega_0^2 = \dfrac{2e\gamma}{NT}$	$t^* = \frac{1}{2N\gamma},\ \delta\omega_0^2 = \dfrac{2e\gamma}{NT}$ (no gain)
Zeno regime, $\Gamma = \kappa t^2$	$t^* = \frac{1}{2\sqrt{\kappa}},\ \delta\omega_0^2 = \dfrac{2\sqrt{e\kappa}}{NT}$	$t^* = \frac{1}{2\sqrt{N\kappa}},\ \delta\omega_0^2 = \dfrac{2\sqrt{e\kappa}}{N^{3/2}T}$

The first row is the classic result that Markovian dephasing erases the quantum advantage of entanglement [82]. The second row is the observation of Matsuzaki, Benjamin and Fitzsimons [83] and of Chin, Huelga and Plenio [84]: when the optimal interrogation time falls inside the initial quadratic (Zeno) regime of $\Gamma(t)$, which it does for a GHZ state at large N because $t^* \propto N^{-1/2}$, the precision scales as $N^{-3/4}$, between the standard quantum limit $N^{-1/2}$ and the Heisenberg limit N^{-1}. In the language of these notes: the short-time behaviour $\Gamma(t) \simeq 2g^2t^2 \cdot (\ldots)$ follows from $K(0) = K^{(2)}(0) \neq 0$ (Section 5.2): a memory kernel that is finite at $\tau = 0$ always produces a quadratic onset, and the crossover time to linear decay is the bath correlation time τ_B. The Markov row applies when $t^* \gg \tau_B$, the Zeno row when $t^* \ll \tau_B$. The numbers in Eq. (35) give $\kappa = 2g^2\cos^2\theta + g^2\sin^2\theta$ (dephasing plus half of the transverse channel) and $\tau_B = 2/\gamma$ for the Lorentzian bath.

10.5 Which formalism for which question

Question	**Natural tool**
Which bath quantity does protocol X measure?	Keldysh rotation: qq lines $\to S$, qc lines $\to \chi$; read off from the diagram (Section 4).
Exact dephasing under arbitrary control	Influence functional with c-number $y(s)$ (Section 7); filter functions.
Rates, steady state, thermometry	Second-order NZ/TCL, secular limit (Section 5); FDT fixes χ''/S.
Short-time (Zeno) behaviour, precision scaling	TCL2 running rates (Section 6); $K(0) \neq 0$.
Strong coupling, incoherent regime	Polaron frame, NIBA and beyond (Section 7).
Linear response of the sensor to a drive	Mori / retarded self-energy (Section 8).
Back-action on the bath (a cavity, a line)	Keldysh Dyson equation for the dressed bath propagator (Section 9).
Benchmarking any of the above	Pseudomode exact solution (Section 2 and Appendix A).

10.6 Exercises

Exercise 10.1. Derive Eq. (116) from the running rate $\Gamma_\downarrow(t) = 2\sin^2\theta\,\mathrm{Re}\int_0^t \mathrm{e}^{i\omega_0\tau}C(\tau)\mathrm{d}\tau$ and show that a periodic modulation of ω_0 shifts the filter. Explain the anti-Zeno effect for a bath whose $C(\omega)$ rises above ω_0.

Exercise 10.2. Derive the Hahn-echo filter $F = 8\sin^4(\omega t/4)$ and Eq. (118). Use the Lorentzian $S(\omega)$ to reproduce the bump in Fig. 7 analytically: at what t does it appear for CPMG with n pulses?

Exercise 10.3. Repeat the Fisher-information analysis for T_1-limited sensing (a transverse sensor) and for a Ramsey sensor with $\Gamma(t) = \Gamma_2 t + \kappa t^2$. Where is the crossover between the two rows of the table, in terms of τ_B and N?

Exercise 10.4. (Capstone.) For the Lorentzian bath with $\theta = \pi/4$, use the pseudomode script to simulate a Hahn echo (insert a π-pulse about $\hat{x}$ at $t/2$) and compare the echo amplitude with (a) the filter-function prediction using only the $\cos^2\theta$ part and (b) the same multiplied by $\mathrm{e}^{-\Gamma_1 t/2}$. Discuss the residual discrepancy in terms of the $\cos\theta\sin\theta$ terms of Eq. (57).

A The numerical companion

All figures are produced by `figures/spin_boson_numerics.py` (NumPy, SciPy, Matplotlib; about one minute on a laptop). This appendix records what it computes, so that each curve can be traced to an equation in the text.

Model and parameters. $\omega_0 = 1$ (units), $S = \cos\theta\sigma_z + \sin\theta\sigma_x$ with $\theta = \pi/4$, Lorentzian bath with $\Omega = 1$, $\gamma = 0.5$, $T = 0$, couplings $g = 0.1$ and 0.3; initial state $|+x\rangle$; time step $\mathrm{d}t = 0.02$, $t \le 40$. Kernels: Eq. (35).

Representation. Column-stacking vectorization, Eq. (5): $A^+ = \mathbb{1} \otimes A$, $A^- = A^T \otimes \mathbb{1}$, $A^{\mathrm{c}} = \frac{1}{2}(A^+ + A^-)$, $A^{\mathrm{q}} = A^+ - A^-$, $\mathcal{L} = -iH^{\mathrm{q}}$. Bloch-basis matrix elements are $M_{\mu\nu} = \frac{1}{2}\,\mathrm{Tr}(\sigma_\mu \mathcal{M} \sigma_\nu)$ (the coefficient convention of Section 1).

Exact reference. The pseudomode master equation (37) with $N = 14$ oscillator levels (checked against $N = 20$: maximal deviation 5×10^{-15} in $\Pi(t)$ for $t \le 4$ at $g = 0.3$). $\Pi(t)$ is obtained by propagating the four basis operators $|i\rangle\langle j| \otimes |0\rangle\langle 0|$ with $\mathrm{e}^{\mathcal{L}_{\mathrm{tot}}\mathrm{d}t}$ and tracing out the oscillator. The exact TCL generator is $\mathcal{G}(t) = \dot{\Pi}\Pi^{-1} - \mathcal{L}_S$ with $\dot{\Pi}$ computed from $\mathcal{L}_{\mathrm{tot}}$ (no numerical differentiation). The exact NZ kernel is computed directly from its definition (64), $K(\tau) = \mathcal{P}\mathcal{L}_I \mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}\mathcal{L}_I\mathcal{P}$, with $\mathcal{P} = \mathrm{Tr}_{\mathrm{osc}}(\cdot) \otimes |0\rangle\langle 0|$ built as a matrix on the $(2N)^2$-dimensional Liouville space of qubit plus oscillator; since the kernel is determined by $\Pi(t)$ alone (Exercise 5.4), it is also the kernel of the physical Lorentzian bath.

Approximations. $K^{(2)}(\tau)$ from Eq. (56); NZ2 solves the Volterra equation with the trapezoidal rule and an implicit treatment of $K(0)\rho_n$; TCL2 uses $\mathcal{G}^{(2)}(t) = \int_0^t K^{(2)}(\tau)\mathrm{e}^{-\mathcal{L}_S\tau}\mathrm{d}\tau$ (cumulative trapezoid) and the midpoint exponential; Markov uses $\mathcal{G}^{(2)}(t_{\max})$.

Checks printed by the script (for $g = 0.3$):

quantity	analytic (Section 5)	numerical ($\mathcal{G}^{(2)}_\infty$)
$\Gamma_\downarrow$, $\Gamma_\uparrow$	0.36000, 0.00554	—
$1/T_1 = -\mathcal{G}_{zz}$	0.36554	0.36552
$1/T_2 = -\frac{1}{2}(\mathcal{G}_{xx} + \mathcal{G}_{yy})$	0.22512	0.22512
$\Gamma_\varphi = -\mathcal{G}_{xx}$	0.04235	0.0424
$\tilde{\omega}_0$ (secular / nonsecular)	1.0222	1.0219
$r_z^{\mathrm{eq}} = \mathcal{G}_{z\mathbb{1}}/(-\mathcal{G}_{zz})$	−0.96970	−0.96970

The small differences are the $\mathrm{d}t$ discretization and the nonsecular versus secular definitions. The nonsecular generator has $\mathcal{G}_{xx} = -\Gamma_\varphi$ and $\mathcal{G}_{yy} = -\Gamma_\varphi - \Gamma_1$, whose average is Γ_2 (Example 5.3), and $\mathcal{G}_{yx} = 2\delta\omega$, $\mathcal{G}_{xy} = 0$.

Check of Theorem 5.4. The function `check_fourth_order` evaluates the fourth-order irreducible kernel $K^{(4)}$ as a double time integral of $\mathcal{P}\mathcal{L}_I\mathrm{e}^{\mathcal{L}_0\cdot}\mathcal{Q}\mathcal{L}_I\mathrm{e}^{\mathcal{L}_0\cdot}\mathcal{Q}\mathcal{L}_I\mathrm{e}^{\mathcal{L}_0\cdot}\mathcal{Q}\mathcal{L}_I\mathcal{P}$ in the pseudomode space and compares $K - K^{(2)}$ with $K^{(4)}$ for $g = 0.05, 0.1, 0.2$ (Example 5.1).

Filter functions. Figure 7 evaluates Eq. (117) with the $1/\pi$ normalization (coupling $\frac{1}{2}\sigma_z \otimes B$) and $g = 0.2$, for Ramsey, Hahn and CPMG-4 sequences.

B Solutions and hints

Exercise 1.1. $(A^{\mathrm{c}})^\dagger$: $(X, A^{\mathrm{c}}Y) = \frac{1}{2}\operatorname{Tr}[X^\dagger(AY + YA)] = \frac{1}{2}\operatorname{Tr}[(AX + XA)^\dagger Y] = (A^{\mathrm{c}}X, Y)$ using $A = A^\dagger$ and cyclicity. $(A^{\mathrm{q}})^\dagger$: $(X, [A, Y]) = \operatorname{Tr}[X^\dagger AY - X^\dagger YA] = \operatorname{Tr}[(AX - XA)^\dagger(-1)\cdots]$, giving $-(A^{\mathrm{q}}X, Y)$. Hence iA^{q} is Hermitian and $\mathcal{L} = -iH^{\mathrm{q}}$ is anti-Hermitian, as the generator of a unitary (isometric) evolution in Liouville space must be.

Exercise 1.2. With vec in the basis $\{|0\rangle\langle 0|, |0\rangle\langle 1|, |1\rangle\langle 0|, |1\rangle\langle 1|\}$, $\mathcal{L}_S = -i(\mathbb{1} \otimes H_S - H_S^T \otimes \mathbb{1}) = \operatorname{diag}(0, -i\omega_0, i\omega_0, 0)$. The change of basis to $\sigma_\mu/\sqrt{2}$ is the unitary with columns $\operatorname{vec}(\sigma_\mu)/\sqrt{2}$; it maps the 2×2 block $\operatorname{diag}(-i\omega_0, i\omega_0)$ on $(|0\rangle\langle 1|, |1\rangle\langle 0|)$ to the real rotation generator $\left(\begin{smallmatrix} 0 & -\omega_0 \\ \omega_0 & 0 \end{smallmatrix}\right)$ on (σ_x, σ_y).

Exercise 1.3. Trace preservation fixes the first row; Hermiticity preservation makes the matrix real. For a unital qubit map with $T = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$, positivity is $|\lambda_i| \le 1$ while complete positivity (Fujiwara–Algoet conditions) is $|\lambda_1 \pm \lambda_2| \le |1 \pm \lambda_3|$; transposition $(1, -1, 1)$ violates the latter.

Exercise 1.4. $(S^{\mathrm{c}})^2 = \frac{1}{4}(S^+ + S^-)^2 = \frac{1}{4}(2 + 2S^+S^-)$ using $S^2 = \mathbb{1}$: it maps $X \mapsto \frac{1}{2}(X + SXS)$. Similarly $(S^{\mathrm{q}})^2 X = 2X - 2SXS$. Hence $(S^{\mathrm{c}})^2 + \frac{1}{4}(S^{\mathrm{q}})^2 = 1$ and $(S^{\mathrm{c}})^2 - \frac{1}{4}(S^{\mathrm{q}})^2 = S^+S^-$, i.e. $X \mapsto SXS$. In matrix form, $(S^{\mathrm{c}})^2$ projects onto $\operatorname{span}(\mathbb{1}, S)$ and $\frac{1}{4}(S^{\mathrm{q}})^2$ onto its complement (the two Pauli directions orthogonal to $\hat{n}$).

Exercise 1.5. $[A^+, B^+]X = ABX - BAX = ([A, B])^+ X$; $[A^-, B^-]X = XBA - XAB = -([A, B])^- X$; $[A^+, B^-]X = AXB - AXB = 0$. For Eq. (11) insert $A^\pm = A^{\mathrm{c}} \pm \frac{1}{2}A^{\mathrm{q}}$ into $(AB)^{\mathrm{c}} = \frac{1}{2}(A^+B^+ + B^-A^-)$: $A^+B^+ = A^{\mathrm{c}}B^{\mathrm{c}} + \frac{1}{2}A^{\mathrm{c}}B^{\mathrm{q}} + \frac{1}{2}A^{\mathrm{q}}B^{\mathrm{c}} + \frac{1}{4}A^{\mathrm{q}}B^{\mathrm{q}}$ and $B^-A^- = B^{\mathrm{c}}A^{\mathrm{c}} - \frac{1}{2}B^{\mathrm{c}}A^{\mathrm{q}} - \frac{1}{2}B^{\mathrm{q}}A^{\mathrm{c}} + \frac{1}{4}B^{\mathrm{q}}A^{\mathrm{q}}$; adding and halving gives the anticommutators and the two commutators, and subtracting gives $(AB)^{\mathrm{q}}$. For $A = \sigma_x$, $B = \sigma_y$ the commutator terms are essential: the two anticommutator terms cancel each other, $\frac{1}{2}\{\sigma_x^{\mathrm{c}}, \sigma_y^{\mathrm{c}}\} = -\frac{1}{8}\{\sigma_x^{\mathrm{q}}, \sigma_y^{\mathrm{q}}\}$: the first swaps $\sigma_x \leftrightarrow \sigma_y$ with weight $\frac{1}{2}$ and annihilates $\mathbb{1}$ and σ_z, the second is its negative (check with the matrices of Example 1.2), while $\frac{1}{4}([\sigma_x^{\mathrm{c}}, \sigma_y^{\mathrm{q}}] + [\sigma_x^{\mathrm{q}}, \sigma_y^{\mathrm{c}}]) = i\sigma_z^{\mathrm{c}}$ exactly, which is the whole answer $(\sigma_x\sigma_y)^{\mathrm{c}} = (i\sigma_z)^{\mathrm{c}}$. The script `spin_boson_numerics.py` contains the vectorized superoperators; the identities hold to machine precision for random A, B.

Exercise 2.1. Straightforward from $b_k(t) = b_k \mathrm{e}^{-i\omega_k t}$, $\langle b_k^\dagger b_k\rangle = \bar{n}_k$, $\langle b_k b_k^\dagger\rangle = \bar{n}_k + 1$.

Exercise 2.2. From Eq. (26), $\chi(\omega) = \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega' J(\omega')\, I(\omega, \omega')$ with $I(\omega, \omega') = \int_0^\infty \mathrm{d}t\, \mathrm{e}^{i\omega t} \sin\omega' t$. This integral does not converge in the ordinary sense; it is defined as a distribution by a convergence factor $\mathrm{e}^{-\epsilon t}$, $\epsilon \to 0^+$ (adiabatic switching; the one-sided range comes from the $\theta(t)$ in χ). *Step 1.* For real k,

$$\int_0^\infty \mathrm{d}t\, \mathrm{e}^{ikt - \epsilon t} = \frac{1}{\epsilon - ik} = \frac{\epsilon}{k^2 + \epsilon^2} + i\,\frac{k}{k^2 + \epsilon^2} \xrightarrow{\epsilon \to 0^+} \pi\delta(k) + i\,\mathrm{P}\frac{1}{k},$$

because the Lorentzian $\epsilon/(k^2 + \epsilon^2)$ has width ϵ and area π, while $k/(k^2 + \epsilon^2)$ is odd and equals $1/k$ away from $k = 0$ (Sokhotski–Plemelj). Compare the two-sided integral $\int_{-\infty}^\infty \mathrm{e}^{ikt}\mathrm{d}t = 2\pi\delta(k)$: the one-sided one is half the delta plus the odd principal-value part that the other half would cancel.

Step 2. Write $\sin\omega' t = (\mathrm{e}^{i\omega' t} - \mathrm{e}^{-i\omega' t})/2i$ and apply Step 1 with $k = \omega \pm \omega'$:

$$I = \frac{1}{2i}\Big[\pi\delta(\omega+\omega') - \pi\delta(\omega-\omega') + i\,\mathrm{P}\frac{1}{\omega+\omega'} - i\,\mathrm{P}\frac{1}{\omega-\omega'}\Big],$$
$$\mathrm{Re}\, I = \frac{1}{2}\Big[\mathrm{P}\frac{1}{\omega+\omega'} - \mathrm{P}\frac{1}{\omega-\omega'}\Big] = -\,\mathrm{P}\frac{\omega'}{\omega^2-\omega'^2}, \qquad \mathrm{Im}\, I = \frac{\pi}{2}\big[\delta(\omega-\omega') - \delta(\omega+\omega')\big].$$

(Without complex exponentials: $\int_0^\infty \sin\omega' t\,\sin\omega t\,\mathrm{e}^{-\epsilon t}\mathrm{d}t = \frac{1}{2}\big[\frac{\epsilon}{(\omega-\omega')^2+\epsilon^2} - \frac{\epsilon}{(\omega+\omega')^2+\epsilon^2}\big]$, two Lorentzians that become the two deltas.) *Step 3.* Insert into $\chi(\omega)$. The deltas are integrated against $J(\omega')$ over $\omega' > 0$, so for $\omega > 0$ only $\delta(\omega-\omega')$ fires and $\chi''(\omega) = J(\omega)$, while for $\omega < 0$ only $\delta(\omega+\omega')$ fires, with its minus sign, and $\chi''(\omega) = -J(-\omega)$: together $\chi''(\omega) = \mathrm{sgn}(\omega)J(|\omega|)$, odd in ω as a dissipative response must be. The real part of the same formula gives $\chi'(\omega) = \frac{2}{\pi}\,\mathrm{P}\int_0^\infty \mathrm{d}\omega'\,\omega' J(\omega')/(\omega'^2-\omega^2)$, the Kramers–Kronig relation of Section 2: one regularized integral yields both the dissipative part (deltas) and the reactive part (principal value), tied together by the causal $\theta(t)$. Two remarks. The sign of $i\epsilon$ is the only place where retarded versus advanced is decided: $\int_{-\infty}^0 \mathrm{e}^{ikt+\epsilon t}\mathrm{d}t = \pi\delta(k) - i\mathrm{P}(1/k)$, same deltas, opposite principal value, hence $\chi^A(\omega) = \chi^R(\omega)^*$. And the finite-time version, $\mathrm{Re}\int_0^t \mathrm{e}^{i(\omega-\omega_0)\tau}\mathrm{d}\tau = \sin[(\omega-\omega_0)t]/(\omega-\omega_0) \to \pi\delta(\omega-\omega_0)$, is the filter of Eq. (116) that turns the running rates of Section 6 into Markov rates: a sharp cutoff in time instead of an exponential one.

For the finite-T Ohmic kernel use $\int_0^\infty \omega\coth(\beta\omega/2)\cos\omega t\,\mathrm{d}\omega = -(\pi/\beta)^2/\sinh^2(\pi t/\beta)$ for $t \neq 0$ (the $t = 0$ singularity carries the positive weight that makes $S(\omega) \geq 0$). $S(t) < 0$ for all $t > 0$ is compatible with $S(\omega) \geq 0$ because $S(\omega) = \int S(t)\mathrm{e}^{i\omega t}\mathrm{d}t$ is dominated by the short-time peak.

Exercise 2.3. The adjoint master equation gives $\dot a = -(i\Omega+\gamma/2)a$ for the Heisenberg operator in the absence of the qubit; with the quantum regression theorem, $\langle a(t)a^\dagger\rangle = \mathrm{e}^{-(i\Omega+\gamma/2)t}$. Gaussianity: the Lindbladian is quadratic in $a, a^\dagger$, so it maps Gaussian states to Gaussian states.

Exercise 2.4. $\mathrm{Tr}_B[(\cos\theta\sigma_z+\sin\theta\sigma_x)B\cdots(\cos\theta\sigma_z+\sin\theta\sigma_x)B]$ contains $\cos\theta\sin\theta\,C(t)$ cross terms between σ_z and σ_x vertices. Two independent baths would give $C_{zz} = \cos^2\theta C$, $C_{xx} = \sin^2\theta C$, $C_{zx} = 0$. The cross terms are the *cs* entries of Eq. (57).

Exercise 3.1. Eight terms $\tilde{\mathcal{L}}_I^{a_1}\tilde{\mathcal{L}}_I^{a_2}\tilde{\mathcal{L}}_I^{a_3}$, $a_i = \pm$, each a contour-ordered three-point function of B, which vanishes for a Gaussian state with $\langle B\rangle = 0$.

Exercise 3.2. For $t > s$: $D^{\mathrm{cq}} = \frac{1}{2}[C(t-s) - C(s-t) + C(t-s) - C(s-t)] = C(t-s) - C(s-t) = \langle[B(t), B(s)]\rangle$. For $t < s$ the later superoperator is $B^{\mathrm{q}}(s)$ and the trace identity kills the term. Without the table: write D^{ab} as $\mathrm{Tr}_B[\mathrm{later}\cdot\mathrm{earlier}\cdot\rho_B]$; if the later one is B^{q} the trace vanishes; if both are B^{c} the result is symmetric in t, s.

Exercise 3.3. $\mathrm{Tr}_S[\tilde S^{\mathrm{c}}(t)\tilde S^{\mathrm{c}}(s)\rho_S] = \frac{1}{2}\langle\{\tilde S(t), \tilde S(s)\}\rangle = \cos^2\theta + \sin^2\theta\cos\omega_0(t-s)$ for the diagonal state (independent of r_z), and $\mathrm{Tr}_S[\tilde S^{\mathrm{c}}(t)\tilde S^{\mathrm{q}}(s)\rho_S] = \langle[\tilde S(t), \tilde S(s)]\rangle = -2i\sin^2\theta\, r_z\sin\omega_0(t-s)$ for $t > s$: the qubit's response function, proportional to its polarization, as a two-level system's susceptibility must be.

Exercise 3.4. See the text; $\chi''(\omega) = J(\omega) - J(-\omega)$ and $S(\omega) = J(\omega) + J(-\omega)$ give $S = \mathrm{sgn}(\omega)\chi''$ at $T = 0$ only for $|\omega|$ where $J(-|\omega|) \ll J(|\omega|)$; the negative-frequency tail is the price of the pseudomode.

Exercise 4.1. $\mathrm{Tr}_S\, K^{(2)}X = -\,\mathrm{Tr}_S[S^{\mathrm{q}}(\cdots)] = 0$ by the first trace identity for S. Hermiticity: with $Y = \mathrm{e}^{\mathcal{L}_S\tau}[S(\tau)S^{\mathrm{q}}X - i\chi(\tau)S^{\mathrm{c}}X]$, $Y^\dagger = \mathrm{e}^{\mathcal{L}_S\tau}[-S(\tau)S^{\mathrm{q}}X^\dagger + i\chi S^{\mathrm{c}}X^\dagger]$, so $(S^{\mathrm{q}}Y)^\dagger = -S^{\mathrm{q}}Y^\dagger = S^{\mathrm{q}}\mathrm{e}^{\mathcal{L}_S\tau}[S\,S^{\mathrm{q}} - i\chi S^{\mathrm{c}}]X^\dagger$ for Hermitian X.

Exercise 4.2. Input σ_z: $S^{\mathrm{c}}\sigma_z = \cos\theta\,\mathbb{1}$, $S^{\mathrm{q}}\sigma_z = -2i\sin\theta\,\sigma_y$. Free rotation: $\mathrm{e}^{\mathcal{L}_S\tau}\sigma_y = \cos\omega_0\tau\,\sigma_y - \sin\omega_0\tau\,\sigma_x$, $\mathrm{e}^{\mathcal{L}_S\tau}\mathbb{1} = \mathbb{1}$. Then $-S^{\mathrm{q}}[\,S_\tau(-2i\sin\theta)(\cos\sigma_y - \sin\sigma_x) - i\chi_\tau\cos\theta\,\mathbb{1}\,]$; use $S^{\mathrm{q}}\mathbb{1} = 0$, $S^{\mathrm{q}}\sigma_y = -2i\cos\theta\sigma_x + 2i\sin\theta\sigma_z$, $S^{\mathrm{q}}\sigma_x = 2i\cos\theta\sigma_y$: $= -S_\tau\sin\theta\,[-4\cos\theta\cos\sigma_x + 4\sin\theta\cos\sigma_z - 4\cos\theta\sin\sigma_y]$, giving the last column of Eq. (57). Input $\mathbb{1}$: $S^{\mathrm{c}}\mathbb{1} = S$, $S^{\mathrm{q}}\mathbb{1} = 0$; $\mathrm{e}^{\mathcal{L}_S\tau}S = \cos\theta\sigma_z + \sin\theta(\cos\sigma_x + \sin\sigma_y)$; then $i\chi_\tau S^{\mathrm{q}}[\cdots] = \chi_\tau[2cs(1-\cos)\sigma_y + 2cs\sin\sigma_x - 2s^2\sin\sigma_z]$, the first column. The σ_x, σ_y columns are analogous. At $\theta = 0$ only the c^2 block survives (dephasing of σ_x, σ_y at rate $4S_\tau$ with the rotation); at $\theta = \pi/2$ only the s^2 entries (relaxation of σ_z, dephasing of σ_y but not of σ_x in the lab frame, and the χ-driven inhomogeneity).

Exercise 4.3. The 15 pairings of six ordered points split into 10 irreducible and 5 reducible: (12)(34)(56), (12)×{2 irreducible 4-point}, {2 irreducible 4-point}×(56). (Irreducible chord diagrams: $1, 1, 2, 10, 74, \ldots$)

Exercise 4.4. $\Gamma(t) = 4\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, g^2\mathrm{e}^{-\gamma(t_1-t_2)/2}\cos\Omega(t_1-t_2)$. Short times: $4g^2t^2/2 = 2g^2t^2$. Long times: $4t\int_0^\infty S(\tau)\mathrm{d}\tau = 2S(\omega{=}0)\,t$ with $S(0) = 2g^2(\gamma/2)/(\Omega^2+\gamma^2/4)$, the zero-frequency value of $J(\omega) + J(-\omega)$.

Exercise 5.1. $\mathcal{PLP} = \mathcal{L}_S\mathcal{P}$ uses $\mathcal{PL}_I\mathcal{P} = 0$; $\mathcal{PL}\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}\mathcal{L}\mathcal{P} = \mathcal{PL}_I\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}\mathcal{L}_I\mathcal{P}$ uses $\mathcal{PL}_0\mathcal{Q} = 0$. With $\langle B\rangle \neq 0$, write $B = \langle B\rangle + \delta B$ and absorb $S\langle B\rangle$ into H_S.

Exercise 5.2. Secular reduction of Eq. (57) with $\mathrm{e}^{-\mathcal{L}_S\tau}$ acting on the input: $\mathcal{G}_{xx} = -4c^2\int S$, $\mathcal{G}_{yy} = -4c^2\int S - 4s^2\int S\cos\omega_0\tau$, $\mathcal{G}_{yx} = 4s^2\int S\sin\omega_0\tau$, $\mathcal{G}_{xy} = 0$, $\mathcal{G}_{zz} = -4s^2\int S\cos\omega_0\tau$, $\mathcal{G}_{z\mathbb{1}} = -2s^2\int\chi\sin\omega_0\tau$. Average $\mathcal{G}_{xx}$ and $\mathcal{G}_{yy}$ (the difference oscillates at $2\omega_0$ in the rotating frame) to get Γ_2; the precession frequency is $\sqrt{\omega_0(\omega_0+\mathcal{G}_{yx})} \simeq \omega_0 + \frac{1}{2}\mathcal{G}_{yx}$. The Lamb shift contains $\int_0^\infty S\sin\omega_0\tau$ only: the χ terms sit in the first column, which is the inhomogeneity, not a frequency. In the operator form, $\Delta(\omega) = \mathrm{Im}\int_0^\infty \mathrm{e}^{i\omega s}C(s)\mathrm{d}s = \int_0^\infty S\sin\omega s - \frac{1}{2}\chi'(\omega)$ and $\delta\omega = \sin^2\theta[\Delta(\omega_0) - \Delta(-\omega_0)]$; χ' is even and cancels.

Exercise 5.3. Secular: $\mathcal{G} = -i[\tilde{H}_S, \cdot] + \Gamma_\downarrow\mathcal{D}[\sigma_-] + \Gamma_\uparrow\mathcal{D}[\sigma_+] + \frac{1}{2}\Gamma_\varphi\mathcal{D}[\sigma_z]$ with $\mathcal{D}[L]X = LXL^\dagger - \frac{1}{2}\{L^\dagger L, X\}$. Nonsecular Redfield for $\theta = \pi/2$ has $\mathcal{G}_{xx} = 0$, $\mathcal{G}_{yy} = -\Gamma_1$: starting from $|+y\rangle$ the Bloch vector initially shrinks along y but not along x, which is fine; the violation appears through the $\mathcal{G}_{yx} = 2\delta\omega$ term combined with the inhomogeneity when $\delta\omega$ is comparable to Γ_1, i.e. when $\gamma \gtrsim \omega_0$ so that $S(\omega)$ is broad.

Exercise 5.4. $K(0) = \mathrm{Tr}_B[\mathcal{L}_I\mathcal{Q}\mathcal{L}_I(X\otimes\rho_B)] = \mathrm{Tr}_B[\mathcal{L}_I\mathcal{L}_I(X\otimes\rho_B)]$ since $\mathcal{PL}_I\mathcal{P} = 0$. Laplace: invert Eq. (65). For $\theta = 0$, $\hat{\Pi}_{01}(z) = \int_0^\infty \mathrm{e}^{-(z+i\omega_0)t-\Gamma(t)}\mathrm{d}t$, and $\hat{K}_{01}(z) = z + i\omega_0 - 1/\hat{\Pi}_{01}$; expanding $\mathrm{e}^{-\Gamma} = 1 - \Gamma + \Gamma^2/2 - \ldots$ with $\Gamma = O(g^2)$ produces all powers of g^2 in $\hat{K}$: the irreducible crossed and nested diagrams do not cancel.

Exercise 6.1. $\Sigma(t) = \int_0^t \hat{G}(t,s)\mathcal{Q}\tilde{\mathcal{L}}_I(s)\mathcal{P}\tilde{G}(s,t)\mathrm{d}s$ is explicitly $O(\mathcal{L}_I)$ and vanishes at $t = 0$. Expanding $\hat{G}$, $\tilde{G}$ and $[1-\Sigma]^{-1}$ and using $\mathcal{P}\tilde{\mathcal{L}}_I\mathcal{P} = 0$ gives Eq. (88) at second order and Eq. (89) at fourth, where the subtracted terms come from $\mathcal{P}\tilde{\mathcal{L}}_I\Sigma^2\mathcal{P}$ and from the expansion of $\tilde{G}(s,t)$ inside Σ. Odd orders have an odd number of B's.

Exercise 6.2. $\int_0^t S(\tau)\mathrm{d}\tau = g^2\,\mathrm{Re}\left[(1-\mathrm{e}^{-(\gamma/2-i\Omega)t})/(\gamma/2-i\Omega)\right]$; negative windows exist when $\Omega > \gamma/2$, the first around $\Omega t \in (\pi, 2\pi)$ for small γ/Ω.

Exercise 6.3. With $\Pi^{(2)}(t) = \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \Pi_0(t-t_1)K^{(2)}(t_1-t_2)\Pi_0(t_2)$ and $\Pi^{-1} = \Pi_0^{-1} - \Pi_0^{-1}\Pi^{(2)}\Pi_0^{-1} + \ldots$, the fourth-order term of Eq. (92) is $\int K^{(4)}(t-s)\Pi_0(s)\Pi_0(t)^{-1} + \int K^{(2)}(t-s)[\Pi^{(2)}(s)\Pi_0(t)^{-1} - \Pi_0(s)\Pi_0(t)^{-1}\Pi^{(2)}(t)\Pi_0(t)^{-1}]$. The bracket, written as a time-ordered integral, is $-\int\int\int$ of the products $\langle\mathcal{L}_I\mathcal{L}_I\rangle\langle\mathcal{L}_I\mathcal{L}_I\rangle$ in the crossed and nested time orderings, which are exactly the last two subtractions of Eq. (89); the reducible product is already absent from $K^{(4)}$.

Exercise 6.4. $\hat\rho_{ee}(z) = \widehat{|c_e|^2}(z)\rho_{ee}(0)$ and $\hat K_{ee}(z) = z - 1/\widehat{|c_e|^2}(z)$; the Laplace transform of $|c_e|^2$ is a rational function with poles at $-\gamma/2 \pm i|d|/2$ and $-\gamma/2$, so $\hat K$ is analytic to the right of them. The generator's rate $\gamma(t)$ has poles on the real t axis, so its Laplace transform does not exist as an ordinary function.

Exercise 7.1. Substitute $s^\pm = s^\mathrm{c} \pm \frac12 s^\mathrm{q}$ and $C = S - \frac{i}{2}\chi$: $(s_1^+ - s_1^-)[Cs_2^+ - C^*s_2^-] = s_1^\mathrm{q}[S(s_2^+ - s_2^-) - \frac{i}{2}\chi(s_2^+ + s_2^-)] = s_1^\mathrm{q}[Ss_2^\mathrm{q} - i\chi s_2^\mathrm{c}]$. Induction: the sum over pairings of $2n$ commuting vertices is $\frac{1}{n!}(\text{one line})^n$.

Exercise 7.2. For a blip $s^\mathrm{q} = \pm 2$ on $[t_a, t_b]$, $-\iint s^\mathrm{q}Ss^\mathrm{q} = -4\int_{t_a}^{t_b}\mathrm{d}t_1\int_{t_a}^{t_1}\mathrm{d}t_2\, S(t_1-t_2) = -Q_2(t_b-t_a)$ using $\ddot Q_2 = 4S$, $Q_2(0) = \dot Q_2(0) = 0$. The blip–sojourn term with the preceding sojourn $s^\mathrm{c} = \pm 1$ on $[t_0, t_a]$ gives $\pm i\cdot 2\int_{t_a}^{t_b}\mathrm{d}t_1\int_{t_0}^{t_a}\mathrm{d}t_2\,\chi(t_1-t_2)$, which for $t_0 \to -\infty$ is $\mp iQ_1(t_b-t_a)$ with $\ddot Q_1 = -2\chi$.

Exercise 7.3. $B_+(\tau) = \prod_k D(\alpha_k \mathrm{e}^{i\omega_k\tau})$ with $\alpha_k = 2g_k/\omega_k$. $D(\alpha_\tau)D(-\alpha) = \mathrm{e}^{-i\alpha^2\sin\omega\tau}D(\alpha(\mathrm{e}^{i\omega\tau}-1))$ and $|\alpha(\mathrm{e}^{i\omega\tau}-1)|^2 = 2\alpha^2(1-\cos\omega\tau)$, so $\langle B_+(\tau)B_-(0)\rangle = \mathrm{e}^{-Q_2-iQ_1}$ and, with the other ordering, $\langle B_-(0)B_+(\tau)\rangle = \mathrm{e}^{-Q_2+iQ_1}$. Detailed balance: expand to first order in Q, $\int_0^\infty \mathrm{e}^{-i\varepsilon\tau}(-Q_2+iQ_1)\mathrm{d}\tau$ picks $+2J(\varepsilon)[\coth(\beta\varepsilon/2)+1]/\varepsilon^2$, hence $w_{\downarrow\leftarrow\uparrow} \propto 1+n$, $w_{\uparrow\leftarrow\downarrow} \propto n$.

Exercise 7.4. (a) $\hat K^s(z)$ for the Ohmic kernel is $\Delta^2\int_0^\infty \mathrm{e}^{-z\tau}(1+\omega_c^2\tau^2)^{-\alpha}\cos(2\alpha\arctan\omega_c\tau)\,\mathrm{d}\tau$; for $z \ll \omega_c$ it behaves as $\hat K^s(z) \simeq \Delta^2\,\Gamma(1-2\alpha)\cos(\pi\alpha)\,z^{2\alpha-1}\omega_c^{-2\alpha}$, giving poles of $\hat P$ at $z \sim \Delta_r\,\mathrm{e}^{\pm i\pi/(2-2\alpha)}$ with $\Delta_r = \Delta(\Delta/\omega_c)^{\alpha/(1-\alpha)} \times O(1)$; for $\alpha \geq \frac12$ the poles are on the negative real axis (incoherent decay) [9, Sec. V]. (b) See Example 7.2.

Exercise 8.1. Apply the identity to $i\mathsf{L}A$: $\dot A(t) = \mathrm{e}^{i\mathsf{L}t}i\mathsf{L}A = \mathrm{e}^{i\mathsf{L}t}[\mathsf{P}+\mathsf{Q}]i\mathsf{L}A$; the P part is $i\Omega A(t)$, the Q part is expanded with the identity into $F(t)$ plus $\int_0^t \mathrm{e}^{i\mathsf{L}(t-s)}i\mathsf{PL}F(s)\mathrm{d}s$, and $i\mathsf{PL}F(s) = A(A,A)^{-1}(A, i\mathsf{L}F(s)) = -A(A,A)^{-1}(i\mathsf{L}A, F(s)) = -AM(s)$ using self-adjointness of L and $(A,F) = 0$.

Exercise 8.2. Project on the *fluctuation* $A = \delta\sigma_z = \sigma_z - \langle\sigma_z\rangle$, with $(\delta\sigma_z, \delta\sigma_z) = 1-\tanh^2(\beta\omega_0/2)$ (projecting on σ_z itself keeps the non-decaying part $\langle\sigma_z\rangle^2$ in $C(t)$ and gives a memory function that vanishes at $z=0$). Then $i\mathsf{L}\delta\sigma_z = i\sin\theta[\sigma_x,\sigma_z]B = 2\sin\theta\,\sigma_y B = F(0)$, $F(t) = 2\sin\theta\,\sigma_y(t)B(t)$ with $\sigma_y(t) = \sigma_y\cos\omega_0 t + \sigma_x\sin\omega_0 t$, and $\langle\sigma_y\sigma_y(t)\rangle = \cos\omega_0 t + i\tanh(\beta\omega_0/2)\sin\omega_0 t$, $\langle\sigma_y(t)\sigma_y\rangle =$ its conjugate. Hence

$$(F, F(t)) = 2\sin^2\theta\big[\langle\sigma_y\sigma_y(t)\rangle C(-t) + \langle\sigma_y(t)\sigma_y\rangle C(t)\big] = 4\sin^2\theta\Big[S(t)\cos\omega_0 t - \tfrac12\tanh\tfrac{\beta\omega_0}{2}\,\chi(t)\sin\omega_0 t\Big],$$

and $\hat{M}(0) = 2\sin^2\theta[S(\omega_0) - \tanh(\beta\omega_0/2)\chi''(\omega_0)]/(1-\tanh^2) = 2\sin^2\theta S(\omega_0) = \Gamma_1$ by the FDT. The response kernel enters with the system's polarization and is what makes the normalization work out; there is no inhomogeneous term because the equilibrium value sits in the reference state and the GLE describes the fluctuation around it.

Exercise 8.3. With $E_\mu = \sigma_\mu/\sqrt{2}$, $(E_\mu, \mathcal{L}^\dagger E_\nu)_B = \frac{1}{2}\operatorname{Tr}[\sigma_\mu(i[H,\sigma_\nu])\rho_B]$, which is the transpose of $\frac{1}{2}\operatorname{Tr}[\sigma_\nu(-i[H,\sigma_\mu])\rho_B]$: Heisenberg generators are the transposes of Schrödinger ones in a real orthonormal basis. The same holds for the kernel.

Exercise 8.4. In the RWA, $\mathsf{Q}i\mathsf{L}$ acting on $F \propto \sigma_- a^\dagger$-type operators closes on $\sigma_- a$ (one excitation), and the second-level memory is the bath's own correlator $\mathrm{e}^{-(i\Omega+\gamma/2)t}$. The continued fraction $\hat{C}(z) = \frac{1}{2}/[z + i\omega_0 + g^2/(z + i\Omega + \gamma/2)]$ has the poles of $c_e(t)$.

Exercise 9.1. Each h^{c} derivative inserts an S^{c} vertex as the *last* operation before the trace in at least one term; by the trace identity the full sum vanishes.

Exercise 9.2. $\sigma_z = -i\eta_x\eta_y$, etc.; $\Theta\eta_x = -i\eta_x\eta_y\eta_z\eta_x = -i\eta_y\eta_z = \sigma_x$. $H_I = S \otimes B$ becomes $\Theta(\cos\theta\eta_z + \sin\theta\eta_x)B$; since Θ is conserved, in each sector $\Theta = \pm 1$ the vertex is linear in one Majorana and in B: the second-order self-energy of η_i is a bath line times a δ_{ij} Majorana loop-free diagram, i.e. $\Sigma_\eta \propto D$.

Exercise 9.3. Follow [64, Sec. 2] with fermionic (Majorana) fields; the result is $\Sigma^R = -i(\Gamma_\downarrow + \Gamma_\uparrow)/2$ and $\Sigma^K = -i(\Gamma_\downarrow - \Gamma_\uparrow)$ for the σ_- propagator up to normalization, so that $\Sigma^K/(\Sigma^R - \Sigma^A) = (\Gamma_\downarrow - \Gamma_\uparrow)/(\Gamma_\downarrow + \Gamma_\uparrow) = \tanh(\beta\omega_0/2)$ iff detailed balance holds.

Exercise 10.1. Insert $C(\tau) = \int \frac{\mathrm{d}\omega}{2\pi} C(\omega)\mathrm{e}^{-i\omega\tau}$ and use $2\,\mathrm{Re}\int_0^t \mathrm{e}^{i(\omega_0-\omega)\tau}\mathrm{d}\tau = 2\sin[(\omega-\omega_0)t]/(\omega-\omega_0)$. A modulation $\omega_0 \to \omega_0 + \delta\cos\nu t$ replaces the sinc by a sum of sincs centred at $\omega_0 + k\nu$ weighted by Bessel functions. If $C(\omega)$ grows above ω_0 the broadened filter samples more spectral weight and the decay accelerates (anti-Zeno).

Exercise 10.2. $\int_0^t y\mathrm{e}^{i\omega s}\mathrm{d}s = -\mathrm{e}^{i\omega t/2}[2\cos(\omega t/2) - 2]/(i\omega)$, so $|\cdot|^2 = 16\sin^4(\omega t/4)/\omega^2$. For the comb, expand the square wave, $y = \frac{4}{\pi}\sum_{k\ \mathrm{odd}} k^{-1}\sin(k\pi s/\tau)$, and use $|\int_0^t \mathrm{e}^{i(\omega - k\pi/\tau)s}\mathrm{d}s|^2 \to 2\pi t\,\delta(\omega - k\pi/\tau)$. The first tooth hits Ω when $n\pi/t = \Omega$, i.e. $t = n\pi/\Omega$ ($= 4\pi$ for $n = 4$).

Exercise 10.3. Maximize $t\mathrm{e}^{-2\Gamma_2 t - 2\kappa t^2}$: the quadratic term matters when $\kappa t^* \gtrsim \Gamma_2$, i.e. $t^* \lesssim \tau_B$; for GHZ states $t^* \sim 1/(2N\Gamma_2)$, so the Zeno regime is reached for $N \gtrsim 1/(\Gamma_2\tau_B)$. For T_1-limited sensing (signal $\propto \mathrm{e}^{-\Gamma_1 t}$ with the phase on the populations) the same algebra applies with Γ_1 in place of $2\Gamma_2$.

Exercise 10.4. Implement the pulse as $\rho \to \sigma_x\rho\sigma_x$ on the qubit factor at $t/2$ in the pseudomode simulation. Expect (b) to be closer than (a), with a residual oscillating discrepancy of relative size $\sim g^2\sin\theta\cos\theta/\omega_0$ at short times and a drift of the echo "baseline" from the χ-driven inhomogeneity.

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